

EACTS/STS/AATS Guidelines on temporary mechanical circulatory support in adult cardiac surgery

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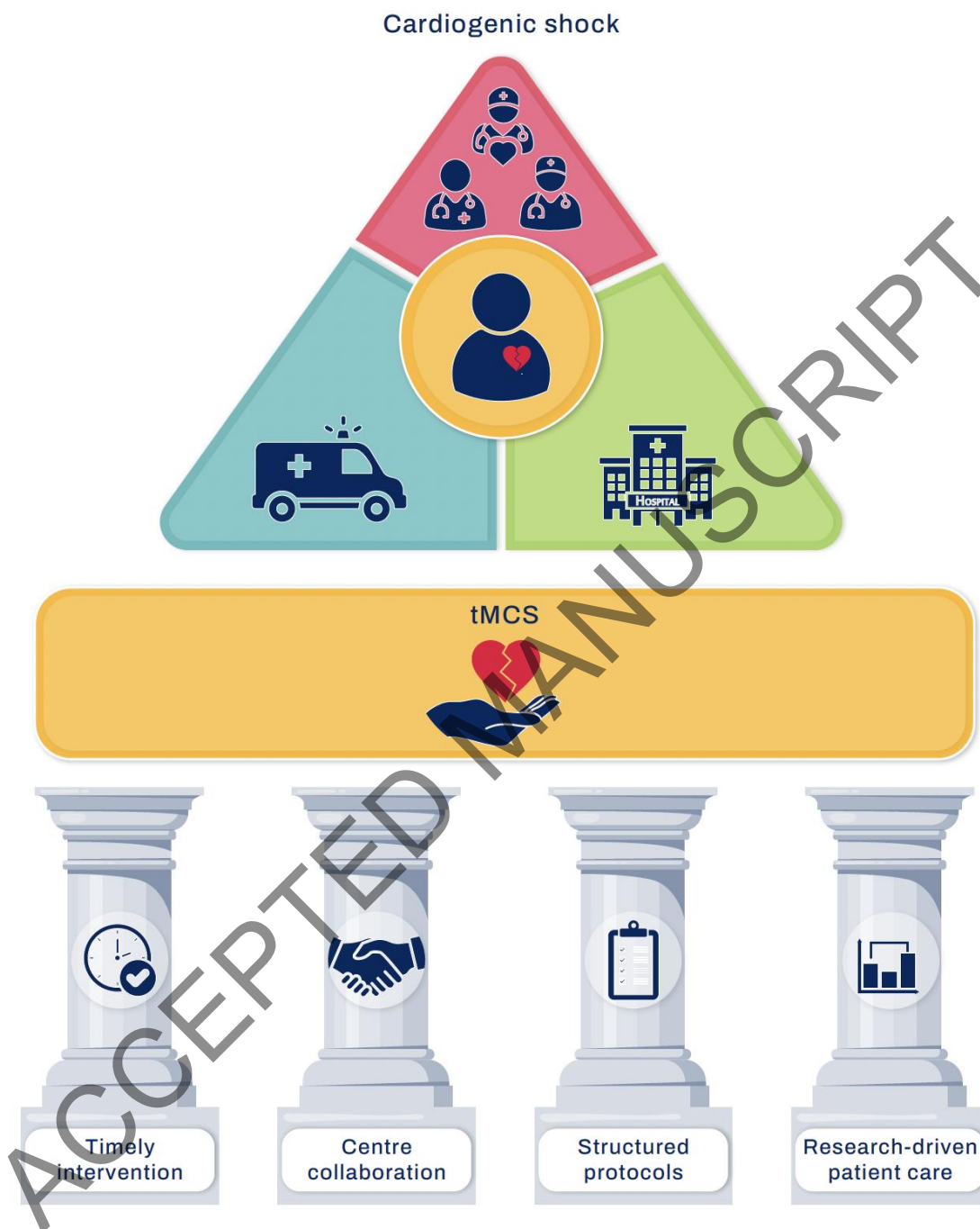
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Disclaimer: Clinical practice guidelines are intended to inform the care of patients with specific conditions. However, individual clinical scenarios may arise where the recommendations are not appropriate. Health-care professionals are encouraged to use these guidelines to support clinical decision making, but they do not replace the responsibility to tailor care to each patient's unique needs and circumstances. Clinical decisions should reflect the most current official recommendations, relevant public health guidance, and applicable laws and regulations. Where possible, such decisions should be made collaboratively with the patient and/or their legal guardian or carer.

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Graphical Abstract



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Central Illustration. Timely Team-Based Temporary Mechanical Circulatory Support As Key to Better Cardiogenic Shock Outcomes in Adult Cardiac Surgery

Keywords: Clinical practice guidelines, adult cardiac surgery, cardiogenic shock, temporary mechanical circulatory support, extracorporeal life support, evidence based practice

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279 ABBREVIATIONS AND ACRONYMS

280	AATS:	American Association for Thoracic Surgery
281	ABI:	Ankle–brachial index
282	ACHD:	Adult congenital heart disease
283	ACS:	Acute coronary syndrome
284	ACT:	Activated clotting time
285	AHA:	American Heart Association
286	AI:	Aortic insufficiency
287	AKI:	Acute kidney injury
288	AMI:	Acute myocardial infarction
289	aPTT:	Activated partial thromboplastin time
290	ARDS:	Acute respiratory distress syndrome
291	AS:	Aortic stenosis
292	BMI:	Body mass index
293	BTR:	Bridge to recovery
294	BTT:	Bridge to a transplant
295	BVAD:	Biventricular assist device
296	CABG:	Coronary artery bypass grafting
297	CAD:	Coronary artery disease
298	CKD:	Chronic kidney disease
299	CO:	Cardiac output
300	CPB:	Cardiopulmonary bypass
301	CPR:	Cardiopulmonary resuscitation
302	CS:	Cardiogenic shock
303	CT:	Computed tomography
304	CVP:	Central venous pressure
305	dLVAD:	Durable left ventricular assist device
306	DM:	Diabetes mellitus
307	dmCS:	Durable mechanical circulatory support
308	DTI:	Direct thrombin inhibitors
309	EACTS:	European Association for Cardio-Thoracic Surgery
310	EBF:	Estimated blood flow
311	ECG:	Electrocardiogram
312	ECLS:	Extracorporeal life support
313	ECPR:	Extracorporeal cardiopulmonary resuscitation
314	ELSO:	Extracorporeal Life Support Organization
315	EN:	Enteral nutrition
316	EOL:	End of life
317	ERAS:	Enhanced Recovery After Surgery
318	HF:	Heart failure
319	HIT:	Heparin-induced thrombocytopenia
320	HKTx:	Heart–kidney transplant
321	HTx:	Heart transplant
322	IABP:	Intra-aortic balloon pump
323	ICU:	Intensive care unit
324	IHCA:	In-hospital cardiac arrest
325	LA:	Left atrium

326	LCOS:	Low cardiac output syndrome
327	LFT:	Liver function test
328	LV:	Left ventricle
329	LVAD:	Left ventricular assist device
330	LVEDP:	Left ventricular end-diastolic pressure
331	LVEF:	Left ventricular ejection fraction
332	mAFP:	Microaxial flow pump
333	MAP:	Mean arterial pressure
334	MCS:	Mechanical circulatory support
335	MR:	Mitral regurgitation
336	NIRS:	Near-infrared spectroscopy
337	OHCA:	Out-of-hospital cardiac arrest
338	OxyRVAD:	A combination of RVAD and oxygenator
339	PA:	Pulmonary artery
340	PAC:	Pulmonary artery catheter
341	PCI:	Percutaneous coronary intervention
342	PE:	Pulmonary embolism
343	PEEP:	Positive end-expiratory pressure
344	PGD:	Primary graft dysfunction
345	POCUS:	Point-of-care ultrasound
346	PN:	Parenteral nutrition
347	PRBC:	Packed red blood cells
348	PVD:	Peripheral vascular disease
349	PVR:	Pulmonary vascular resistance
350	QoL:	Quality of life
351	RCT:	Randomized controlled trial
352	ROSC:	Return of spontaneous circulation
353	RRT:	Renal replacement therapy
354	RV:	Right ventricle
355	RVAD:	Right ventricular assist device
356	RVSP:	RV systolic pressure
357	STS:	Society of Thoracic Surgeons
358	SvO ₂ :	Venous oxygen saturation
359	TAPSE:	Tricuspid annular plane systolic excursion
360	TAVI:	Transcatheter aortic valve implant
361	tLVAD:	Temporary left ventricular assist device
362	tMCS:	Temporary mechanical circulatory support
363	TOE:	Transoesophageal echocardiography
364	TTE:	Transthoracic echocardiography
365	tVAD:	Temporary VAD
366	UFH:	Unfractionated heparin
367	VA-:	Veno-arterial-
368	VA-V-:	Veno-arterial-venous-
369	VAD:	Ventricular assist device
370	VIS:	Vasoactive inotropic score
371	VF:	Ventricular fibrillation
372	VSD:	Ventricular septal defect

373 VT: Ventricular tachycardia

374 VV-: Veno-venous-

375

376 The term *Extracorporeal Life Support* (ECLS) is used in this manuscript as it accurately reflects
377 both the function and configuration of the system. The previously used term *Extracorporeal*
378 *Membrane Oxygenation* (ECMO), which refers only to the type of oxygenator employed, will
379 not be used. The term *differential oxygenation* is used in place of *Harlequin syndrome*.
380 Additionally, the term *OxyRVAD* has been defined and applied in accordance with the
381 terminology established in the EACTS/ELSO/STS/AATS Expert Consensus Document.¹

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PREAMBLE

The *Guidelines on temporary mechanical circulatory support in adult cardiac surgery* evaluate the impact of various diagnostic and therapeutic strategies on patient outcomes and assess the associated risk–benefit profiles. Although these guidelines are not intended to replace comprehensive textbooks, they offer supplementary, clinically relevant insights that serve as a practical resource for physicians. Their primary aim is to support clinical decision making in daily practice.

It is important to emphasize that these recommendations are intended to inform, not dictate, clinical care. They are meant to assist, not replace, the physician's clinical judgement, which remains rooted in professional expertise, experience, and a comprehensive understanding of the individual patient's condition.

The European Association for Cardio-Thoracic Surgery (EACTS), the Society of Thoracic Surgeons (STS), and the American Association for Thoracic Surgery (AATS) jointly assigned a multidisciplinary task force composed of experts in mechanical circulatory support. All mechanical circulatory support task force members involved in the development and review of these guidelines submitted conflict of interest declarations. Any changes to these declarations during the development process were promptly reported to the sponsoring societies. This work was solely funded by EACTS, STS, and AATS, with no financial support or involvement from the health-care industry or other external parties.

The initial draft underwent comprehensive external validation by the designated review team of EACTS, STS, and AATS and was further refined based on feedback from an independent panel of subject matter experts. The final version was approved by all task force members, designated reviewers, and the leadership of the 3 societies, allowing for simultaneous publication in the *European Journal of Cardio-Thoracic Surgery*, *The Annals of Thoracic Surgery*, and *The Journal of Thoracic and Cardiovascular Surgery*.

These guidelines represent the official, up-to-date consensus on this topic. Regular updates are planned to ensure continued relevance in this rapidly evolving field of clinical practice.

1. INTRODUCTION

Temporary mechanical circulatory support (tMCS) has evolved significantly over the past decade. The distinction between temporary and durable mechanical circulatory support (dMCS) is based on discharge potential: Patients supported with tMCS are not eligible for hospital discharge. Modern tMCS devices demonstrate improved haemocompatibility and significantly reduced device-related complications compared to earlier generations. Enhanced safety and broader clinical applicability have led to wider adoption. Notably, use of microaxial flow pumps (mAFPs) and extracorporeal life support (ECLS) has increased substantially.^{2,3} In the United States, changes to the donor heart allocation system have led to a sharp rise in tMCS use as a bridge to a transplant (BTT),^{4,5} and the number of patients bridged to a heart transplant (HTx) with tMCS has tripled since 2018.⁵ Similar trends have been observed internationally.⁶⁻⁸

tMCS is now a standard of care for managing acute cardiogenic shock (CS),⁹⁻¹¹ acute-on-chronic heart failure (HF),¹²⁻¹⁴ and circulatory failure following cardiac procedures.^{1,15} Moreover, broader availability of mAFP devices and growing clinical expertise have enabled tMCS use in increasingly complex scenarios, including in elderly and paediatric patients,¹⁶ those with adult congenital heart disease (ACHD), and individuals undergoing complex cardiac operations or presenting with advanced CS. This progress has been made possible through increased research efforts. Although conducting a randomized controlled trial (RCT) in this setting remains extremely challenging, particularly due to the short time frame between patient presentation and the need for intervention, ethical concerns, and the heterogeneity of clinical scenarios, several recently conducted trials in acute myocardial infarction (AMI) have demonstrated that well-designed multicentre studies are not only feasible but also essential to better understand the safety and efficacy of available therapeutic modalities.^{17,18} Today, the field has evolved from a “one-size-fits-all” ECLS model to a modular, scenario-specific use of tMCS devices with varying mechanisms and levels of invasiveness. Although this shift has facilitated individualized care, it has also increased complexity across all phases of management, from indication to postoperative care. This guideline addresses these challenges and supports standardized approaches regarding indication, implanting techniques, management, weaning, and end-of-life (EOL) treatment.

Following its previous publications on long-term MCS and post-cardiotomy ECLS,^{1,19} the EACTS, in partnership with STS and AATS, now presents these joint guidelines on tMCS. A

companion document on "protected cardiac surgery" was recently published,²⁰ making this guideline a natural extension of ongoing collaborative efforts to improve patient care in this rapidly evolving field.

These guidelines do not cover tMCS use in elective interventional procedures such as "protected percutaneous coronary intervention" (PCI) or "protected transcatheter mitral valve replacement" in stable patients. They also exclude veno-venous (VV)-ECLS for isolated lung failure, which is addressed in anaesthesia-led guidelines.²¹ tMCS use in paediatric populations will be the focus of a future joint document.

Management of patients with tMCS requires collaboration among surgeons, anaesthesiologists, intensive care specialists, and cardiologists. This interdisciplinary approach is a key pillar of the present expert consensus, which includes contributors from all relevant specialties. However, it is critical to emphasize that delayed or absent deployment of tMCS remains a major threat to patients with CS. While interdisciplinary decision making is ideal, it must never delay life-saving therapy.

This document includes both surgical guidance (for implants and explants) and medical management (device selection, intensive care, transport, weaning, and palliative care), providing a comprehensive framework for all aspects of tMCS therapy.

2. METHODOLOGY

A systematic literature review was then performed, guided by Population, Intervention, Comparison, Outcome, and Time (PICOT) questions, with support from a medical informatics specialist. The search included all study types, RCTs, registries, cohort and case series, comparative and descriptive studies, systematic reviews, meta-analyses, and expert opinions that focused on human studies in English. Databases searched included MEDLINE, EMBASE, and the Cochrane Library, covering literature published between September and December 2023. Relevant studies published after this period, including during the writing and review process, were also incorporated. References are representative but not exhaustive. (See search strings for each chapter in the supplementary material.)

The guideline recommendations are, whenever possible, evidence based. Recommendations were formulated using established methodological principles, with particular attention to assessing the risks and benefits of each intervention (see [Table 1](#) and [Table 2](#)).^{22,23} Where

robust evidence was lacking, expert consensus recommendations were developed to address critical issues in daily clinical practice. Preliminary consensus was achieved through conference calls and in-person meetings, requiring at least 75% agreement among participants to advance draft recommendations. Subsequently, an anonymous electronic survey was conducted to vote on each recommendation and assign the appropriate Class of Recommendation) and Level of Evidence. Consensus was confirmed with a minimum response rate of 80% and at least 75% affirmative agreement for each recommendation. Each participating society appointed a peer review committee, comprising a lead reviewer and up to 5 additional members, to evaluate the document. Following acceptance by the review committees, the final manuscript was thoroughly reviewed and formally endorsed for publication by the governing bodies of the EACTS, the STS, and the AATS.

Table 1: Levels of Evidence

Level of Evidence A	Data derived from multiple randomized clinical trials or meta-analyses
Level of Evidence B	Data derived from a single randomized clinical trial or from large non-randomized studies
Level of Evidence C	The consensus of expert opinion and/or small studies, retrospective studies, and registries

Table 2: Classes of Recommendations

Classes of recommendations	Definition	Suggested wording
Class I	Evidence and/or general agreement that a given treatment or procedure is beneficial, useful, and effective	Is recommended/is indicated
Class II	Conflicting evidence and/or a divergence of opinion about the usefulness/efficacy of the given treatment or procedure	
Class IIa	Weight of evidence/opinion is in favour of usefulness/efficacy	Should be considered

Class IIb	Usefulness/efficacy is less well established by evidence/opinion	May be considered
Class III	Evidence/general agreement that the given treatment/procedure is not useful/effective and may sometimes be harmful	Is not recommended

3. EPIDEMIOLOGY AND PATIENT PROFILES

Appropriate candidates for tMCS are patients with CS in whom cardiac and circulatory recovery is expected or who are eligible for an HTx or a dMCS implant.²⁴ Durable MCS includes implantable left ventricular assist devices (LVADs), biventricular assist devices (BVADs), and total artificial hearts (TAHs). In addition to patients with CS, tMCS may also be used to optimize preoperative conditions and to provide circulatory support during and after a high-risk cardiac surgery or intervention.

Despite the increased use of tMCS, the in-hospital survival of patients with CS remains sub-optimal.^{25,26} However, recent trends in outcomes for CS complicating AMI suggest improvements in early survival.²⁷ The types of tMCS used have evolved over the years, with a decline in intra-aortic balloon pump (IABP) use and a shift towards mAFP and veno-arterial extra-corporeal life support (VA-ECLS).^{28,29} Data sharing and ongoing analysis of large registries, along with upcoming RCTs, will further enhance the knowledge base necessary for selecting and treating this patient population.^{30,31} Notably, the design and interpretation of studies in CS have been challenging due to the wide clinical variability among affected patients. Standardized classification systems, such as the Society for Cardiovascular Angiography and Interventions (SCAI) shock classification, have been proposed to promote more consistent diagnosis, risk stratification, and decision making in both clinical and research settings.³²⁻³⁴

Given the complexity and heterogeneity of patients requiring tMCS, careful and comprehensive assessment is essential. Input from an interdisciplinary Shock Team is therefore strongly recommended. Such a strategy, which incorporates diverse clinical perspectives, may improve pre-implant evaluation, device selection, implant technique, and post-implant management.

3.1 Patient profiles

Previously, more than 80% of CS cases resulted from acute coronary syndrome (ACS). However, although the number of patients with CS has increased, the proportion of patients with ACS has decreased over the past 2 decades, paralleling an increase in CS of other aetiologies. Currently, only about 30%-45% of CS are related to ACS.²⁶ Mechanical complications of AMI (ventricular free-wall rupture, ventricular septal rupture, and papillary muscle rupture with subsequent acute mitral regurgitation [MR]) are relatively rare causes of CS, but patients with those complications have even higher mortality risks.³⁵ Causes of non-ACS CS include acute myocarditis, atrial and ventricular arrhythmia, valvular disease, acute decompensation of chronic HF, other cardiomyopathies (eg, peripartum, toxic aetiology, and Takotsubo), tamponade, or post-cardiotomy shock.^{25,36-38} In addition, tMCS can play an important role in massive pulmonary embolism (PE) in whom conventional interventions have failed.³⁹ Furthermore, the COVID-19 pandemic has influenced the utilization of tMCS, particularly VA-ECLS for a subset of patients with concomitant haemodynamic instability and cardiac dysfunction⁴⁰ or oxyRVAD (a combination of right ventricular assist device [RVAD] and an oxygenator) for patients with pulmonary and right ventricular (RV) failure.⁴¹⁻⁴³

Within the last decade, new indications for tMCS have developed, including protected PCI, protected or transcatheter aortic valve implantation (TAVI),⁴⁴⁻⁴⁷ protected “high risk” cardiac operations, and tMCS-protected ventricular tachycardia (VT) ablation.^{48,49} The timing of tMCS placement prior to high-risk PCI is associated with improved survival compared to post-PCI.⁴⁷ In extremely high-risk populations undergoing TAVI, emergency use of tMCS during the procedure leads to high short- and long-term mortality rates compared to prophylactic utilization.⁵⁰

Although originally developed for CS, tMCS indications have broadened to include prophylactic insertion before high-risk cardiac procedures, treatment of postcardiac-surgery circulatory collapse, and bridging end-stage patients with HF to a left ventricular assist device (LVAD) or an HTx^{51,52} or to recovery (bridge to recovery, BTR).⁵³ Typical clinical scenarios for application of tMCS are post-cardiotomy HF,^{4,36,54} fulminant myocarditis,^{55,56} stunned myocardium after acute ischaemia,⁵⁷ Takotsubo cardiomyopathy,⁵⁸ peripartum cardiomyopathy,⁵⁹ or massive PE.³⁹ In extreme situations with uncertain neurologic or end-

organ prognosis, tMCS may serve as a bridge to decision (BTD),⁶⁰ a scenario frequently encountered during extracorporeal cardiopulmonary resuscitation (ECPR).⁶¹ Where brain death is confirmed or irreversible post-anoxic damage is confirmed per local regulations, organ donation remains possible after tMCS withdrawal.

Recommendation Table 1: Recommendations for Data Collection and Sharing in Patients Receiving tMCS Support

Recommendations	Class ^a	Level ^b	Ref ^c
It is recommended that data be included in local, national, and international registries.	I	C	-
Data sharing within and between registries, along with continuous data analysis, is recommended for quality and outcome improvement.	I	B	(30, 31)

Abbreviation: tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

4. GOALS OF tMCS

Significant progress has been made in the use of tMCS for managing CS.^{24,62,63} Recent studies highlight 2 key advances: (1) earlier deployment of support, which appears to improve survival, and (2) newer devices with better safety profiles.⁶⁴

The goals of tMCS are (1) prevention or reversal of CS; (2) prevention or reversal of multiorgan failure; (3) stabilization of haemodynamics during a cardiac procedure (surgical or interventional); (4) preoperative optimization for a cardiac operation including implanting a dMCS (bridge to bridge; BTB); (5) stabilization of haemodynamics during the waiting period for an organ transplant (BTT); (6) use as a BTD; and (7) use as a bridge to organ donation (BTOD).

These goals may be achieved through partial or complete replacement of cardiac pump function, left, right, or biventricular, as well as pulmonary function, , or both.^{62, 63} Importantly,

a key objective of tMCS is to minimize associated hypoperfusion-related organ using patient-tailored device strategies and optimized combinations.⁶⁵

Recommendation Table 2: Recommended Goals of tMCS

Recommendations	Class	Level	Ref
It is recommended that the therapeutic goals of tMCS be discussed via an interdisciplinary team with the patient and the family when feasible.	I	C	-

Abbreviation: tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

5. DEVICE SELECTION AND CONFIGURATIONS

The selection and use of a specific tMCS device is multifactorial. Key considerations include the clinical indication, availability of devices, patient-specific comorbidities (eg, peripheral vascular disease or bleeding diathesis), likelihood of cardiac recovery, and the potential need for transition to dMCS. Patients with profound cardiocirculatory compromise typically require cardiopulmonary not just cardiac support, such as VA-ECLS. In contrast, patients with less severe haemodynamic instability may be adequately supported using uni- or biventricular devices alone.

Recent advances in cannulation technology have expanded available options. For example, access to the main pulmonary artery (PA) via a percutaneous right internal jugular vein approach has enabled new configurations for RV support, significantly broadening the scope of tMCS-based strategies.

Table 3 outlines currently available tMCS systems and their functional profiles, highlighting the type and extent of support offered, along with their respective advantages and limitations.

Table 3. Currently Available tMCS Systems and Related Actions in Terms of Type and Extent of Support, Including Known Advantages and Disadvantages

Type of MCS	Mode of action	Target of support	Level of support	Access	Advantages	Disadvantages and potential complications
IABP	Cardio-circulatory	LV RV (indirect)	*a *a	Percutaneous peripheral arteries	Ease of insertion; very limited costs; ease of performance and monitoring; ability to place in upper body for ease of ambulation	Limited support; limb ischaemia; supra-aortic embolization; bowel ischaemia; if inappropriately positioned, potential for vascular injury (rupture, dissection)
Venoarterial extracorporeal life support (VA-ECLS)						
Continuous flow VA-ECLS	Cardio-Circulatory-Respiratory	RV LV Lungs	*****e **b *****e	Percutaneous and surgical peripheral and central vessels	Full right ventricular support; powerful cardiac and circulatory support; effective respiratory support; central or peripheral (femoral and axillary) approach	LV distension (often LV unloading required); differential oxygenation; limb ischaemia; spinal cord ischaemia; SIRS; bleeding/embolization with potential of vascular injury (rupture, dissection), risk of thrombiformation, and systemic arterial embolization access site bleeding
Pulsatile VA-ECLS	Cardio-Circulatory	RV LV Lungs	*****d ***c *****d	Percutaneous peripheral arteries	Pulsatile support	Still limited experience;

						complications similar to those of VA-ECLS; potential for vascular injury (rupture, dissection); risk of thrombi formation and systemic arterial embolization; access site bleeding
Left-sided microaxial flow pump (mAFP)						
Partial support mAFP	Cardio-Circulatory	LV RV (indirect)	***c **b	Percutaneous and surgical peripheral arteries	Fast initiation during intervention; aortic root washout; LV unloading and related advantages	Dislodgement; haemolysis; limited ventricular support; limb ischaemia; positional potential of vascular injury (rupture, dissection); risk of thrombi formation and systemic arterial embolization; access site bleeding
Full support mAFP	Cardio-Circulatory	LV RV (indirect)	*****e **b	Surgical, preferably via axillary artery; ascending aorta or carotid artery also possible	Full left ventricular support; prolonged support possible; aortic root washout; LV unloading and related advantages;	Surgical access required; haemolysis; dislodgement; infection; thrombi at the prosthetic access; positional potential of vascular injury (rupture, dissection); risk of thrombi formation and

					ability to place in upper body for ease of ambulation; enhanced patient mobility	systemic arterial embolization; access site bleeding
Right-sided microaxial flow pump (mAFP)						
mAFP	Cardio-Circulatory	RV	*****e	Percutaneous right femoral vein	Fast insertion in catheterization laboratory	No respiratory support; haemolysis; dislocation
mAFP	Cardio-Circulatory	RV	*****e	Percutaneous right jugular vein	Fast insertion in catheterization laboratory; enhanced patient mobility	No respiratory support; haemolysis; dislocation
Left-sided centrifugal flow pump						
Percutaneous transseptal tLVAD	Cardio-Circulatory Respiratory	LV RV Lungs	****d ***c *****e	Percutaneous femoral veins	Left ventricular support and unloading; possible in patients with mechanical aortic valve	Mainly for the femoral approach; limb ischaemia; spinal cord ischaemia; access site bleeding Risk of life-threatening inflow transseptal cannula migration; risk of thrombi formation and systemic arterial embolization

tLVAD employing magnetically levitated extracorporeal centrifugal pump	Cardio-Circulatory	LV	*****e *****e	Surgical central cannulation	Full support and ventricular unloading; prolonged support possible; low anticoagulation required; incorporation of oxygenator possible; biventricular support possible	Central or minimally invasive cannulation; haemorrhage; infection; embolic stroke; risk of thrombi formation and systemic arterial embolization
Pump with bidirectional valve (obtained CE Mark; not FDA approved)	Cardio-Circulatory	LV	***c **b	Percutaneous peripheral arteries	Pulsatile support	Limited LV support; no comprehensive and conclusive experience; mainly adopted in "Protected PCI"; potential of vascular injury (rupture, dissection); risk of thrombi formation and systemic arterial embolization; access site bleeding
Paracorporeal RVAD						
RA-PA OxyRVAD	Cardio-Circulatory Respiratory	RV & Lungs	*****e *****e	Percutaneous with single-lumen or double-lumen cannula or	Enhanced patient mobility	With an open-chest approach: infection and chest reopening to remove the cannula; bleeding.

				central cannulation employing vascular prosthesis on the PA		With percutaneous approach, thrombosis and risk of lung embolization from the circuit; impingement of the pulmonary or tricuspid valve with subsequent regurgitation.
Right atrium–left atrium RVAD	Cardio-Circulatory Respiratory	RV Lungs	***** *****	Surgical central cannulation	Full RV support	Only RV support; surgical approach requiring a sternotomy or thoracotomy, with risk for infection and need for chest reopening to remove the cannula; bleeding

Abbreviations: CE, *Conformite Europeenne*; ECLS, extracorporeal life support; FDA, Food and Drug Administration; IABP, intra-aortic balloon pump; LV, left ventricle; mAFP, microaxial flow pump; MCS, mechanical circulatory support; PA, pulmonary artery; PCI, percutaneous coronary intervention; RA, right atrium; RV, right ventricle; RVAD, right ventricular assist device; SIRS, systemic inflammatory response syndrome; tLVAD, temporary left ventricular assist device; tMCS, temporary mechanical circulatory support; VA-ECLS, veno-arterial extracorporeal life support.

^aMinimal support (*), progressing through increasing levels of support (** to ****), to ^eFull support (*****).

5.1 Intra-aortic balloon pump

The haemodynamic support provided by an IABP is modest, augmenting cardiac output (CO) by approximately 0.5-1.0 L/min. Because IABP relies on preserved native left ventricular (LV)

contractility to promote forward flow, it is generally insufficient for treating patients with profound CS. The standard insertion route is via the common femoral artery. However, alternative approaches, including axillary artery or a direct transaortic implant through a prosthetic graft, may be considered in patients with inadequate peripheral access. Notably, axillary IABP placement has gained popularity in transplant candidates to allow continued ambulation while awaiting HTx.

Despite the neutral findings of the IABP-SHOCK and SHOCK II trials, IABP remains the most widely used form of tMCS, particularly in the post-cardiotomy setting.⁶⁶⁻⁶⁸ Nonetheless, its global use has declined significantly in recent years.^{69,70} The limitations of these trials, including the timing of implant and patient selection, have raised concerns regarding their generalizability. Consequently, international guidelines vary in their recommendations: Some assign a Class IIb rating for selected use in AMI-related CS, whereas others maintain a Class III recommendation advising against routine use.^{10,71,72}

Recent multicentre studies have reported reduced vascular complications with sheathless IABP implant techniques.^{71,73} Additionally, when combined with VA-ECLS, IABP may facilitate partial LV unloading, thereby enhancing its role as part of a multimodal tMCS strategy in selected patients.⁷⁴

5.2 Veno-arterial extracorporeal life support

Veno-arterial extracorporeal life support, provides circulatory support via a centrifugal pump coupled with a membrane oxygenator. This system enables blood drainage from the right atrium, oxygenation, and CO₂ removal before return to the arterial circulation. It remains the most effective form of support for profound cardiopulmonary failure from a pathophysiological perspective.

Its use has extended to cardiac arrest, but recent RCTs evaluating VA-ECLS in out-of-hospital cardiac arrest (OHCA) have shown mixed results.⁷⁵⁻⁷⁷ The most recent ECLS-Shock trial failed to demonstrate superiority of VA-ECLS over conventional therapy in post-AMI CS.¹⁷ However, patient selection may have impacted these findings (eg, 77.7% had prior cardiac arrest, and 26% of deaths were due to brain injury), underscoring the importance of determining the appropriate criteria to identify those most likely to benefit from the intervention.¹⁷

5.3 Microaxial flow pump

The left-sided mAFP consists of a microaxial impeller that propels blood from the LV into the ascending aorta, while the right-sided version moves blood from the RV across the pulmonary valve into the PA. These devices are available in partial (up to 3.5 L/min) and full haemodynamic support configurations (up to 5.5 L/min). The most commonly used partial-support mAFP devices typically involve a 9 Fr catheter driving a 14 Fr motor, delivered percutaneously via the femoral artery or surgically via the axillary artery.^{78,79} Full-support versions generate up to 5.5 L/min flow, using a 21 Fr motor and a 9 Fr catheter, requiring surgical access to the supra-aortic vessels or ascending aorta via a vascular graft.⁸⁰ This configuration facilitates patient mobility and reduces risks associated with prolonged intensive care unit (ICU) immobilization. Although U.S. Food and Drug Administration-approved for 14 days and CE-marked for 28 days, longer durations of successful support have been reported.⁸¹

The left-sided mAFP generates several beneficial physiologic effects. By drawing blood directly from the LV, it unloads the ventricle, reducing end-diastolic volume (EDV) and end-diastolic pressure. This unloading decreases myocardial wall stress, microvascular resistance, and oxygen demand^{82,83} while increasing transmural pressure gradient. The aortic root outflow enhances mean arterial pressure (MAP), diastolic pressure, CO, and cardiac power output. Collectively, these changes improve coronary perfusion and myocardial oxygen supply. Reduced LV end-diastolic pressure (LVEDP), pulmonary capillary wedge pressure, and pulmonary vascular resistance (PVR) also lower RV afterload and enhance RV function during left-sided support.

Relative contraindications for mAFP include severe aortic regurgitation and hypertrophic cardiomyopathy. Absolute contraindications include mechanical aortic valve prostheses, floating LV thrombi or ruptured papillary muscles, and severe ascending aortic calcification or dissection of ascending aorta. Although post-AMI ventricular septal defect (VSD) was previously considered a contraindication (per IFU), emerging evidence suggests patients may benefit from support.^{84,85}

Potential complications of percutaneous mAFP include haemolysis with resultant acute kidney injury (AKI), infection, and limb ischaemia. Device malposition, particularly displacement into the aorta or towards submitral apparatus or septum, must be closely monitored. Reported complications include stroke (2%–8%), ventricular perforation, arrhythmias, and reversed flow through a VSD.

The right-sided mAFP via percutaneous femoral or right internal jugular vein access is designed for RV support. The right-sided trans-pulmonary mAFP provides up to 4 L/min of support, using an 11 Fr catheter and a 22 Fr microaxial pump.

The mAFPs are increasingly employed for uni- or biventricular support or in combination with VA-ECLS to facilitate LV unloading. One emerging approach combines axillary artery access with a Y-graft to enable both VA-ECLS return and mAFP placement through the same arteriotomy, simplifying implantation, management, physiotherapy, and weaning.^{12,13,65}

Related devices and configurations are summarized in Table 4.

Table 4. Overview of Available mAFP Types and Features

	Partial support left-sided mAFP	Full support left-sided mAFP	Femorally implanted right-sided mAFP	Jugularly implanted right-sided mAFP	Expandable partial support left-sided mAFP ^a
Catheter size	9 Fr	9 Fr (70 cm length)	11 Fr	11 Fr	9 Fr
Maximal size	14 Fr	21 Fr	22 Fr	22	9 Fr
Peak flow	4.3 L/min	5.5 L/min	4.0 L/min	4 L/min	>3.5 L/min
Access site	Percutaneous femoral or axillary	Surgical cutdown of supra-aortic vessels or direct aortic access	Single venous access through right femoral vein	Percutaneous, right internal jugular insertion	Percutaneous femoral
Ventricle	LV unloading	LV unloading	RV unloading	RV unloading	LV unloading
Approval	CE mark FDA-approved	CE mark FDA-approved	CE mark FDA-approved	Not yet certified for clinical use in Europe	Not yet certified for clinical use

Abbreviations: CE, *Conformite Europeenne*; FDA, Food and Drug Administration; Fr, French scale; LV, left ventricle; mAFP, microaxial flow pump; min, minute; RV, right ventricle.

^aECP (expandable cardiac pump) just in pre-clinical studies.

5.4 Pumps for extracorporeal VAD

The fully magnetically levitated centrifugal pump is one of the most efficient temporary ventricular assist devices (tVAD) devices. Its design allows good haemocompatibility. The system supports flows of up to 10 L/min at 5,500 rpm and is particularly suitable for patients requiring prolonged support (approved for up to 30 days), due to its reliability and safety profile. It is typically used for univentricular support (LVAD or RVAD) but can also be configured for BiVAD support. When an oxygenator is added to the pump circuit, the system can provide combined cardiopulmonary support. Some pumps are equipped with integrated oxygenators for transport purposes, while others are compatible with various external oxygenators and cannulas. In some cases central cannulation via sternotomy or thoracotomy is necessary.

5.5 Pulsatile ECLS with a centrifugal pump

Pulsatile VA-ECLS, also known as synchronized cardiac support, utilizes electrocardiogram (ECG)-triggered pulsatility as an adjunct to standard VA-ECLS (Figure 1). The core principle involves synchronizing pump speed, and thus estimated blood flow (EBF), with the cardiac cycle based on R-wave detection. During systole, EBF is reduced, while during diastole, EBF is augmented.

The proposed clinical benefits of this synchronized approach include (1) reduced cardiac afterload during systole due to decreased pump speed⁸⁶; (2) improved coronary perfusion through diastolic augmentation,^{87,88} and (3) improved end-organ microcirculation from more efficient pulsatile energy transfer.^{89,90} These benefits have been validated in several preclinical in vivo studies.⁸⁶⁻⁹⁰

The first clinical application was reported in a prospective observational study of patients with CS following AMI undergoing high-risk PCI, demonstrating the technical feasibility of synchronized cardiac support.⁹¹ Further prospective trials are needed to identify optimal clinical scenarios in which this novel tMCS approach may offer the most benefit.

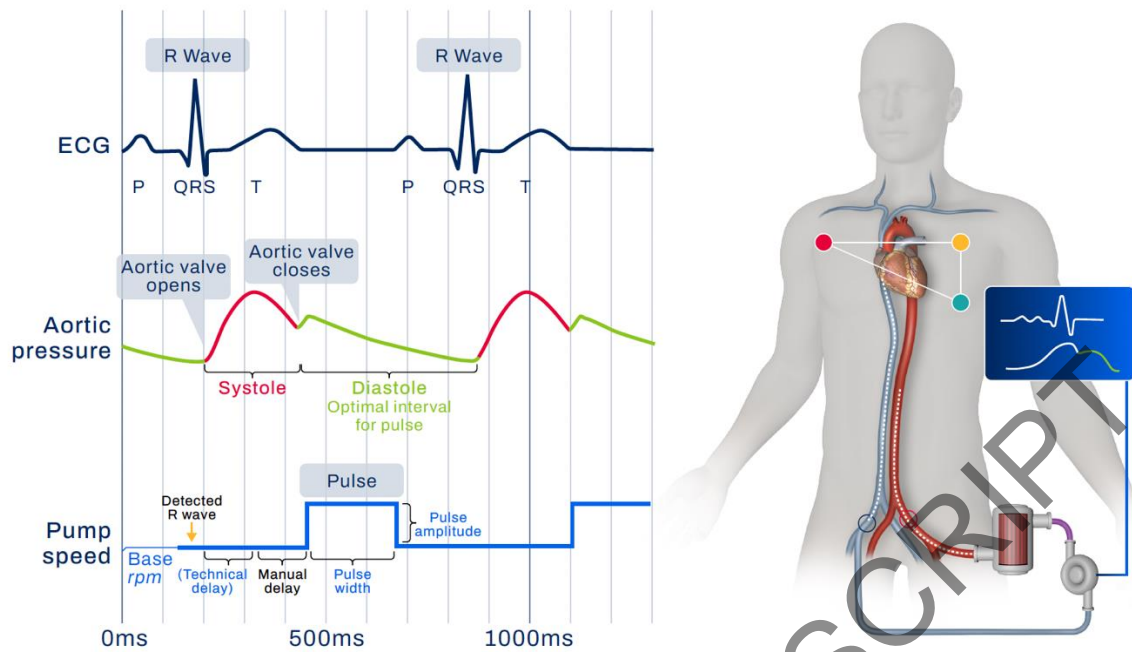


Figure 1. Synchronized Peripheral VA-ECLS Operation. ECG indicates electrocardiogram; VA-ECLS, veno-arterial extracorporeal life support.

5.6 Pulsatile pumps

A percutaneously inserted 17F transaortic cannula is designed to directly unload the LV but provides limited circulatory support (1.0 to 1.5 L/min). The device functions as an extracorporeal pulsatile pump, driven by a conventional IABP console and equipped with a rotating two-way valve (Table 4). Approximately 600 cases have been performed, mainly in the context of protected PCI, though published clinical data remain limited.^{92,93} The device has also been used for LV unloading during VA-ECLS and in preclinical models of CS.⁹⁴ However, robust clinical series evaluating its role in CS are lacking, and current evidence does not allow for definitive conclusions about its effectiveness or clinical benefit.

5.7 Transseptal cannula

The transseptal extracorporeal temporary left ventricular assist device (tLVAD) system uses a drainage cannula percutaneously inserted via the femoral vein transeptally into the left atrium (LA), enabling LA drainage and, indirectly, LV unloading. Blood is returned to the femoral artery. When inserted into the RA instead of the LA, the device functions as a standard VA-ECLS. Clinical experience and literature regarding this approach remain limited

738 compared to other devices.^{95,96} A major limitation of this configuration is the tendency for the
739 cannula to migrate back into the RA, at which point it functions merely as conventional VA-
740 ECLS. Moreover, successful placement requires a team expert in transseptal puncture and
741 manipulation of guidewires and catheters/cannulas across the interatrial septum.

742 743 **5.8 Temporary RVAD**

744 RV support can be achieved by routing venous drainage to a centrifugal pump and returning
745 the blood into the PA.^{97, 98} This configuration constitutes a pure RVAD. When an oxygenator
746 is added to the circuit, the system becomes an "OxyRVAD", a combined RV and pulmonary
747 support configuration.

748 Percutaneous RV support can be established using either 2 single-lumen cannulas or a double-
749 lumen PA cannula, representing a notable advance. The double-lumen cannula, available in
750 29 and 31 Fr sizes, is inserted via the right internal jugular vein. This configuration has
751 demonstrated favourable results in COVID-19 patients with refractory ARDS as an OxyRVAD,
752 as well as in cases of isolated RV failure.^{99,100} Advantages of this single-site cannulation
753 approach include reduced infection risk, preserved patient mobility, facilitated extubation,
754 and support for rehabilitation during prolonged use, while also preserving physiologic blood
755 flow direction.

756 Another double-lumen cannula, available in 24, 26, and 31 Fr sizes, is currently under clinical
757 investigation in Europe. This cannula has multiple openings in the outer catheter to drain
758 blood from both the RA and RV, thereby reducing the likelihood of unoxygenated venous
759 return to the lungs. The 2 designs differ in length, 520 mm versus 488 mm, and ongoing
760 preclinical and clinical studies aim to define their specific haemodynamic and perfusion
761 performance, including risks of displacement and circulation-related challenges.

762 Additionally, the right-sided mAFP device, as mentioned above, has been specifically
763 designed for percutaneous RV support and enables rapid deployment in cases of acute RV
764 failure.¹⁰⁰ However, it is not able to support an oxygenator.

765 766 **5.9 Alternatives to tMCS**

767 Alternatives to tMCS include optimal medical therapy, urgent or emergency surgical
768 intervention, and palliative care.

5.9.1 Medical therapy

Inotropic and vasoactive agents remain the first-line treatment for all patients with CS. These drugs can be titrated to provide circulatory support and act as a bridge to decision regarding tMCS initiation. Some patients stabilize with medical therapy alone, whereas others are relegated to medical therapy when deemed unsuitable for escalation to tMCS.

5.9.2 Urgent/emergency surgery

Surgical intervention should be promptly considered in the presence of correctable structural pathology (eg, emergency coronary artery bypass grafting [CABG], or mitral valve replacement for acute MR due to papillary muscle rupture). Early stabilization with IABP, mAFP, or percutaneous VA-ECLS may facilitate hemodynamic support while preparing for surgery, followed by intraoperative conversion to cardiopulmonary bypass (CPB).⁵¹ These tMCS devices may also be continued postoperatively for haemodynamic support.

5.9.3 Palliative care

Palliative care consultation should be integrated early in the management of patients in profound CS. The hospital course in these patients is often prolonged and unpredictable. Early involvement facilitates the development of realistic expectations, improves communication with families, and ensures better emotional support in cases of clinical deterioration or futility. This approach also promotes patient dignity, respects autonomy, and ensures responsible use of health-care resources.

6. PATIENT EVALUATION AND IMPLANT TIMING AS A LIFE-SAVING PROCEDURE

6.1 Assessment of acute CS

Initial management of CS relies on vasoactive and inotropic drugs to augment CO and maintain perfusion pressure. Escalating doses, or the need for multiple agents, often signal severe or rapidly worsening CS, with mortality increasing in proportion to total vasoactive/inotropic support.^{101,102} A practical way to quantify this pharmacologic burden is the vasoactive-inotropic score (VIS), calculated as follows^{103,104}:

- dopamine dose ($\mu\text{g}/\text{kg}/\text{min}$) + dobutamine dose ($\mu\text{g}/\text{kg}/\text{min}$) + $100 \times$ epinephrine dose ($\mu\text{g}/\text{kg}/\text{min}$) + $10 \times$ milrinone dose ($\mu\text{g}/\text{kg}/\text{min}$) + $10\,000 \times$ vasopressin dose (unit/kg/min) + $100 \times$ norepinephrine dose ($\mu\text{g}/\text{kg}/\text{min}$)

An alternative is the norepinephrine equivalent score.¹⁰⁵ Both scales use similar weighting, but VIS is far more widely referenced. A full, updated VIS formula has also been recently published.¹⁰⁶ Furthermore, the SCAI shock staging system further stratifies risk by integrating blood pressure, lactate, and vasoactive requirements, showing a clear step-wise rise in mortality with advancing stage.^{33,107} Although higher shock severity correlates directly with worse outcomes, evidence remains mixed regarding the survival benefit of ECLS across all SCAI stages.^{17,108-110} Therefore, careful patient selection and a timely implant remain critical.

6.1.1 Systematic markers of end organ dysfunction

Organ-specific markers provide insight into impaired forward flow, venous congestion, and microcirculatory disturbances. Although the reversibility of organ injury cannot be predicted in the acute setting, such markers are valuable indicators of CS severity. Multiple retrospective studies have shown a direct relationship between the degree of organ failure and shock severity.^{30,111-113} Current clinical practice supports the use of tMCS to restore organ perfusion and alleviate venous congestion. Accordingly, laboratory values, such as lactate, creatinine, and liver function tests, should be used to assess both the seriousness of shock and the impact of circulatory support. Although these markers may worsen in the first hours after shock onset, clinical signs such as anuria or changes in mental status can immediately help identify whether perfusion is adequate.^{32,114}

6.1.2 Lactate and pH

Lactate is produced in many tissues in response to tissue ischaemia and serves as a non-specific marker of poor perfusion.¹¹⁵ Elevated lactate levels correlate with increased mortality in patients with CS not receiving tMCS,¹¹⁶⁻¹¹⁸ and there is universal support for the use of lactate as a diagnostic and prognostic marker.^{66,119} Lactate has also been evaluated for its role in monitoring response to tMCS.¹²⁰⁻¹²⁷ Rising lactate levels are associated with higher mortality, suggesting that early initiation of tMCS is warranted to reduce risk.

Two retrospective studies demonstrated that lactate >15 mmol/L was associated with 100% mortality.^{109,125} However, due to variability in published data, no definitive lactate threshold for futility can be established. In shock, metabolic derangements and organ injury are closely linked. Thus, metabolic markers (eg, lactate and pH) and end-organ markers (eg, creatinine and liver enzymes) should be interpreted together with clinical factors such as age.¹²⁸

In more severe CS, combined support strategies (eg, VA-ECLS with LV unloading via mAFP) may be required, particularly in cases with high lactate levels.¹²⁹ Furthermore, acidosis at the time of VA-ECLS initiation is associated with lower survival.^{125,130-132} Therefore, patients showing signs of ongoing hypoperfusion (eg, rising lactate) should be considered for tMCS.

6.2 ACS with CS

Acute coronary syndrome is a common cause of CS. Emerging evidence supports the benefit of tMCS in conjunction with coronary revascularization.^{133,134} A meta-analysis of 11 observational studies showed improved survival with mAFP initiation prior to PCI,^{52,135} and similar benefits have been reported with ECLS cannulation before PCI or surgical revascularization.^{46,136-139} The randomized DANGER Shock Trial found improved survival in patients with ST segment elevation myocardial infarction-CS treated with mAFP.¹⁸ However, mAFP was also associated with an increased rate of adverse safety events, including bleeding, limb ischaemia, haemolysis, device failure, and worsening AKI. Nonetheless, initiation of tMCS before revascularization is recommended in patients with MI-related CS. Additional details are discussed in [Section 8](#).

6.3 Cardiac arrest and ECPR

The use of VA-ECLS for ECPR has grown since 2012, although survival outcomes remain mixed.¹⁴⁰ Thus, careful patient selection is crucial to identify those most likely to benefit. End-tidal CO₂ should be monitored as a prognostic indicator.

Outcomes differ significantly based on arrest location: in-hospital cardiac arrest (IHCA) is associated with better survival than OHCA,¹⁴¹⁻¹⁴⁴ and cardiopulmonary resuscitation (CPR) duration is a critical determinant.

In an analysis from the American Heart Association (AHA) Get-With-The-Guidelines Registry, median CPR duration was significantly longer in IHCA non-survivors (44 minutes) compared with IHCA survivors (23 minutes).¹⁴⁵ Another study of 137 patients with IHCA found median low-flow times of 30 versus 50 minutes among survivors and non-survivors, respectively ($P<0.01$).¹⁴⁶ Higher ECPR case volumes are associated with better outcomes.¹⁴² Targeted temperature management is not recommended in the ECPR settings.¹⁴⁷

In OHCA, the Prague OHCA RCT demonstrated improved 30-day, but not 6-month, survival with ECPR.⁷⁶ Post hoc analysis showed a survival benefit among patients who had not

achieved return of spontaneous circulation (ROSC) upon arrival at the hospital.¹⁴⁸ A meta-analysis of 4 RCTs found that ECPR improved survival in OHCA, especially in patients with shockable rhythms.¹⁴⁹ Thus, ECPR should be considered in patients with OHCA who receive immediate bystander CPR, do not achieve ROSC within 10 minutes, and can receive tMCS within 50–60 minutes.¹⁵⁰ In patients with a limited history, whole-body computed tomography (CT) after tMCS initiation may help exclude occult conditions.¹³⁸ Elderly patients benefit much less from ECPR, and the decision to initiate ECPR in this population is particularly challenging, because the urgency of the situation limits the opportunity for interdisciplinary decision making. Although not an absolute contraindication, elderly patients should be carefully evaluated before ECPR is initiated. The DanGer trial showed a cut-off of 77 years for benefit of treatment with a partial support mAFP in patients with CS complicating AMI. Timely LV unloading on VA-ECLS is associated with lower early mortality and higher weaning rates.^{151–154} In overt biventricular failure, combining VA-ECLS (for RV unloading) with mAFP (for LV unloading) should be considered first-line support.^{14,155,156} Certain arrhythmias, such as refractory VT or ventricular fibrillation (VF), may also necessitate combined support.^{157–162} Pulsatile flow generated by some VA-ECLS systems may further improve ECPR outcomes.^{163,164}

6.4 Contraindications

6.4.1 Contraindications for mAFP

Implanting an mAFP is contraindicated in patients with a mechanical aortic valve or a large, mobile LV thrombus. Severe aortic stenosis (AS) may also complicate the implant procedure.

6.4.2 Contraindication for VA-ECLS and IABP

Severe arterial calcification, aortic valve insufficiency, and untreated aortic dissection are contraindications to VA-ECLS and IABP.

6.4.3 Contraindication for venous cannulation

The presence of a thrombus in the right atrium or venae cavae precludes venous cannulation. Pulmonary embolism is a contraindication for right-sided mAFP or percutaneous RV support.

6.4.4 Special contraindications for temporary RVAD

Pulmonary haemorrhage and a primary pulmonary hypertension crisis are specific contraindications for temporary RVAD support.

6.4.5 tMCS contraindications

897 tMCS should not be initiated in patients with irreversible neurologic injury or non-
898 cardiopulmonary illness that limits life expectancy to ≤ 1 year. Extended resuscitation or
899 profound metabolic derangement may also make recovery unlikely. Where possible, patient
900 autonomy should be respected, including review of advance wishes.

901 **Figures 2 and 3** outline expert-recommended algorithms for tMCS selection in CS related and
902 unrelated to AMI.

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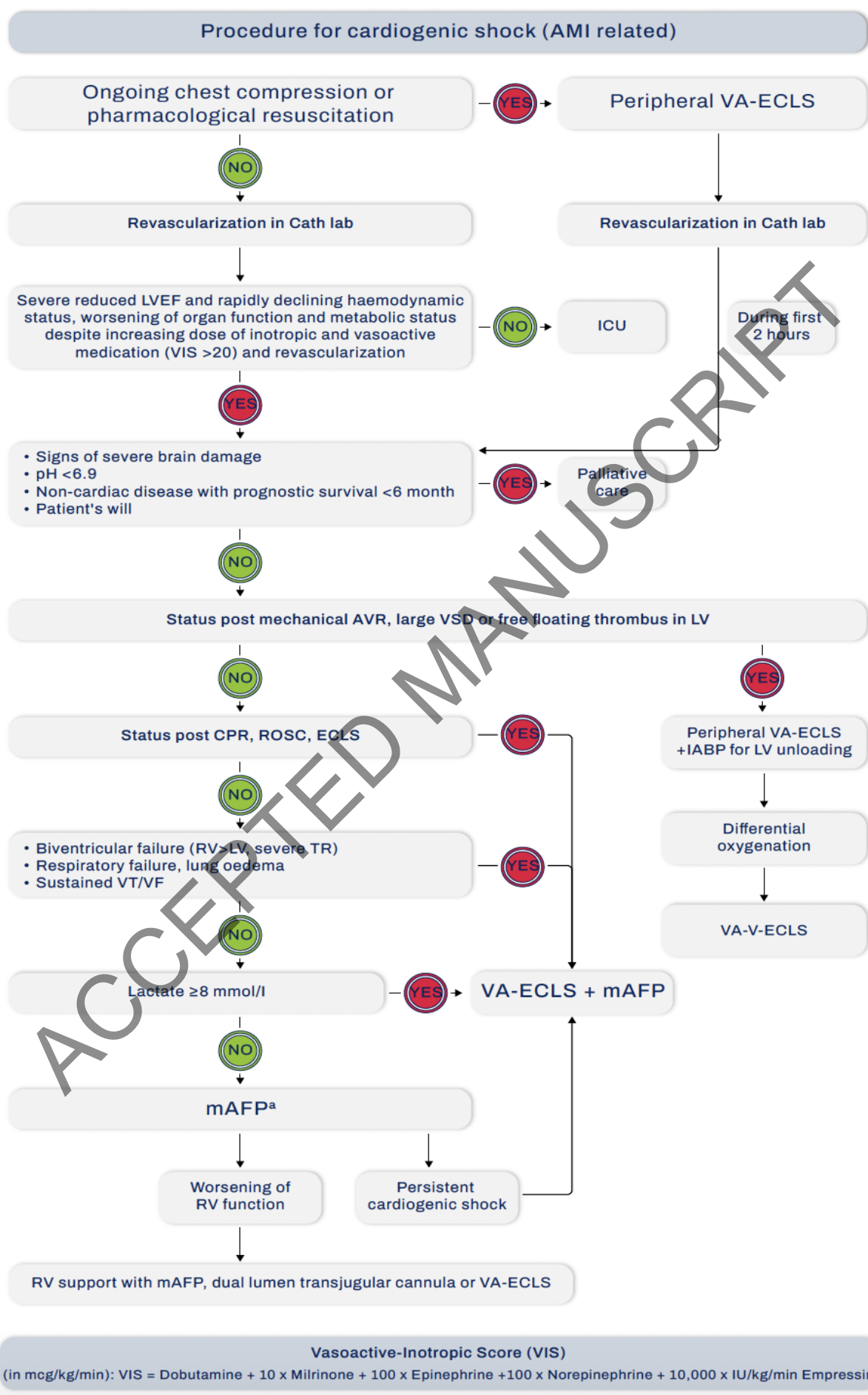


Figure 2. Procedure for Cardiogenic Shock Related to Acute Myocardial Infarction. In patients undergoing CPR, tMCS should be initiated before attempting revascularization. AVR indicates aortic valve replacement; CPR, cardiopulmonary resuscitation; ECLS, extracorporeal life support; IABP, intra-aortic balloon pump resuscitation; ICU, intensive care unit; LV, left ventricle; LVEF, left ventricular ejection fraction; mAFP, microaxial flow pump; ROSC, return of spontaneous circulation; RV, right ventricle; TR, tricuspid regurgitation; VA-veno-arterial; VA-V-, veno-arterial-venous; VF, ventricular fibrillation; VIS, vasoactive inotrope score; VSD, ventricular septal defect; VT, ventricular tachycardia. Adapted from Ott et al.¹⁴ with permission from Multidisciplinary Digital Publishing Institute. ^aIn patients with profound shock, there is a clear benefit of full support over partial support mAFP

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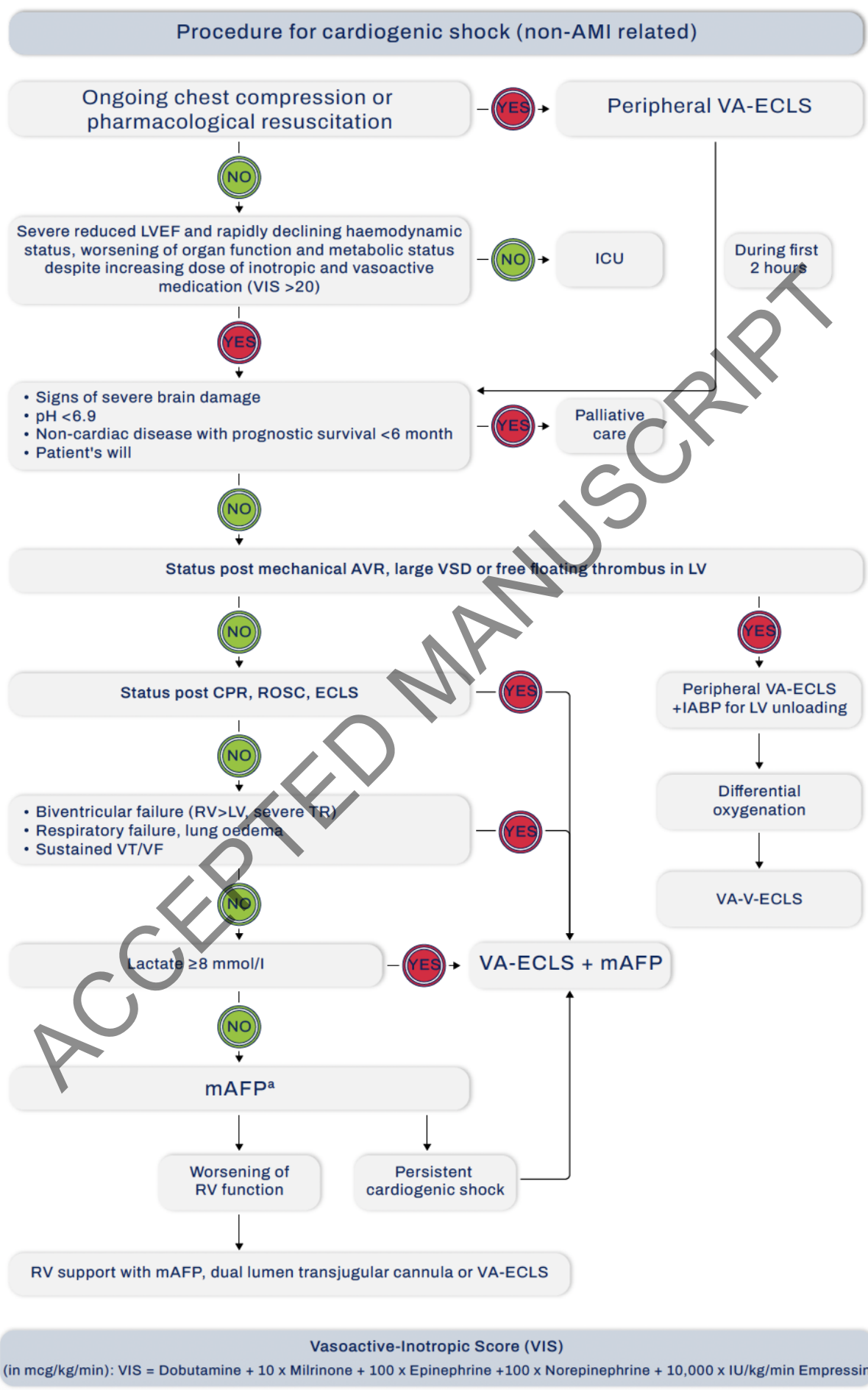


Figure 3. Procedure for Cardiogenic Shock Not Related to Acute Myocardial Infarction. AMI indicates acute myocardial infarction; AVR, aortic valve replacement; CPR, cardiopulmonary resuscitation; ECLS, extracorporeal life support; IABP, intra-aortic balloon pump; ICU, intensive care unit; LV, left ventricle; LVEF, left ventricular ejection fraction; mAFP, microaxial flow pump; ROSC, return of spontaneous circulation; RV, right ventricle; TR, tricuspid regurgitation; VA, veno-arterial; VA-V, veno-arterial-venous; VF, ventricular fibrillation; VIS, vasoactive inotropic score; VSD, ventricular septal defect; VT, ventricular tachycardia. Adapted from Ott et al.⁴ with permission from Multidisciplinary Digital Publishing Institute. ^aIn patients with profound shock, there is a clear benefit of full support over partial support mAFP.

Recommendation Table 3: Recommendations for the Initiation of tMCS in Patients with Cardiogenic Shock

Recommendations	Class ^a	Level ^b	Ref ^c
In patients with cardiogenic shock, it is recommended to initiate tMCS before the onset of severe organ dysfunction.	I	B	(114, 122, 126, 156)
In patients with cardiogenic shock, escalating vasoactive and/or inotropic drugs should be considered as an indication for tMCS.	IIa	B	(14, 34, 101, 108, 110)
In patients with cardiogenic shock, clinical signs of tissue hypoperfusion combined with rising lactate levels should be considered as an indication for tMCS.	IIa	B	(14, 114, 121, 125-127)
In patients in cardiogenic shock on high-dose vasoactive and/or inotropic drugs, early initiation of tMCS should be considered to reduce AKI.	IIa	B	(165)
For patients with a witnessed cardiac arrest undergoing immediate bystander CPR without ROSC, ECPR should be considered if it can be accomplished within 60 min.	IIa	B	(126, 127, 141, 148, 149)
In patients in cardiogenic shock due to acute MI, tMCS may be considered prior to revascularization.	IIb	B	(18, 51, 133-136, 166)
The initiation of tMCS is not recommended in patients with: life expectancy < 1 year,	III	C	-

• severe, irreversible neurologic damage, • unwitnessed arrest, • known patient or family wishes against the use of tMCS.			
In patients with a mechanical aortic prosthesis, or the presence of a mobile LV thrombus, or a large VSD, the institution of an mAFP is not recommended.	III	C	-

Abbreviations: AKI, acute kidney injury; CPR, cardiopulmonary resuscitation; ECPR, extracorporeal resuscitation; LV, left ventricle; mAFP, microaxial flow pump; MI, myocardial infarction; ROSC, return of spontaneous circulation; tMCS, temporary mechanical circulatory support; VSD, ventricular septal defect.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

7. TEMPORARY MCS FOR ADVANCED HF

There is a critical shortage of donor hearts for patients with end-stage HF awaiting HTx. In many countries and regions, wait times are prolonged, often leading to clinical deterioration during the waiting period. Implantable durable left ventricular assist devices (dLVADs) used as BTT therapy have stabilized many patients while allowing them to maintain an acceptable quality of life (QoL). However, despite their reliability, many patients are not candidates for a dLVAD due to anatomical challenges (eg, unfavourable ventricular geometry or a hostile apex), biventricular failure, hypercoagulable states, central nervous system disorders, valvular disease, or other comorbidities. In such cases, tMCS offers an alternative strategy for haemodynamic stabilization and bridging to an HTx.

7.1 Current tMCS trends and outcomes

Since the 2018 revision of the United States organ allocation policy, which lowered the priority of patients with a dLVAD, there has been a notable increase in tMCS use. The trend has shifted away from the dLVAD toward VA-ECLS, IABP, and mAFP. In contrast, in most European countries, IABP has largely been abandoned as a tool for waiting list prioritization. Notably, over 5% of current HTx candidates in the United States are now bridged with VA-ECLS, compared to less than 1% prior to the policy change.¹⁶⁷ No difference in transplant

deaths between VA-ECLS and a dLVAD has been observed,^{8,168} although graft loss has increased.^{8,169} Nonetheless, there has been a decline in waiting list deaths and removals, with higher transplant rates and improved 6-month survival (91.2%) compared to the pre-policy-change era when a dLVAD was the preferred BTT strategy. A recent study comparing 118 patients bridged with VA-ECLS versus 415 patients who received direct transplants showed no difference in post-HTx survival, although the VA-ECLS group required more packed red blood cell (PRBC) transfusions during the first postoperative week.

mAFP used as a BTT has demonstrated a 94.8% 1-year survival rate. After risk adjustment, the dLVAD was associated with a 4-fold increase in 1-year deaths compared to mAFP. Postoperative complication rates (eg, stroke, AKI, permanent pacemaker implant) were similar.¹⁷⁰ Patients BTT using tLVADs, and, in particular, mAFP have shown superior survival compared to those with VA-ECLS, with no differences in rates of rejection or need for renal replacement therapy (RRT).¹⁷¹ However, such comparisons are challenging due to differences in baseline acuity.

Despite survival benefits, tMCS is associated with significant morbidity, including bleeding, stroke, and renal failure. Axillary mAFP or upper-body VA-ECLS configurations are preferable, because they improve rehabilitation potential and pre-HTx conditioning.

7.2 Complications associated with tMCS as a bridge to dMCS

Factors such as elevated bilirubin levels, high C-reactive protein levels, VA-ECLS duration >7 days, body mass index (BMI) >30, and female sex are associated with worse outcomes in patients undergoing dLVAD implants after VA-ECLS. These risk factors are similarly relevant in HTx candidates bridged with VA-ECLS.^{172,173} Overall, VA-ECLS prior to a dLVAD implant is associated with poorer survival, whereas other forms of tMCS showed outcomes comparable to those patients directly undergoing dLVAD implants without bridging.¹⁷⁴

7.3 Complications associated with tMCS as a bridge to an HTx

Outcomes for patients bridged to an HTx with VA-ECLS have been inconsistent. For example, in Spain, this strategy has been associated with increased 1-year post-transplant deaths compared to patients supported with isolated LV support.

tMCS-related complications significantly affect post-transplant outcomes. A multicentre registry study reported that nearly 2% of HTx recipients with preoperative tMCS experienced

a stroke, with VA-ECLS identified as an independent risk factor.¹⁷⁵ Similarly, in a recent registry of 4,257 patients, those bridged with VA-ECLS had the highest stroke risk compared to dLVAD recipients.¹⁷⁶ Some reports show that surgical tLVAD support leads to longer operative and CPB times and worse long-term survival versus a dLVAD, despite a similar adverse event profile.¹⁷⁶

Risk profiles vary among bridging strategies. For instance, patients bridged with VA-ECLS showed higher rates of surgical re-exploration and infection compared to those supported with other tVADs, although 1-year survival did not differ.¹⁷⁷ A 2023 United Network for Organ Sharing registry study found that patients supported with VA-ECLS or tMCS had the highest operative and 30-day mortality rates, as well as increased bleeding and sepsis compared to those undergoing a primary HTx.¹⁷⁸

Given these findings, patient instability and the degree of haemodynamic compromise should be carefully assessed before initiating tMCS. Both the decision to initiate tMCS and the choice of the device should be tailored to minimize end-organ dysfunction and optimize outcomes.

Recommendation Table 4: Recommendations for Timing of tMCS Implants in Patients with Chronic Heart Failure

Recommendations	Class ^a	Level ^b	Ref ^c
In patients who are deteriorating ^d while waiting for dMCS/HTx, tMCS is recommended.	I	B	(8, 170, 171, 174)
Prior to the placement of a dMCS or an HTx, optimization of end-organ function with tMCS should be considered.	IIa	B	(172, 173)
In patients scheduled for a dLVAD, optimization of RV function with tMCS may be considered.	IIb	C	-

Abbreviations: dLVAD: durable left ventricular assist device; dMCS: durable mechanical circulatory support; HTx: heart transplant; RV: right ventricle; tMCS: temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

^dDespite guideline-directed medical therapies, including inotropes.

8. PATIENT EVALUATION AND PLANNING FOR TMCS: PROTECTED CARDIAC SURGERY CONCEPT IN HIGH-RISK PATIENTS

Patients undergoing cardiac surgery may present with significant haemodynamic and end-organ compromise and/or may be at high risk for postoperative low cardiac output syndrome (LCOS).¹⁷⁹ LCOS can also develop unexpectedly, often triggered by intraoperative complications that prevent successful weaning from CPB.¹⁸⁰⁻¹⁸² Preoperative cardiac dysfunction, whether acute or chronic, increases the likelihood of a complicated postoperative course. In this high-risk population, early consideration of tMCS, rather than escalating vasoactive and inotropic support, may provide superior haemodynamic stability and improved outcomes while minimizing the consequences of postoperative cardiopulmonary failure (Figure 4).^{51,179-182}

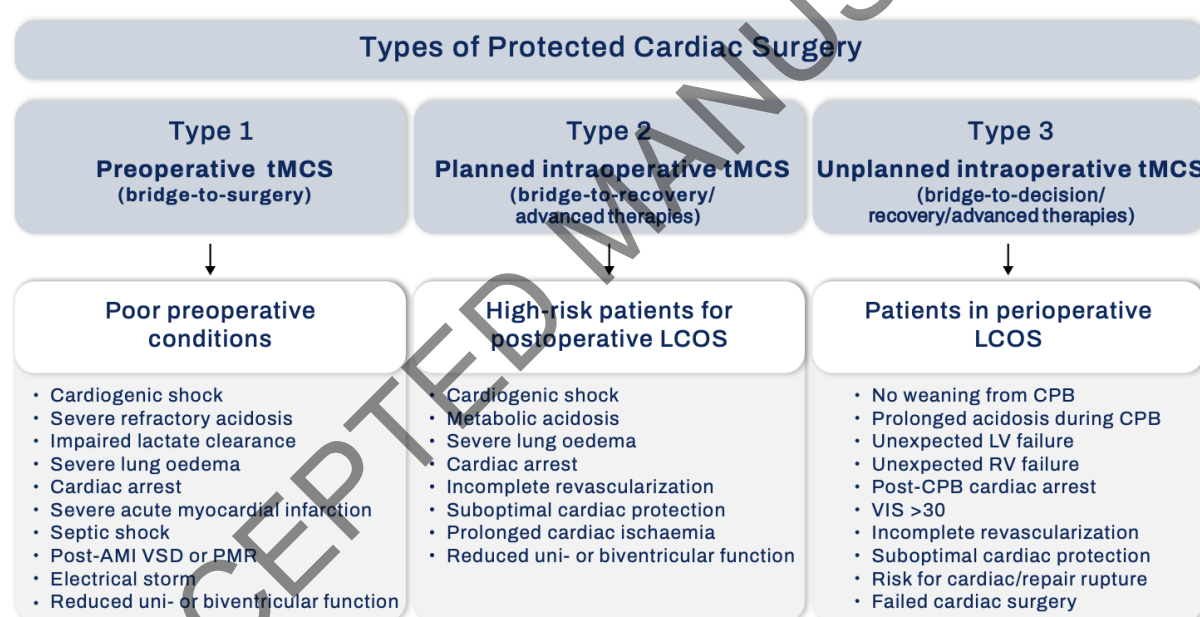


Figure 4. Surgical settings for protected cardiac surgery management using pre-emptive tMCS. AMI indicates acute myocardial infarction; CPB, cardiopulmonary bypass; LCOS, low cardiac output syndrome; LV, left ventricle; PMR, polymyalgia rheumatica; RV, right ventricle; tMCS, temporary mechanical circulatory support; VIS, vasoactive inotropic score; VSD, ventricular septal defect.

The application of perioperative tMCS, both pre- and post-surgery, represents a significant advancement and is now referred to as protected cardiac surgery. This concept is analogous to that of protected PCI, which has been used with increasing frequency in interventional

cardiology.¹⁸³ The use of different tMCS strategies should be considered based on institutional resources, availability, and expertise.

8.1 Pre-procedural tMCS planning and application in patients at high risk for peri-procedural LCOS (protected cardiac surgery)

Even when surgical indications are well established, an unfavourable risk–benefit ratio may argue against operative management, favouring either optimal medical therapy, interventional alternatives or a palliative treatment. A range of preoperative factors, including age, frailty, metabolic and haemodynamic disturbances, and end-organ vulnerability, must be carefully considered when assessing the need for prophylactic tMCS support. Heart Team discussions involving the patient and family should inform the decision-making process.^{179, 184–188} In patients with severe clinical compromise and peripheral organ dysfunction, preoperative optimization with tMCS may be critical to improving outcomes (Figure 5).^{139,185,189,190}

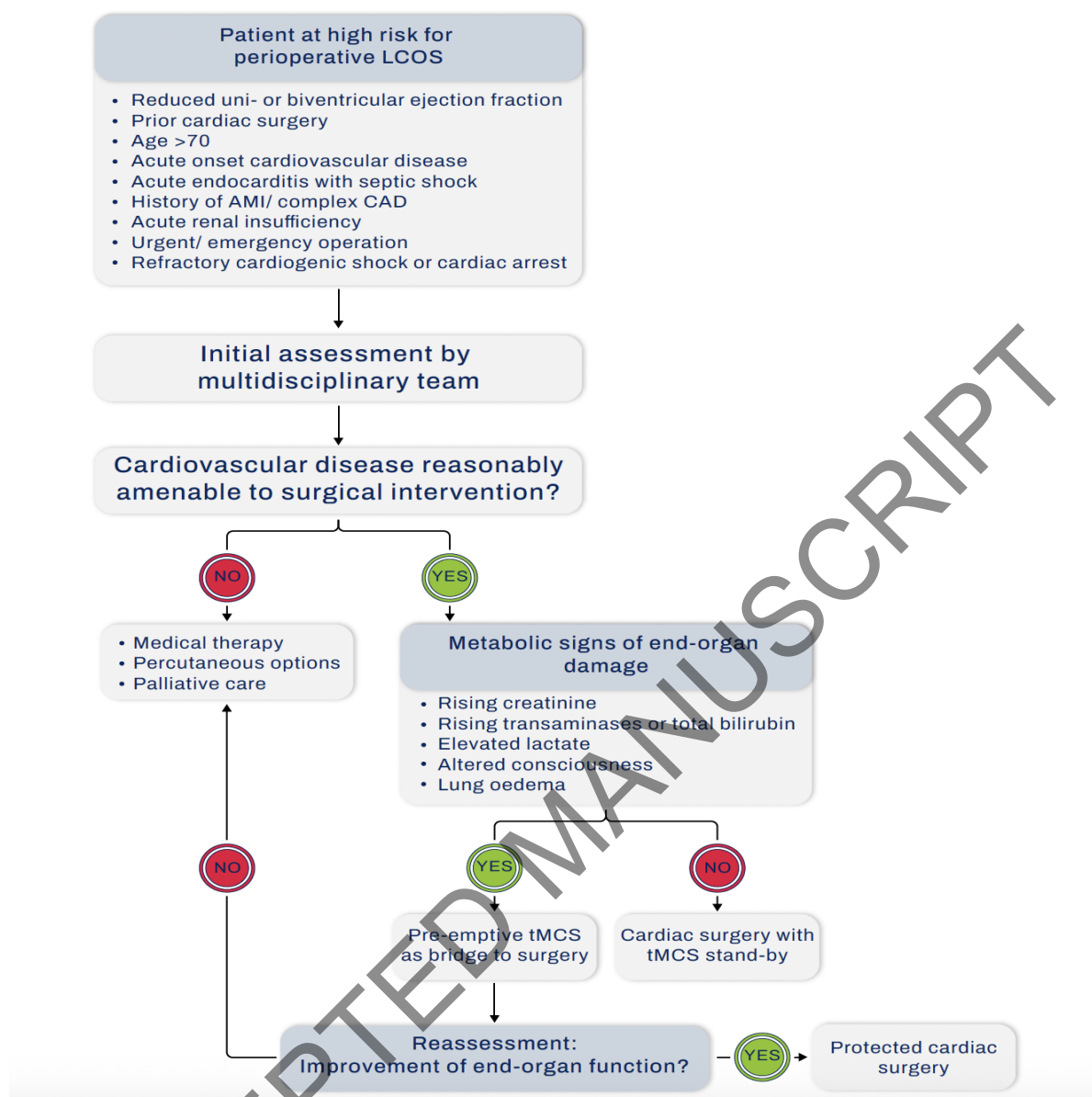


Figure 5. Algorithm for Assessing Preoperative Mechanical Circulatory Support Indication. AMI indicates acute myocardial infarction; CAD, coronary artery disease; LCOS, low cardiac output syndrome; MCS, mechanical circulatory support. Modified from Stein and Silvestry SC185 and adapted from Lorusso et al.²⁰

In patients with ischaemic mechanical complications post-AMI, such as ventricular septal rupture (VSR), papillary muscle rupture (PMR), or ventricular free wall rupture, who exhibit haemodynamic, respiratory, or metabolic instability, preoperative tMCS may be used as a bridge to corrective surgery.^{54,191-193} Similarly, in patients at high risk of developing postoperative LCOS, an intraoperative tMCS implant should be considered to facilitate CPB weaning and support recovery.¹⁹⁴⁻¹⁹⁷ Prophylactic MCS decisions should involve the Heart

Team. Appropriate device selection and meticulous haemodynamic monitoring are essential to optimize the direction of post-implant recovery.^{192,198-201} Moreover, maintaining tMCS into the early postoperative period appears beneficial in preventing the poor outcomes typically associated with post-cardiotomy shock.^{192,198,200-202} For example, patients with CABG with severely reduced LV function may benefit from mAFP support,^{203,204} whereas patients with acute or chronic RV dysfunction undergoing cardiac surgery may require tRVAD. In these high-risk patients, a “protected” strategy using tailored RV support should be considered.^{70,179,205-214}

8.2 Protected cardiac surgery and the ERAS cardiac framework

The principles of protected cardiac surgery align closely with the recommendations of the Enhanced Recovery After Surgery (ERAS) cardiac guidelines.²¹⁵ The use of tMCS should be considered to facilitate improved perioperative courses in high-risk populations, including older or frail patients, and patients with malnutrition or chronic end-organ dysfunction.

Recommendation Table 5: Recommendations for a tMCS Implant in Protected Cardiac Surgery for Patients at High Risk of Periprocedural LCOS

Recommendations	Class ^a	Level ^b	Ref ^c
It is recommended that the type and modality of tMCS should be discussed by an interdisciplinary team (Shock Team) based on haemodynamic conditions, patient characteristics, and prognosis.	I	B	(186-188)
Significant co-morbidities, advanced age, elevated lactate levels, and renal injury are risk factors associated with death and should be considered prior to initiation of tMCS.	IIa	B	(186, 194, 197)
In patients with structural cardiac anomalies, post-AMI mechanical complications, and other haemodynamic, respiratory, or metabolic compromise refractory to optimal medical therapy, implanting a preoperative tMCS may be considered as a bridge to the corrective procedure.	IIb	B	(54, 191-193)

In patients at high risk of post-procedural LCOS, intraoperative implantation of tMCS should be considered to facilitate CPB weaning and the perioperative course as a bridge to recovery.	IIa	C	(194)
When the likelihood of myocardial recovery is low, post-procedural tMCS is recommended in patients eligible for HTx or a dMCS.	I	C	(186, 194)
When experiencing difficulty weaning from CPB, early implementation of tMCS is recommended.	I	B	(37, 186, 197)

Abbreviations: AMI, acute myocardial infarction; CPB, cardiopulmonary bypass; dMCS, durable mechanical circulatory support; HTx, heart transplant; LCOS, low cardiac output syndrome; tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

9. PATIENT EVALUATION AND TIMING OF A tMCS IMPLANT FOR POST-CARDIOTOMY SHOCK

9.1 Post-cardiotomy LCOS

Post-cardiotomy CS is defined as persistent LCOS either manifesting as failure to wean from CPB or developing in the immediate postoperative period. LCOS may arise from pre-existing heart failure, worsened by surgical trauma, or occur acutely due to intraoperative events. These include technical complications, inadequate myocardial protection, iatrogenic myocardial ischaemia, or pre-existing myocardial dysfunction. Additional risk factors include advanced age, high BMI, recent MI, renal insufficiency, emergency surgery, reoperations, and prolonged CPB or aortic cross-clamp times.^{38,194,216,217}

Post-cardiotomy LCOS is associated with high mortality, substantial morbidity, and increased health-care resource utilization.^{54,218,219} Incidence varies from 2% to 20%, depending on patient comorbidities and procedure type.^{109,186,220-224} Although persistent postoperative LCOS carries a poor prognosis,²²⁵ survival rates of 25% to 42% can be achieved with timely initiation of tMCS.^{54,226-228} Consequently, early identification of high-risk patients (eg, reduced

left ventricular ejection fraction [LVEF], redo surgery, and pulmonary arterial hypertension [PAH]) is essential.

According to the 2020 EACTS/ELSO/STS/AATS Expert Consensus Document,¹⁷⁹ tMCS should be considered intraoperatively without delay to prevent acidosis, ongoing myocardial damage, excessive vasopressor use, pulmonary oedema, and end-organ hypoperfusion.³⁷ Some patients may benefit from prophylactic intraoperative tMCS, an approach central to the concept of protected cardiac surgery (Table 5).

Historically, the IABP has been the most frequently used device in this setting and has demonstrated reductions in in-hospital mortality, complication rates, and length of stay in the ICU,²²⁹⁻²³² although this has not been replicated in non-operative settings.²³³

Table 5 outlines clinical scenarios and corresponding tMCS modalities warranting consideration in the post-cardiotomy setting.

Table 5. Clinical Scenarios That Warrant Consideration for tMCS and Possible tMCS Type^a

Type of surgical setting	Extent of periprocedural risk ^b	Type of tMCS
Poor preoperative surgical candidate status		
Severe metabolic acidosis	****	VA-ECLS + mAFP
Severe acute pulmonary oedema	***	VA- or VV- ECLS
Severe cardiogenic shock	****	VA-ECLS + mAFP
Cardiac arrest	*****	VA- ECLS
CABG surgery		
Coronary artery bypass grafting with preoperative LVEF (<30%)	**	IABP mAFP
Coronary artery bypass grafting in patients with AMI-related cardiogenic shock or cardiac arrest	****	VA-ECLS + mAFP
Coronary artery bypass grafting with low LVEF and expected incomplete or complex revascularization	****	IABP mAFP

Coronary artery bypass grafting with low LVEF and diffuse coronary artery disease	***	IABP mAFP
Coronary artery bypass grafting in Kawasaki syndrome with aneurysmatic (giant) coronary arteries	***	IABP mAFP
Post-AMI mechanical complications		
Ventricular septal rupture	***	IABP VA-ECLS + mAFP
Ventricular septal rupture with RV dysfunction	****	VA- ECLS VA-ECLS + mAFP
Papillary muscle rupture	**	VA-ECLS + mAFP
Ventricular reconstruction for extensive LV aneurysm	***	mAFP VA-ECLS + mAFP
Valve surgery		
Patients undergoing TAVI with severely depressed uni- or biventricular function	***	mAFP VA-ECLS + mAFP
Acute mitral and/or aortic endocarditis with massive regurgitation, pulmonary oedema, and uni- or biventricular dysfunction	****	mAFP VA-ECLS + mAFP
Mitral valve surgery with low preoperative LVEF	***	IABP mAFP VA-ECLS + mAFP
Mitral valve surgery with high pulmonary hypertension and signs of RV dysfunction	****	VA-ECLS + mAFP
Tricuspid valve surgery in the presence of preoperative RV dysfunction	*****	tRVAD OxyRVAD – VA-ECLS –
Aortic valve replacement for massive regurgitation and chronic poor preoperative LVEF	***	mAFP VA-ECLS + mAFP

Aortic and mitral valve surgery for severe insufficiency combined with CABG and severe LV dysfunction (LVEF < 30%)	*****	VA-ECLS + mAFP
Heart failure surgery		
Heart transplant with prolonged ischaemic time and uni- or biventricular failure after ECC weaning	***	VA-ECLS + mAFP tRVAD OxyRVAD
LVAD implant with preoperative moderate-to-severe RV dysfunction	****	tRVAD OxyRVAD
Pulmonary artery-related surgery		
Pulmonary artery endo-endarterectomy for chronic embolization and with preoperative chronic RV dysfunction	*****	RVAD OxyRVAD VA-ECLS
Pulmonary artery embolectomy for acute pulmonary embolism and preoperative RV dysfunction	***	RVAD OxyRVAD–VA-ECLS
Aortic surgery		
Type A dissection with coronary artery compromise and preoperative myocardial ischaemia	****	VA-ECLS + mAFP
Prolonged surgery with long cardiac ischaemic time and difficult ECC weaning (particularly for RV dysfunction)	***	VA-ECLS + mAFP
Post-trauma cardiovascular surgery		
Cardiac or major vessel injury requiring cardiac surgery with lung contusion	***	VA-ECLS
Constrictive pericarditis surgery		

Pericardiectomy with preoperative moderate-to-severe RV dysfunction, with preoperative TVI requiring intervention associated with RV dysfunction	****	RVAD OxyRVAD VA-ECLS VA-ECLS + mAFP
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Abbreviations: AMI, acute myocardial infarction; CABG, coronary artery bypass grafting; ECC, extracorporeal circulation; ECLS, extracorporeal life support; IABP, intra-aortic balloon pump; LV, left ventricle; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction; mAFP, microaxial flow pump; oxyRVAD, combination of RVAD and oxygenator; RV, right ventricle; RVAD, right ventricular assist device; TAVI, trans-catheter aortic valve implant; tMCS, temporary mechanical circulatory support; tRVAD, temporary right ventricular assist device; TVI, time velocity integral; VA-, veno-arterial; VV-, veno-venous.

^aAdapted from EACTS/ELSO/STS/AATS expert consensus on post-cardiotomy extracorporeal life support in adult patients¹ with permission from Oxford University Press.

^b“Low periprocedural risk (*)” through incremental levels (** to ****) to “High periprocedural risk (*****)”.

Implementation of tMCS should occur prior to the onset of irreversible systemic injury, ideally pre- or intraoperatively. This is particularly important at the onset of intraoperative or immediate post-procedural LCOS,²³⁴ especially when separation from CPB is uncertain.²³⁵

Device selection is a critical component of the decision-making process for protected cardiac surgery, because the chosen support modality may need to serve as a bridge to more advanced therapies.^{208,210}

9.2 Patient evaluation and timing

Two principal clinical scenarios may necessitate tMCS in post-cardiotomy LCOS: (1) Inability to wean from CPB and/or (2) development of refractory LCOS in the ICU. The critical task is identifying patients who will benefit from early tMCS initiation (see [Sections 8, 10, and 11; Figure 6](#)). In patients with LCOS following cardiac surgery, full-support mAFP should be prioritized over partial support when feasible.

In the face of the inability to wean the patient from CPB, comprehensive reassessment of potential contributors is warranted^{1,236}:

- ECG: Rhythm disturbances, ischaemic changes

- Pulmonary artery catheter or alternate device (eg, the pulse index continuous cardiac output device [PiCCO]): Haemodynamic parameters
- Transoesophageal echocardiography (TOE): Valve competence, ventricular function, wall motion abnormalities
- Ventilation/bronchoscopy: Tidal volumes, compliance, pulmonary congestion, atelectasis
- Blood gas analysis: Acid-base balance, oxygenation/ventilation status
- Haemostasis: Active bleeding, coagulopathy, need for surgical revision
- Reperfusion parameters: Duration, adequacy of myocardial recovery

If LCOS persists despite optimal correction of the above, immediate transition to tMCS should be strongly considered to support recovery or serve as a bridge to dMCS or HTx.^{1,14,37,186,187,191,194-196,236,237} Importantly, delays in escalation should be avoided (see **Sections 4 and 5**).

9.3 Predicting outcomes

Predicting the outcome of post-cardiotomy LCOS requiring tMCS remains challenging, with only a few scoring systems currently available. Existing scores, such as the Survival after Veno-Arterial ECMO (SAVE) and ENCOURAGE,^{199,238} have not been specifically validated for post-cardiotomy shock. The REMEMBER score,²³⁹ developed for patients undergoing CABG procedures, has been proposed for use in this setting, although formal validation is still lacking.

The SAVE (Survival After Veno-arterial ECMO) score was originally developed to predict survival to hospital discharge in patients supported with VA-ECLS, only a subset of whom were post-cardiotomy.¹⁹⁹ It incorporates factors such as age, body weight, aetiology of cardiogenic shock, and pre-ECLS assessments of respiratory, cardiac, and end-organ function. The REMEMBER score (pRedicting mortality in patients undergoing veno-arterial Extracorporeal MEMbrane oxygenation after coronary artEry bypass gRafting) was specifically designed to predict in-hospital mortality in post-cardiotomy patients requiring VA-ECLS, but its validation is limited to patients who undergo CABG.²³⁹ It is based on pre-ECLS parameters—some overlapping with SAVE—and includes additional factors such as coronary artery disease (CAD) subtype, VIS, and selected laboratory values. This score has demonstrated better

1195 performance than SAVE, SOFA, ENCOURAGE, or EuroSCORE, but again, its use may not be
1196 generalizable beyond CABG cases.
1197 Notably, all scoring systems highlight the importance of timing in determining treatment
1198 success, with early tMCS initiation associated with improved outcomes. In all cases,
1199 assessment of the potential for myocardial recovery or eligibility for bridging strategies should
1200 guide the decision to initiate tMCS, with palliative care as an alternative when appropriate.

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Algorithm for employment of tMCS for cardiogenic shock after heart surgery

- CPB weaning not possible^a
- CPB weaning possible, chest is closed and acute deterioration in OR - re-opening and causal treatment^a, or
- Patient is in the OR or on the ICU and meets one of the criteria:
 - VIS score >30
 - VIS score >20 and
 - Cardiac Index <2.2 L/min/m²
 - PCWP >16 mmHg and/or CVP >20 mmHg (in euvoaemic status^b)
 - Pulmonary oedema with PCWP >16 mmHg (caution: TRALI with low PCWP)
 - >50% reduction of LV/RV ejection fraction compared to preoperative measurements
 - Lactate increasing by 50% in 2 consecutive measurements over 30 min or >72 mg/d
 - SvO₂ or ScvO₂ <50% in 2 consecutive measurements over 30 min

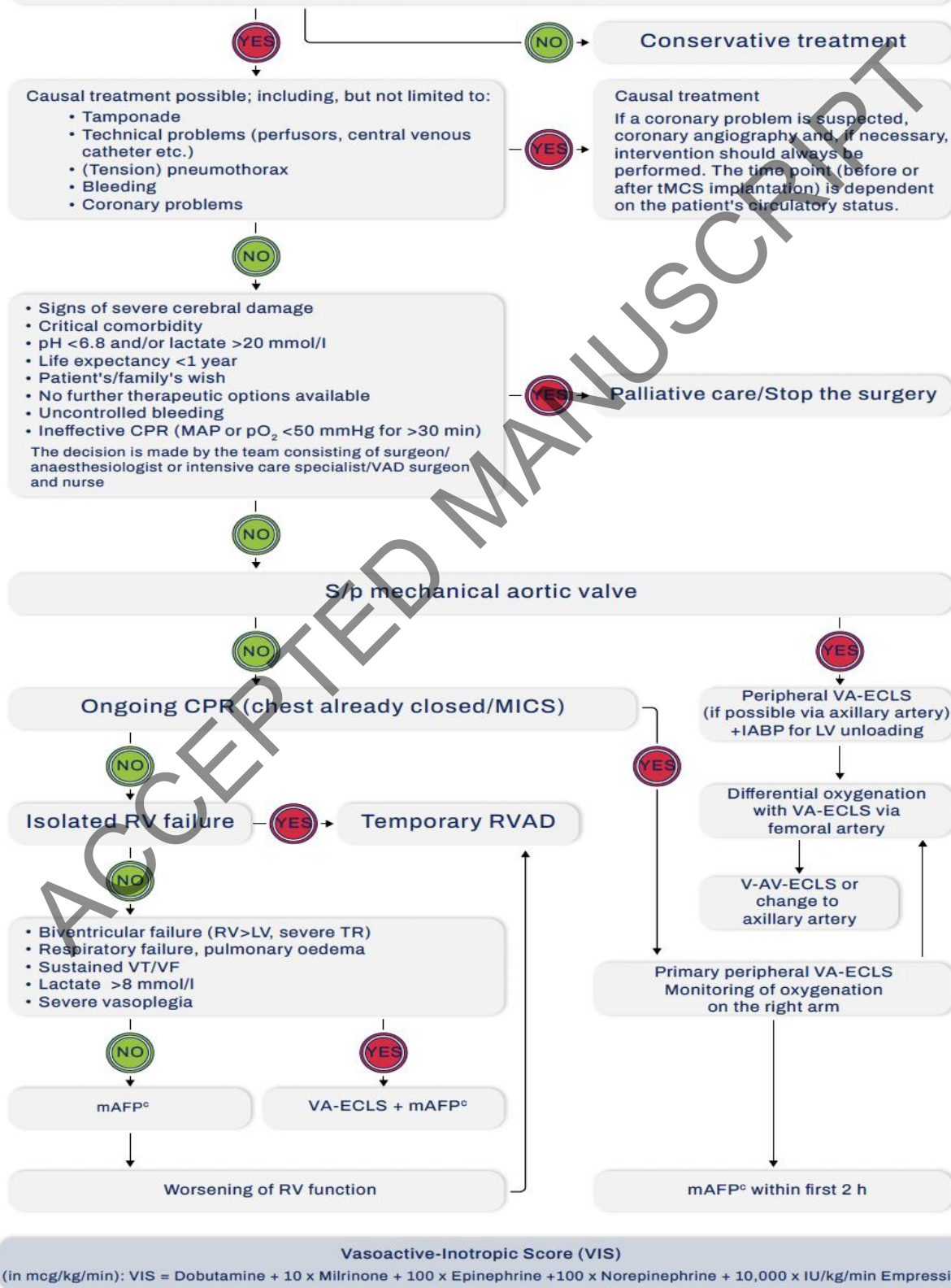


Figure 6. Algorithm for Employment of Temporary Mechanical Circulatory Support for Cardiogenic Shock After a Cardiac Surgery. In this context, there is a clear benefit of full-support mAFP over partial-support mAFP, and full-support mAFP should be preferentially used whenever feasible. CPB indicates cardiopulmonary bypass; CPR, cardiopulmonary resuscitation; CVP, central venous pressure; ECLS, extracorporeal life support; ICU, intensive care unit; LV, left ventricle; MAP, mean arterial pressure; mAFP, microaxial flow pump; MICS, minimal invasive cardiac surgery; OR, operating room; PCWP, pulmonary capillary wedge pressure; pO₂, partial pressure of oxygen; preop, preoperative; RV, right ventricle; RVAD, right ventricular assist device; S/P, status post; SvO₂, venous oxygen saturation; ScvO₂, central venous oxygen saturation; tMCS, temporary mechanical circulatory support; TR, tricuspid regurgitation; TRALI, transfusion-related acute lung injury; V-AV, venous-arterial-venous; VA-ECLS, veno-arterial extracorporeal life support; VAD, ventricular assist device; VF, ventricular fibrillation; VT, ventricular tachycardia; VIS, vasoactive inotropic score. Adapted from Nersesian et al.¹⁵ with permission from Multidisciplinary Digital Publishing Institute. ^aPrimary chest closure should be anticipated. ^bV. cava < 2.2 cm, collapsed > 50%, stable haematocrit; lung compliance > 50 ml/mbar. ^cIn patients with profound shock, there is a clear benefit of full support over partial support mAFP.

Recommendation Table 6: Recommendations for a tMCS Implant in Post-procedural LCOS

Recommendations	Class ^a	Level ^b	Ref ^c
In patients with post-procedural LCOS, tMCS is recommended.	I	B	(195, 196, 228, 240)
In patients with no likelihood of myocardial recovery and not eligible for dMCS/HTx, tMCS is not recommended.	III	C	(37, 54)
For tMCS-eligible patients who cannot be weaned from CPB despite optimal therapeutic measures, immediate transition to tMCS is recommended.	I	B	(195, 197, 228, 240)

Abbreviations: CPB, cardiopulmonary bypass; dMCS, durable temporary mechanical circulatory support; HTx, heart transplant; LCOS, low cardiac output syndrome; tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

10. SPECIAL CONSIDERATIONS FOR A TMCS IMPLANT IN ACHD

Temporary MCS in ACHD is an evolving and increasingly relevant field. The growing population of children surviving complex congenital heart defects is now transitioning into a wave of adult patients with ACHD (eg, those with Fontan circulation), the majority of whom will eventually develop HF. This trend will continue due to improvements in paediatric cardiac care.²⁴¹ The field of tMCS is also evolving, with tVADs now used in various support strategies.²⁴² These devices are frequently employed as a BTT, especially in patients with ACHD for whom a durable VAD implant poses challenges (eg, multiple prior sternotomies, collateral circulation, complex anatomy, and bleeding risk).²⁴³

The indications for tMCS in ACHD are similar to those in other adult populations, including CS due to myocarditis, acute allograft failure, and acute decompensation of chronic HF precipitated by intercurrent illness (eg, respiratory infections) or procedures. Physiologic indications remain comparable.²⁴⁴ Contraindications are similar, but one unique consideration is the higher prevalence of advance directives for patients with ACHD, even among those in their 20s and 30s, and this must be checked in patients presenting with shock or cardiac arrest.

The challenges lie in applying tMCS, given the anatomical and physiologic complexities of congenital heart diseases. Systemic vessel anatomy and patency may be unreliable, with anomalies such as aberrant aortic arch vessels, interrupted inferior vena cava (IVC), absent right superior vena cava (SVC), bilateral SVCs (1 ligated), or cavopulmonary connections. Femoral and cervical vessels may be occluded or narrowed from prior interventions (eg, shunts, operations, catheterizations, central lines, shunts sewn to the innominate artery, prior bypass cannulation of femoral vessels, ECLS).

Generally, patients with non-single ventricle ACHD, including those with systemic RV failure (eg, congenitally corrected transposition of the great arteries), should receive acute tMCS support similar to that received by patients without ACHD.²⁴⁵⁻²⁴⁸ However, for patients with a single ventricle, one must consider the high collateral flow (20%–50% of CO), which may require flows >5 L/min/m² to ensure adequate support. This should influence tMCS device selection.

In otherwise functional patients with ACHD who deteriorate acutely, tMCS may serve as a BTR or a transplant.²⁴³ In contrast, in deconditioned patients with chronic HF or failing Fontan physiology, tMCS is typically insufficient for rehabilitation; these patients should be

transitioned early to dMCS. Outcomes following an HTx in deconditioned patients with ACHD with end-organ dysfunction are often poor.

10.1 ECLS and ACHD

In ECPR scenarios where time constraints prevent confirmation of vessel patency, alternative access sites should be considered if cannulation proves difficult. ECLS in patients with ACHD is similar to that in other adults, but Fontan physiology warrants specific attention.²⁴⁶ In these patients, the systemic ventricle must be directly decompressed; fenestration-based decompression is often inadequate in the presence of ventricular dysfunction.^{249,250}

In adult Fontan patients, 20%–50% of CO may return to the systemic ventricle via aorto-pulmonary collaterals. This volume load in the setting of systolic dysfunction and restrictive physiology leads to high atrial pressures and pulmonary oedema. Traditionally, after initiating peripheral VA-ECLS, the safest next step is central cannulation via a Glenn connection, IVC, and systemic atrium. Peripheral arterial cannulation may be continued if tolerated.

The mAFP offers a promising option for systemic ventricular decompression²⁴² but is often technically challenging to use in “Fontan patients” due to difficult arterial access and anatomical distortions (eg, large DKS or “aneurysmal” arch).

In patients with a single ventricle (eg, post-Fontan or Glenn procedures) or those with systemic-to-pulmonary shunts (eg, classic or modified Blalock-Taussig, Potts, or Waterson shunts), high cardiac index targets (>4 L/min/m²) may be required. This requirement impacts cannula size selection.²⁵¹

10.2 tVAD support

Experience with continuous-flow tVAD support, such as magnetically levitated extracorporeal centrifugal pumps, is limited.²⁵⁰ This strategy can serve as a BTR in patients with ACHD with post-cardiotomy dysfunction, myocarditis, or allograft rejection. Although a thoracotomy may avoid a re-sternotomy, it can be hazardous due to collaterals and adhesions. A sternotomy approach is generally preferred.

In patients with the “Fontan circulation”, extracorporeal tVAD may help assess candidacy for transitioning from VA-ECLS to a durable ventricular assist device. However, co-existing pulmonary pathology (eg, pneumonia, single lung, high PVR) may reduce the preload and limit VAD effectiveness.

A thorough anatomical assessment is crucial, especially for a right aortic arch, aberrant subclavian arteries, or prior shunts sewn to the innominate artery. These may have been placed for systemic-to-pulmonary flow or prior surgical palliation (eg, antegrade cerebral perfusion).

Most patients with ACHD now receive mAFP support when feasible. With a careful understanding of the patient's underlying defect and the need for supranormal CO, a tMCS implant is largely similar to that in non-ACHD populations.²⁵¹

If a Fontan fenestration is present, there is no need to close it during mAFP support. In fact, some centres intentionally create a fenestration when establishing dVAD support. The resulting cyanosis created by the right-to-left shunting can enhance VAD filling, and oxygen saturation typically improves with VAD-driven cardiac output.

A rare but critical scenario involves patients with ACHD with residual banded physiology. In such cases, decompression of the systemic ventricle can reduce pulmonary blood flow, making mAFP use inappropriate.

Finally, IABP is suitable for "non-Fontan circulation" patients with ACHD, similar to non-ACHD use. However, in patients with "Fontan circulation", IABP use is limited due to abnormal anatomy, small aortic size, or poor compliance of the aortic wall. The use of IABP in this setting should proceed with caution.²⁵²

Recommendation Table 7: Recommendations for tMCS in Patients with ACHD

Recommendations	Class ^a	Level ^b	Ref ^c
In patients with good functional status and preserved end-organ function, tMCS is recommended as a BTT dMCS or cardiac recovery.	I	C	(243, 244, 246)
In patients with biventricular anatomy and physiology, including those with systemic right ventricles (eg, cc-TGA; d-TGA s/p atrial switch), a tMCS implant is recommended in a manner similar to that for adults without ACHD.	I	C	(245-248)

In patients with Fontan circulation supported with central ECLS, direct ventricular decompression should be considered. In patients with Fontan circulation supported with peripheral ECLS, mAFP should be considered for ventricular decompression.	IIa	C	(249, 250)
In patients with Fontan circulation supported with ECLS, direct systemic ventricular decompression via a fenestration is not recommended.	III	C	(249, 250)
In patients with Fontan circulation requiring tMCS, the presence of collaterals and persistent arterial-to-pulmonary shunts should be considered when determining pump flow.	IIa	C	(251)

Abbreviations: ACHD, adult congenital heart disease; BTT, bridge to a transplant; CCTGA, congenitally corrected transposition of the great arteries; dMCS, durable mechanical circulatory support; d-TGA, dextro-transposition of the great arteries; ECLS, extracorporeal life support; mAFP, microaxial flow pump; tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

11. CONCOMITANT CARDIAC CONDITIONS

11.1 Aortic valve

Aortic valve replacement or repair is indicated for patients with severe pathology who are candidates for an operation.^{253,254} In patients presenting with CS, tMCS can serve as a bridge to a definitive intervention.

11.1.1 Aortic stenosis

In patients with severe AS, priority should be given to surgical aortic valve replacement or TAVI. In cases of CS due to severe AS, VA-ECLS may be considered for haemodynamic stabilization and end-organ perfusion.^{255,256} Peripheral cannulation is often the fastest option and can improve perfusion, but it does not decompress the LV. Up to 50% of patients may require LV venting to avoid LV distension.²⁵⁷ The decision to vent should be guided by

evidence of LV overload, pulmonary hypertension, or RV compromise. Balloon atrial septostomy or percutaneous transseptal LA drainage can be used to decompress the LA, although in severe AS this may increase the risk of LV thrombus formation.²⁵⁵

Transaortic mAFPs have been used as standalone therapy or LV vents in this setting. Abraham et al. described a case series in which they successfully bridged patients with severe AS with mAFP to TAVI.²⁵⁸ Notably, full-support mAFP requires a minimum aortic valve area (AVA) >0.6 cm².^{255,259}

11.1.2 Aortic regurgitation

In patients with severe aortic insufficiency (AI) and depressed LVEF, urgent surgical intervention is typically indicated. However, for patients in CS with moderate-to-severe AI, tMCS with active LV unloading may be considered.²⁶⁰ IABP and VA-ECLS alone may exacerbate LV distension, while mAFP can provide effective decompression. The primary goal is to improve end-organ perfusion and reverse the shock state while closely monitoring for pulmonary congestion and LV overload. Surgical correction of AI should not be delayed in cases of persistent LV distension.

11.2 Mitral valve

In patients with severe mitral valve disease, surgical repair or replacement remains the definitive treatment. For patients presenting with acute decompensation or for whom surgery is not immediately available, tMCS may serve as a BTT.

11.2.1 Mitral regurgitation

In acute severe MR with CS, tMCS with active LV unloading should be considered. VA-ECLS alone may restore end-organ perfusion but can increase afterload and worsen MR, leading to pulmonary congestion.¹⁹³ An mAFP may actively unload the LV and can be used alone or with VA-ECLS.^{261,262} Percutaneous transseptal LA drainage, with or without oxygenation, is also an option for LV unloading.²⁶³ IABP may be useful for short-term augmentation of forward flow but should not be the sole support modality in patients with profound shock.

11.2.2 Mitral stenosis

In patients with critical mitral stenosis and CS, VA-ECLS or transseptal tLVAD configurations with LA and RA drainage and arterial return may be considered.^{260,264,265}

11.2.3 Prior mitral prosthesis

Patients with a mitral valve replacement requiring tMCS represent a unique challenge. Lack of pulsatile transmitral flow can lead to prosthetic valve or LA thrombosis.²⁶⁶⁻²⁶⁸ VA-ECLS can be used if sufficient preload is maintained to allow forward flow, as confirmed by serial echocardiography. For severely impaired LV function, mAFP or apical cannulation for extracorporeal tLVAD may be the preferred treatment modality.²⁶⁶

11.3 Acute myocardial infarction

The incidence of CS in patients who present with AMI is 5% to 10%.²⁶⁹ These patients are increasingly managed with early revascularization (PCI or CABG).²⁶⁹ The use of IABP in patients with AMI-CS was historically supported by United States and European guidelines.^{270,271} However, recent data, including the multicentre randomized IABP-SHOCK II trial demonstrated no survival benefit in the IABP group. Consequently, routine use of IABP is no longer recommended for patients with AMI-CS.⁶⁶ Nevertheless, IABP may still be considered in select patients with AMI-CS as a bridge to definitive revascularization strategies.

The ECMO-SHOCK trial found no benefit of VA-ECLS over medical therapy¹⁷ in patients with AMI-CS. However, few patients received active unloading, and the majority had suffered cardiac arrest prior to VA-ECLS initiation. In a separate study, Lemor et al. analysed over 6,000 patients admitted with AMI-CS who underwent PCI and were treated with either mAFP or VA-ECLS. Their findings demonstrated improved clinical outcomes in the mAFP group.²⁷² When used in combination, VA-ECLS with mAFP unloading in post-MI CS patients demonstrated a survival benefit compared to those treated with VA-ECLS alone or VA-ECLS plus IABP, further suggesting that active unloading is critical for improving outcomes in the post-MI state.^{273,274} The DanGer study also showed a clear benefit of even partial LV unloading in this setting using mAFP alone.¹⁸

11.3.1 Post-infarction VSD

Post-infarction VSD is a well-documented mechanical complication of MI, associated with high morbidity and mortality (19%-60%).¹⁹² CS in this context is driven by the extent of infarction related to occlusion of the culprit vessel and left-to-right shunting through the VSD. Echocardiography aids in assessing biventricular function and determining the size and location of the VSD. Patients with moderate to large VSDs are at risk of acute RV deterioration and circulatory collapse, necessitating prompt initiation of tMCS.^{192,275,276}

More than 80% of patients with post-MI VSD present in CS.¹⁹² tMCS may help mitigate shock and allow for delayed VSD closure. Given the variability in presentation, selection of appropriate tMCS requires careful balancing of therapeutic strategy and evolving haemodynamics. The haemodynamic effects of tMCS in this setting are illustrated by Ronco et al. in [Figure 7](#).

Post-infarct VSDs can be surgically repaired depending on their size and location.^{85,277} The timing of intervention remains a matter of debate. Advocates of early repair point to the benefits of prompt shunt closure, which can be difficult to manage with conventional tMCS if left unaddressed. However, an early operation carries a higher risk of mortality, likely due to persistent shock and the friability of the infarcted tissue. As a result, delayed closure has increasingly become the preferred strategy.²⁷⁸

IABP may not provide sufficient support in cases of RV failure or profound LV dysfunction.¹⁹² Alternatively, mAFP has been employed as a bridge to closure, as described in some case series.^{85,277} However, caution must be exercised to avoid creating an iatrogenic right-to-left shunt, which may lead to systemic hypoxaemia in the presence of a large VSD. The high mortality rate observed in this series (46%) was primarily attributable to RV failure and complications of CS, emphasizing the importance of continuous monitoring of RV function and timely escalation of support.²⁷⁷ VA-ECLS has also been used to support patients in CS before VSD repair, with Vondran et al. reporting improved survival in patients preoperatively supported with VA-ECLS.^{275,279}

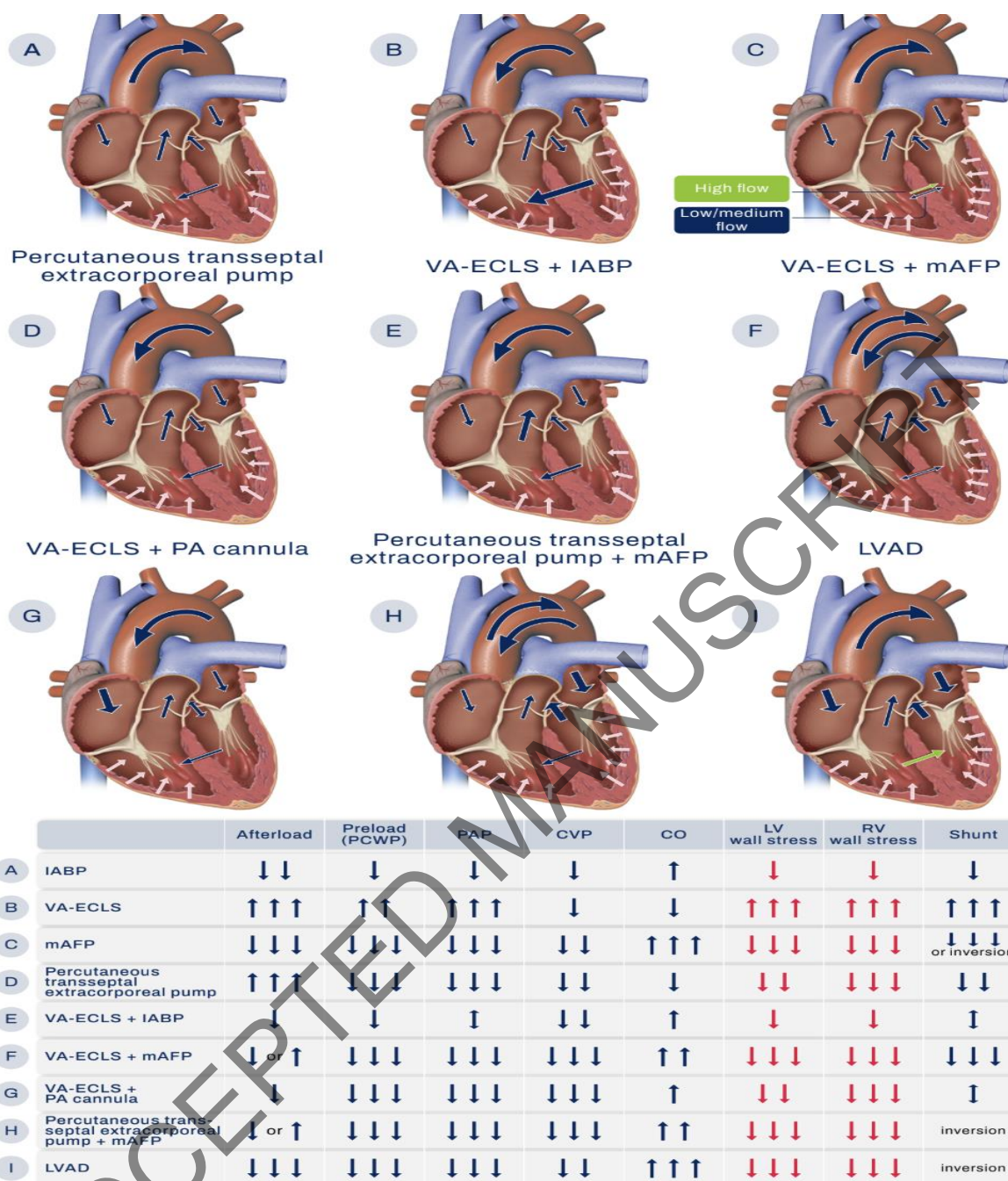


Figure 7. Baseline Haemodynamic Effects of Mechanical Circulatory Support on the Pathophysiology of Post-infarction Ventricular Septal Rupture. CO indicates cardiac output; CVP, central venous pressure; IABP, intra-aortic balloon pump; LV, left ventricle; LVAD, left ventricular assist device; mAFP, microaxial flow pump; PA, pulmonary artery; PAP, pulmonary artery pressure; PCWP, pulmonary capillary wedge pressure; RV, right ventricle; VA-ECLS, veno-arterial extracorporeal life support. Modified from Ronco et al.¹⁹² with permission from Elsevier.

11.4 Pulmonary embolism

A massive PE is defined as an acute PE with associated haemodynamic instability, cardiac arrest, or persistent bradycardia with signs and symptoms of shock without other apparent causes.²⁸⁰ It results in an acute increase in RV afterload, RV strain, impaired LV filling, and a subsequent decrease in CO.³⁹ In-hospital mortality rates range from 25% for patients with a PE and CS to 65% for those who require CPR.^{281,282}

Treatment strategies include systemic thrombolysis, pulmonary embolectomy, and catheter-directed therapies. Given the heterogeneity of presentations, many centres have developed Pulmonary Embolism Rapid Response Teams, comprising medical, surgical, and interventional specialists who collaborate to expedite decision making in a patient-centred manner. In this population, increasing evidence supports the use of VA-ECLS in cases of cardiac arrest or CS,²⁸³ where rapid initiation can decompress the RV and restore adequate circulation.

In a retrospective comparison of patients with confirmed massive PEs treated at their centre, Moon et al. found that survival was significantly improved in the VA-ECLS group compared to those receiving medical therapy.²⁸¹ Similarly, a literature review by Bhalla and Attaran involving 164 patients with a PE treated with VA-ECLS showed that 74% were successfully weaned from support, with a 73% survival to discharge.³⁹ Al-Bawardy et al. reported patients with cardiac arrest due to a massive PE who received VA-ECLS support; despite the high-risk profile, 30-day mortality was 31%, with a mean support duration of 5.5 days.²⁸⁰

11.5 Heart transplant

11.5.1 Primary graft dysfunction

Primary graft dysfunction (PGD) is a poorly understood complication following HTx, affecting 2% to 28% of all transplants.²⁸⁴ According to The International Society of Heart and Lung Transplantation (ISHLT) consensus guidelines, PGD must be diagnosed within 24 hours of the operation and is categorized as a PGD-LV group and a PGD-RV group.²⁸⁴ Mechanical support is indicated when medical therapy fails to sustain the newly transplanted heart in patients with CS.²⁸⁴ The type and timing of support have been the focus of several studies.

Takeda et al., in a cohort of 597 patients who received transplants, reported superior survival with VA-ECLS compared to tVAD for patients with severe PGD.²⁸⁵ DeRoo et al. further explored timing, showing in their study of 362 HTxs that 10.5% of the patients developed severe PGD. Patients who received early VA-ECLS (mean 1.9 hours postoperatively) had significantly better

1-year survival compared to those with delayed initiation (7.3 hours) (90% vs 67%, respectively).²⁸⁶

Olivella et al. reviewed all HTx cases from the Spanish Registry between 2010 and 2020.²⁸⁷ PGD occurred in 21% of patients who had a primary HTx, with 48% requiring tMCS. Seventy percent received early support (<3 hours). Although mortality did not significantly differ between early (28%) and late (31%) groups, late initiation was associated with increased renal failure. Notably, peripheral cannulation was linked to better survival compared to central access. Interestingly, DeRoo et al. also used peripheral access and partial flow in 95% of patients. Both studies emphasized that early initiation and centre experience, particularly with peripheral cannulation, may contribute to improved survival in post-HTx patients requiring VA-ECLS.

11.5.2 Vasoplegia

Vasoplegia is common occurrence following an HTx.²⁸⁸ Evidence supporting the use of tMCS is largely anecdotal, and initial treatment remains pharmacologic. However, in cases of refractory vasoplegia where hypotension persists and adequate perfusion cannot be maintained, early initiation of tMCS may be considered.²⁸⁹

11.6 Durable left ventricular assist device

Severe RV failure requiring tMCS occurs in 4% to 6% of patients undergoing an LVAD implant and is associated with increased morbidity and mortality.²⁹⁰ Beller et al. reviewed the STS Database of isolated LVAD implants from 2011 to 2019.²⁹¹ In this study of over 18,000 patients, they found that 5.1% of patients required either RVAD or VA-ECLS therapy for severe RV dysfunction after a dLVAD implant. The overall operative mortality for patients requiring tMCS for RV support after LVAD was 43.3%. VA-ECLS was associated with a significantly higher risk of mortality compared to tRVAD. Additionally, VA-ECLS was linked to increased rates of vascular injury and infectious complications compared to tRVAD in this cohort.²⁹¹

Although tRVAD use has shown benefits in treating severe RV dysfunction post-dLVAD, there are situations in which VA-ECLS may be the only feasible option.²⁹⁰ In these circumstances, the interaction between ECLS and the LVAD must be closely monitored. VA-ECLS increases afterload and decreases preload, which may negatively affect the performance of pre- and afterload-sensitive devices. Therefore, partial ECLS flow and LVAD speed adjustments should

be considered. Patients with peripheral VA-ECLS cannulation, including those with dLVADs, must also be monitored for differential hypoxia.

The timing of tMCS initiation for severe RV failure after an LVAD implant has been shown to significantly affect survival. Data strongly support initiating tMCS at the time of the index procedure. In a large cohort study, patients with delayed tMCS had a 54.6% unadjusted mortality rate compared to 36.5% in those who received tMCS during the index procedure.²⁹¹ These findings align with those of Fitzpatrick et al., who reported increased deaths when the RVAD implant was delayed beyond the index procedure.²⁹² Therefore, for patients experiencing severe RV failure following a dLVAD implant, VA-ECLS should be avoided. tRVAD is the preferred modality and should be placed at the time of the index procedure.

11.7 Arrhythmias

Patients with refractory unstable ventricular arrhythmia have improved outcomes with rapid deployment of tMCS compared to conventional non-tMCS therapies.^{48,293,294} Prophylactic tMCS may be indicated in high-risk patients undergoing ablation, especially when salvage tMCS would be associated with poorer outcomes or when prolonged ablation procedures are anticipated. Risk scores such as the PAINESD score can help identify patients who may benefit from tMCS during ablation. This score includes factors such as pulmonary disease, age >60, VT, ischaemic cardiomyopathy, New York Heart Association functional class III or IV, ejection fraction <25%, VT storm, and diabetes mellitus (DM).^{49,295} Advocates of tMCS-supported ablation cite an improved ability to map VT foci.^{48,296,297} However, tMCS-supported ablation has been associated with increased procedural complications, and some studies found no significant difference in 1-year mortality or VT recurrence.²⁹⁶

11.8 Intracardiac thrombus

Intracardiac thrombus is associated with both acute and chronic HF conditions.²⁹⁸ In a study by Zhai et al., left-sided thrombi were most commonly found in the LV of patients with CAD, ventricular noncompaction, and dilated cardiomyopathy. Patients with atrial arrhythmias were more likely to have thrombus localized in the LA.²⁹⁸ Floating intracardiac thrombi carry a heightened risk of embolization or device-related complications such as pump ingestion and stroke.^{299,300} Therefore, active suction devices are not recommended in these cases, and VA-

ECLS is the preferred modality for circulatory support. If LV unloading is necessary, mAFP is contraindicated, and IABP should be considered instead. If the thrombus is confined to the LV, transseptal LA venting may be an alternative. For right-sided thrombi, tRVAD are not recommended. VA-ECLS remains the only viable option, though close monitoring for thrombosis of the inflow cannula in the RA and for haemolysis is essential.

In patients supported with VA-ECLS who lack an effective ventricular ejection due to severe ventricular dysfunction or persistent malignant arrhythmias, the risk of intracardiac thrombosis increases due to blood stasis. In such scenarios, LV unloading with mAFP or conversion to BiVAD support should be considered to mitigate thrombotic risk.^{255,266}

11.9 Pulmonary arterial hypertension

Patients with severe PAH and associated RV failure present a particularly challenging clinical scenario. In selected patients, VA-ECLS should be considered as a BTT or for RV recovery.³⁰¹

The use of isolated tRVAD in this setting has been associated with a heightened risk of pulmonary haemorrhage and oedema and is therefore not recommended.³⁰¹ Although some case reports have described isolated RVAD use in PAH, the prevailing evidence suggests that this approach is contraindicated.⁴¹⁻⁴³

11.10 Takotsubo cardiomyopathy

Takotsubo cardiomyopathy is a transient form of ventricular dysfunction typically triggered by physical or emotional stress. It is characterized by apical and anteroapical dyskinetic ballooning of the LV with hyperkinetic basal segments.³⁰² Approximately 10% of patients with Takotsubo cardiomyopathy progress to CS, which is associated with in-hospital death rates as high as 15%. Managing CS in this population is challenging. Although inotropic support is often required initially, there is concern that it may exacerbate the underlying condition. For patients with severe CS, mAFP may provide effective LV unloading; however, evidence supporting this strategy remains limited and is primarily derived from case series.^{58,303-305}

Recommendation Table 8: Recommendations for the Use of tMCS in Patients with Cardiogenic Shock and Specific Concomitant Cardiac Conditions

Recommendations	Class ^a	Level ^b	Ref ^c
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Aortic valve			
In patients with cardiogenic shock and a mechanical aortic valve, a transaortic valve mAFP is not recommended.	III	C	(255)
In patients with cardiogenic shock and moderate-to-severe aortic regurgitation, tMCS with active LV unloading should be considered.	IIa	C	-
In patients with cardiogenic shock and severe aortic stenosis, VA-ECLS as a bridge to AVR/TAVI may be considered.	IIb	C	(255, 256)
In patients with cardiogenic shock and severe aortic stenosis, mAFP may be considered.	IIb	C	(255)
Mitral valve			
In patients with severe acute mitral regurgitation and cardiogenic shock, tMCS with active LV unloading should be considered as a bridge to interventional or surgical treatment of the mitral valve.	IIa	C	(193, 261, 262)
In patients with a prosthetic mitral valve on tMCS, strategies to maintain pulsatile transmitral valvular flow are recommended.	I	C	(266-268)
In patients with cardiogenic shock and critical mitral stenosis, ECLS or transseptal pVAD should be considered.	IIa	C	(260)
Acute MI			
In patients with post-MI cardiogenic shock, the use of tMCS with active unloading should be considered.	IIa	B	(17, 272, 273)
The use of IABP may be considered in carefully selected patients ^d with post-MI cardiogenic shock, particularly in settings where access to advanced tMCS is limited.	IIb	B	(66)
VSD			

In patients at risk for deterioration, close monitoring of RV function and venous congestion should be considered for prompt initiation of tMCS.	Ila	C	(192, 275)
In patients with post-MI VSD, the use of VA-ECLS to support patients in cardiogenic shock prior to VSD closure is recommended.	I	B	(192, 275, 276, 278, 279)
In patients with post-MI VSD, the use of mAFP to support patients in cardiogenic shock prior to VSD closure may be considered.	Ilb	C	-
In patients with post-MI VSD on mAFP, serial monitoring of RV function should be considered for escalation of support.	Ila	C	(275)
Pulmonary embolism			
In patients with a massive PE who present with cardiac arrest or cardiogenic shock, VA-ECLS is recommended.	I	B	(39, 280, 281)
Heart transplant			
In patients with severe primary graft dysfunction following an HTx, early use of tMCS should be considered.	Ila	B	(286, 287)
In patients with refractory vasoplegia following an HTx, early use of tMCS may be considered.	Ilb	C	-
dLVAD			
In patients with severe RV failure immediately after an LVAD implant, a tRVAD implant is recommended at the time of the index procedure.	I	B	(291)
In patients with severe RV failure after an LVAD implant, early tRVAD should be considered over VA-ECLS.	Ila	B	(291)
Arrhythmias			
In a high-risk patient with cardiac arrhythmia, a tMCS may be considered to provide haemodynamic stability during mapping and/or ablation.	Ilb	B	(48, 49, 294, 296, 297)
In patients with refractory unstable ventricular arrhythmia, biventricular tMCS is recommended.	I	C	(293)

Intracardiac thrombus			
In patients with intra-atrial thrombus, transseptal pVAD is not recommended.	III	C	(299)
In patients with acute intraventricular thrombus, an mAFP is not recommended.	III	C	(300)
Pulmonary hypertension			
In patients with RV failure due to primary, idiopathic pulmonary hypertension, the use of tRVAD is not recommended.	III	C	(301)
In patients with RV failure due to primary, idiopathic pulmonary hypertension, the use of VA-ECLS should be considered as a BTT.	IIa	C	(301)
Takotsubo cardiomyopathy			
In patients with Takotsubo cardiomyopathy who are developing cardiogenic shock, tMCS with active LV unloading is recommended.	I	B	(302-305)

Abbreviations: AVR, aortic valve replacement; BTT, bridge to a transplant; dLVAD, durable left ventricular assist device; ECLS, extracorporeal life support; HTx, heart transplant; IABP, intra-aortic balloon pump; LV, left ventricle; LVAD, left ventricular assist device; mAFP, microaxial flow pump; MI, myocardial infarction; PE, pulmonary embolism; pVAD, percutaneous ventricular assist device; RV, right ventricle; TAVI, transcatheter aortic valve implant; tMCS, temporary mechanical circulatory support; tRVAD, temporary right ventricular assist device; VA, veno-arterial; VSD, ventricular septal defect.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

^dPatients with certain conditions precluding the use of active left ventricular unloading, such as ventricular thrombus, mechanical aortic valve, or large ventricular septal defect, may particularly benefit from IABP.

12. NON-CARDIAC COMORBIDITIES

Non-cardiac pathologies are a critical consideration when evaluating candidacy for tMCS, because they can significantly impact the clinical course and prognosis. Some comorbidities,

such as frailty, may go unrecognized due to the emergency circumstances under which tMCS is often initiated. Whereas clinical history is informative, a focused bedside assessment may be the only diagnostic resource available in emergencies, despite its inherent limitations. If the clinical situation allows, chest X-ray, angiography, and whole-body CT imaging should be performed to exclude underlying malignancy and identify anatomical variants that could influence cannulation strategy.¹⁹ In patients undergoing elective tMCS, a thorough pre-implant evaluation, including screening for non-cardiac comorbidities and frailty, is recommended to guide the viability of dMCS or HTx as exit strategies.³⁰⁶

12.1 Age

Advanced age is associated with an increased prevalence of medical comorbidities and frailty.³⁰⁷ Whereas some studies comparing patients aged ≥ 65 vs < 65 years report similar survival rates, others indicate worse outcomes in the older group, leading to mixed conclusions overall.³⁰⁸⁻³¹⁰ Several retrospective studies have shown that elderly patients can achieve acceptable outcomes following a dMCS implant.^{19,311,312} For VA-ECLS specifically, 2 single-centre studies reported survival to discharge rates between 36%–44% in octogenarians.^{313,314} Additionally, an analysis of the Extracorporeal Life Support Organization (ELSO) registry found no significant association between age and adverse outcomes among patients older than 70 years.³⁴⁵ In contrast, a subgroup analysis of the randomized DanGer trial suggests that older patients (cut-off 77 years) with CS complicating AMI have high mortality and may derive less benefit from the routine use of a partial-flow mAFP than younger patients.³¹⁶ Taken together, age alone should not be considered a contraindication to tMCS, but associated comorbidities, functional capacity, and frailty must be integrated into the decision-making process.^{19,310,317,318}

12.2 Frailty

Frailty, characterized by features such as weight loss, exhaustion, reduced physical activity, slowness, and weakness, reflects diminished physiologic reserve and heightened vulnerability to clinical stressors.³¹⁹ In patients with HF, frailty is highly prevalent, with a meta-analysis of 26 observational studies estimating a rate of approximately 45%.³²⁰ Multiple comorbidities are present in 72% of patients meeting frailty criteria.³²¹

Although data on the impact of frailty in tMCS recipients are limited, it has been associated with poor outcomes following a dMCS implant.³²²⁻³²⁴ In acute, life-saving settings, comprehensive frailty assessment is rarely feasible.^{325,326} In such cases, surrogate markers from imaging, such as diminished psoas or erector spinae muscle mass, may offer additional insights.

When tMCS is planned in advance, structured frailty screening tools (eg, Clinical Frailty Scale, Katz Index), combined with imaging-based sarcopenia assessment, should be utilized.³¹² Ultimately, moderate-to-severe frailty should be regarded as a relative contraindication to tMCS, while in emergency scenarios, frailty assessment must rely on clinical judgement.

12.3 Malnutrition and cardiac cachexia

Malnutrition is independently associated with increased mortality among hospitalized patients with HF.^{327,328} A review of the INTERMACS database found that 5% of patients undergoing a dLVAD implant were diagnosed with cardiac cachexia, and 2.5% had a BMI <18.5 kg/m².³²⁹ Although malnourished dLVAD recipients exhibited higher mortality, those who achieved >5% postoperative weight gain experienced improved outcomes.³²⁹ Low preoperative pre-albumin levels were associated with increased mortality and complication rates after a dMCS implant, though they did not appear to impact the success of a BTT.³³⁰ Evidence on the impact of malnutrition in the setting of tMCS is limited. In a cohort of 150 patients, 10 of whom had a BMI <18.35 kg/m², no significant association was found between underweight status and clinical outcomes.³³¹ A case series by Salas et al. described improvements in caloric intake, strength, and mobility among cachectic patients supported with mAFP as a bridge to dLVAD.³³² Taken together, existing data do not support malnutrition as an absolute contraindication to tMCS. In acute scenarios, assessment of malnutrition and cardiac cachexia should rely on clinical judgement.

12.4 Obesity

Obesity increases the risk of comorbid conditions, though some have proposed an “obesity paradox” in the context of HF.^{333,334} Uncertainty remains regarding the influence of obesity on outcomes in decompensated chronic HF and CS.³³⁵ Similarly, the effect of obesity on VA-ECLS outcomes is debated. One regression analysis found no correlation between BMI and VA-ECLS outcomes,³³⁶ and 2 retrospective studies of patients with CS supported with VA-ECLS

found no significant mortality difference between those with a BMI above or below 30 kg/m².^{336,337}

However, a subsequent retrospective study of 283 patients who underwent ECPR for OHCA reported higher mortality among the 133 patients with a BMI ≥ 30 kg/m².³³⁸ Further analysis identified a BMI ≥ 35 kg/m² as associated with higher mortality,³³⁹ and increased complications with peripheral cannulation.³⁴⁰ Currently, obesity should not be considered an absolute contraindication to ECLS. However, a BMI ≥ 35 kg/m² should be recognized as a risk factor for poorer outcomes with ECLS support.

12.5 Sepsis

Sepsis has traditionally been considered a contraindication to MCS in adults, primarily due to concerns about the use of vascular prostheses in the setting of bloodstream infections. Nevertheless, accumulating evidence suggests that selected patients with refractory septic shock, particularly those with sepsis-induced cardiomyopathy or hyperdynamic shock, may benefit from MCS, including VA-ECLS.³⁴¹⁻³⁴⁵

Several retrospective studies have evaluated VA-ECLS in this context. Brechot et al. analysed 212 patients with sepsis-related cardiomyopathy and reported 90-day survival of 51% among the 82 patients treated with VA-ECLS, compared to 14% in unsupported patients.³⁴⁶ Similarly, Cheng et al. conducted a retrospective study of 151 septic patients, of whom 101 received VA-ECLS, and found a 42% survival to discharge.³⁴⁷ A 2021 meta-analysis of 14 observational studies found that patients supported with ECLS who had an LVEF $< 20\%$ had the best survival (62%), compared to those with an LVEF of 20%–35% (42.3%) or $> 35\%$ (32.1%).³⁴⁸

12.6 Removal of intracardiac devices

In some cases, intracardiac devices must be removed to eradicate device-related infections. Many of these patients cannot haemodynamically tolerate a support-free interval before a transplant or a device re-implant, especially if the optimal heart rhythm or re-synchronization cannot be maintained. Small case series and case reports have described the use of mAFP or ECLS as a bridge to a re-implantation or HTx.^{81,349-351} Each patient's ability to tolerate device removal should be carefully evaluated, and the use of tMCS may be considered to provide haemodynamic support during the device-free interval.

12.7 Diabetes mellitus

Type 2 DM is a well-established risk factor for cardiovascular disease and is associated with worse outcomes and an increased risk of complications in CS.³⁵² Two groups used meta-analyses to examine the association between DM and outcomes following a dLVAD implant. Although one group found that DM was associated with higher 30-day mortality,³⁵³ no difference was observed in 1-year mortality, all-cause mortality, device-related infections, or ischaemic/haemorrhagic stroke.^{353,354} Thus, DM should not be considered a contraindication to tMCS.

Hyperglycaemia is common in patients with CS: A recent study revealed that 56% of 393 patients supported with mAFP or ECLS had a blood glucose level >140 mg/dL.³⁵⁵ Patients with significantly abnormal glucose levels at the time of VA-ECLS cannulation that presented as either >400 mg/dL or <80 mg/dL were associated with worse outcomes compared to patients with normoglycaemia or moderate hyperglycaemia.³⁵⁶ Due to the high prevalence of peripheral vascular disease (PVD) in DM, clinicians should maintain a high index of suspicion for vascular pathology.³⁵⁷ Patients with long-standing or poorly controlled DM are at increased risk of microvascular disease and impaired wound healing, which may complicate cannulation. Early glucose optimization protocols may improve outcomes in patients with tMCS; however, future studies should determine whether improved glycaemic control independently contributes to better clinical outcomes.

Diabetic ketoacidosis, which may occur in both type 1 and type 2 DM, can lead to severe electrolyte and metabolic disturbances, potentially precipitating cardiac arrest or stress-induced cardiomyopathy.^{358,359} Case reports support the use of VA-ECLS in patients with diabetic ketoacidosis-associated cardiomyopathy,³⁶⁰⁻³⁶² suggesting that it may be a reasonable option in selected cases.

12.8 Peripheral vascular disease

Peripheral vascular disease is present in up to 6% of patients with AMI complicated by CS.^{363,364} Several studies have identified PVD and vascular complications as predictors of increased mortality in tMCS-supported patients.^{365,366} Vascular complications occur in 10% to 44% of patients on ECLS and are strongly associated with worse outcomes.³⁶⁷⁻³⁶⁹

Preprocedural imaging is essential to assess vascular anatomy, guide cannulation, and minimize complications. Insertion of a cannula into a diseased vessel can lead to distal flow

obstruction, embolization, or retrograde dissection. Among patients supported with VA-ECLS, PVD is associated with a higher risk of limb ischaemia and death.³⁷⁰⁻³⁷²

Whenever possible, a thorough evaluation of vessel size, atherosclerotic burden, and prior interventions (eg, stents, endarterectomy, or bypass grafts) should be conducted before cannulation. Point-of-care ultrasound (POCUS) enhances the safety of peripheral access.^{373,374}

Non-diseased, non-instrumented vessels are preferred. Use of a distal perfusion cannula significantly reduces the risk of critical limb ischaemia (see [Section 17](#)). In patients with severely calcified or critically stenotic femoral/iliac arteries, alternative approaches such as single-artery VA-ECLS with mAFP via supra-aortic vessels may be considered.⁶⁵ Axillary cannulation is a reasonable alternative to femoral cannulation and facilitates mobilization.³⁷⁵⁻

³⁷⁷ However, patients with axillary cannulation should be closely monitored for signs of distal arm hyper- or hypoperfusion.³⁷⁸ In extreme cases, central cannulation (eg, via the brachiocephalic artery or the ascending aorta/aortic arch) may be necessary.^{375-377,379}

Cannulation of previously stented or bypassed vessels should be avoided.

12.9 Heparin-induced thrombocytopenia

Retrospective analyses report the prevalence of heparin-induced thrombocytopenia (HIT) among patients supported with VA-ECLS to be between 0.4% and 3.6%, with an associated mortality rate approaching 40%.³⁸⁰⁻³⁸³ Although the persistence or de novo development of HIT type II is not a contraindication to tMCS, it necessitates modified anticoagulation strategies (see [Section 15.1.1 Anticoagulation agents](#)).

The use of heparin-coated circuits in patients with HIT remains controversial. A single retrospective study showed that discontinuing heparin while continuing to use heparin-bonded circuits resulted in outcomes similar to those in patients without HIT.³⁸¹ However, evidence is insufficient to provide clear guidance on the safety of heparin-coated circuits in patients with a history of HIT.

Plasmapheresis has been reported in the context of a dLVAD implant to remove antiplatelet antibodies both before and during the procedure.^{384,385} Nonetheless, evidence supporting the use of plasma exchange specifically for ECLS in HIT is sparse.³⁸⁶

12.10 Malignancy

There are multiple scenarios in which patients with malignancies may benefit from tMCS. In evaluating the appropriateness of tMCS in oncology patients, it is essential to consider their overall oncologic prognosis and whether bridging to advanced cardiovascular therapies is feasible. When possible, the patient's preferences must be factored into the decision-making process. The available data on tMCS in patients with cancer are limited and largely based on case reports and small series.

ECLS has been successfully used as a bridge to oncologic surgery in selected patients.³⁸⁷⁻³⁹² Case reports and 1 series describe the application of VA-ECLS for patients experiencing cardiopulmonary instability due to hormonal disturbances or paraneoplastic syndromes from endocrine tumours, serving as a bridge to definitive therapy.³⁹⁰⁻³⁹⁵

For patients with pheochromocytoma and haemodynamic instability, VA-ECLS may be employed to support circulation during preoperative alpha- and beta-blockade.³⁹¹⁻³⁹⁴ tMCS may also be considered in patients with haematologic malignancies involving direct cardiac infiltration, especially during chemotherapy-associated cardiovascular compromise.^{396,397} Neoplasms of the anterior mediastinum may cause compressive effects on the heart, great vessels, or airways, resulting in obstructive shock, with or without respiratory failure. In these cases, VA-ECLS has been used as a bridge to chemotherapy or surgical resection, as reported in isolated cases and small series.^{389-391,393-395,398-400} Although the data remain limited, the use of tMCS may be a reasonable option in selected cancer patients experiencing haemodynamic collapse due to mass effect, tumour-related cardiovascular complications, or chemotherapy-related heart failure, when used as a BTT and a BTR.

12.11 Liver disease

Liver dysfunction is frequently observed in patients with decompensated HF and CS undergoing evaluation for tMCS. The bidirectional interplay between cardiac and hepatic function complicates the management of acute haemodynamic deterioration.^{401,402} Reduced CO can cause centrilobular hepatocellular injury and necrosis due to hypoperfusion, while right heart failure with venous congestion may result in hepatic fibrosis and dysfunction. In a post hoc analysis of the SURVIVE trial, 46% of patients with acute decompensated HF exhibited abnormal liver function test (LFT) results at least once.⁴⁰³ Conversely, chronic liver conditions such as cirrhosis and non-alcoholic fatty liver disease may themselves contribute

to cardiac dysfunction.⁴⁰¹ In many patients, restoration of haemodynamics may lead to improvement of hepatic function.

12.11.1 Patients with ascites or known chronic liver disease

Evidence on outcomes of patients with chronic liver disease undergoing tMCS remains limited. Several studies have linked markers of hepatic fibrosis or dysfunction to worse outcomes after dMCS or an HTx.^{330,404,405} In a retrospective study by Chou et al., 233 patients with cirrhosis received either VA-ECLS or VV-ECLS.⁴⁰⁶ Among the 107 patients supported with VA-ECLS, in-hospital mortality was 67% and 2-year mortality was 80%.⁴⁰⁶ Mortality was significantly higher in patients over 65 years of age, with hypoalbuminaemia, respiratory failure, or post-liver transplant status. In addition, cirrhotic patients were at higher risk for bleeding and RRT.⁴⁰⁶ According to the ELSO Red Book, Child–Pugh Class B or C cirrhosis constitutes a contraindication to VA-ECLS.⁴⁰⁷ These patients face significantly higher in-hospital mortality, increased risk of renal failure and transfusion requirements, and lower 1-year survival.⁴⁰⁶ tMCS may be cautiously considered in carefully selected patients with Child–Pugh B or C cirrhosis.

12.11.2 Patients with laboratory evidence of acute liver dysfunction

Due to its high metabolic demands and dual vulnerability to poor inflow and venous congestion, the liver is particularly susceptible to injury in the setting of low CO and venous hypertension.⁴⁰⁸ Hepatic dysfunction often results from a combination of acute hypoperfusion, chronic or acute-on-chronic congestion, and underlying liver disease. Several studies have identified total bilirubin as a prognostic marker in tMCS. In 1 cohort of 223 patients on VA-ECLS treated between 2012 and 2016, total bilirubin levels strongly correlated with death: No patient with total bilirubin >30 mg/dL survived, and levels >11 mg/dL had 90% specificity for predicting death.⁴⁰⁹ Similar findings were reported in another study of 89 patients, where elevated bilirubin levels were significantly associated with in-hospital mortality.⁴¹⁰ Freundt et al. reported an inverse correlation between initial bilirubin levels and survival.⁴¹¹ Persistent or rising bilirubin levels after day 2 of VA-ECLS were associated with worse prognosis.⁴¹¹ Multiple studies have identified day 3 bilirubin ≥ 6 mg/dL and elevated bilirubin on post-implant day 5 as independent predictors of inpatient mortality.^{412–414}

12.12 Renal disease

Acute kidney injury in the setting of cardiogenic shock is frequently reversible and is often an indication, or a contributing factor, for initiating tMCS. Chronic kidney disease (CKD), including dialysis dependence, should not preclude tMCS, because renal function may improve after MCS initiation⁴¹⁵ or patients may be bridged to a combined heart–kidney transplant (HKTx). An analysis of the United Network for Organ Sharing database including 457 HKTx recipients reported that IABP and mAFP appear safe as bridging strategies in dual-organ candidates, with comparable post-transplant heart and kidney graft function, complications, and survival up to 2 years.⁴¹⁶ By contrast, another study noted higher mortality among VA-ECLS–bridged patients, although renal outcomes were similar.⁴¹⁷ INTERMACS data show that preoperative CKD is associated with approximately 3-fold higher mortality and greater morbidity in dLVAD recipients. Nevertheless, about one-third of patients on preoperative dialysis recover renal function after receiving an implant, achieving long-term survival and QoL similar to those without dialysis.⁴¹⁸ Although AKI commonly prompts tMCS initiation, advanced CKD should be considered in the context of the likelihood of myocardial and renal recovery and/or candidacy for HKTx. tMCS as a bridge to dMCS may be considered in carefully selected patients.

12.13 Cognitive and psychological disorders

Data on the use of tMCS in patients with pre-existing cognitive or psychological disorders are limited. Severe dementia is considered a contraindication to tMCS when intended as a bridge to a dMCS or HTx. In all cases, decision making should be informed by a multidisciplinary team evaluation and aligned with the patient’s and family’s preferences.

Recommendation Table 9: Recommendations for Use of tMCS in Patients with Non-cardiac Comorbidities

Recommendations	Class ^a	Level ^b	Ref ^c
Age and frailty			
It is recommended that advanced age not be considered as an absolute contraindication for tMCS.	I	B	(308-310, 313-315, 317, 318)

Moderate and severe frailty should be considered as a relative contraindication to tMCS.	IIa	C	(319, 322-324)
Patients who are obese or malnourished			
It is recommended that malnutrition alone not be considered as an absolute contraindication to tMCS.	I	C	(313, 330, 331)
It is recommended that obesity alone not be considered as an absolute contraindication to tMCS.	I	B	(331, 336-338)
A BMI > 35 kg/m ² may be considered as an additional risk factor for tMCS.	IIb	B	(338, 339)
In obese patients (BMI > 35 kg/m ²), imaging-guided cannulation for peripheral tMCS is recommended.	I	B	(340, 419)
Patients who are septic			
In patients with refractory septic shock and sepsis-related cardiomyopathy, tMCS should be considered.	IIa	B	(346-348, 420)
In patients who require durable device removal to treat a related infection, tMCS may be considered as a bridge to device exchange or a transplant.	IIb	C	(81, 349-351)
Patients with diabetes			
It is recommended that diabetes mellitus not be considered a contraindication for tMCS.	I	B	(352, 355)
tMCS should be considered in patients in cardiogenic shock due to diabetic ketoacidosis.	IIa	C	(360, 362)
Peripheral vascular disease			
In patients with PVD, it is recommended that adjustments to cannulation strategies be guided by available imaging.	I	C	-
Malignancy			
In oncology patients based upon an interdisciplinary prognosis of ≤ 1 year, tMCS is not recommended.	III	C	-
Liver disease			
It is recommended that chronic liver disease not be considered an absolute contraindication for tMCS.	I	B	(404-406, 408)

It is recommended that acute liver injury not be considered an absolute contraindication for tMCS.	I	C	(409-414)
Renal disease			
It is recommended that AKI not be considered a contraindication for tMCS.	I	B	(415, 416)
Patients with CKD should be considered for tMCS if HKTx or anticipated myocardial recovery is the primary goal.	IIa	B	(416)
In patients with CKD, tMCS as a bridge to dMCS may be considered.	IIb	B	(418)
Cognitive and psychological disorders			
In patients with dementia or cognitive or psychological disorders, their wishes, as communicated directly by the patient or family, are recommended to be considered in the decision for tMCS.	I	C	-

Abbreviations: AKI, acute kidney injury; BMI, body mass index; CKD, chronic kidney disease; dMCS, durable mechanical circulatory support; HKTx, heart–kidney transplant; PVD, peripheral vascular disease; tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

13. OPERATIVE TECHNIQUES

Device selection should be guided by the clinical scenario and individual patient characteristics. Echocardiography and invasive haemodynamic assessment are essential to determine the presence of uni- or biventricular failure and any pulmonary involvement.⁴²¹ Image guidance is recommended for cannula placement. In emergency settings, however, the need for imaging should not delay cannulation.

13.1 Isolated LV support

LV support can be achieved using mAFP, transseptal LVAD, or paracorporeal LVADs. Whenever possible, cannulation strategies should be considered in a way to facilitate early patient mobilization.^{14, 422}

13.2 Biventricular support

Biventricular support can be achieved through a combination of isolated LV and RV support devices or by using a VA-ECLS system with LV unloading via mAFP or IABP. In patients with biventricular failure, it is recommended to support both ventricles using one of the following approaches: (1) VA-ECLS combined with active LV unloading using an mAFP or IABP; (2) a combination of right- and left-sided mAFP devices; and (3) a paracorporeal RVAD combined with a tLVAD.

13.3 Cardiopulmonary support

Cardiopulmonary support may be achieved with VA-ECLS or a BVAD equipped with an integrated oxygenator (tLVAD+oxyRVAD). In patients requiring cardiopulmonary support, it is recommended that VA-ECLS or tLVAD+oxyRVAD be utilized. Whenever the femoral artery is used for arterial cannulation, it is recommended to ensure distal leg perfusion.⁴²³⁻⁴²⁸

13.4 LV unloading

Unlike dedicated LVADs, VA-ECLS increases afterload and does not actively unload the LV, which may exacerbate LV distension and pulmonary congestion. Although no universally accepted thresholds exist, accumulating evidence supports the use of LV unloading during VA-ECLS. Therefore, active LV unloading should be considered in patients receiving VA-ECLS.^{274,429-433} Experimental studies suggest benefits such as reduced myocardial wall stress and improved pulmonary decongestion. A recent randomized trial confirmed a survival benefit of LV unloading in patients with AMI.¹⁸

Inotropes, though commonly used, may be detrimental due to increased myocardial oxygen consumption and β -receptor downregulation. Alternative unloading techniques, such as the atrial septostomy procedure, are not always available.

A recent study showed that LV unloading during VA-ECLS improved in-hospital survival, although mAFP use was associated with higher rates of bleeding and AKI compared to IABP.⁴³⁴ Schrage et al. demonstrated reduced 30-day mortality and higher likelihood of ventilator weaning in VA-ECLS patients undergoing LV unloading,⁴³² with benefits being inversely related to the time elapsed before initiation of unloading.⁴³⁵

Radakovic et al. reported improved 30-day survival and myocardial recovery with prophylactic mAFP compared to reactive (“bailout”) unloading.⁴³⁰ A meta-analysis of 12 studies (5,063 patients) found that combining VA-ECLS with IABP lowered 30-day mortality and increased successful weaning.⁴³⁶ Another meta-analysis focusing on patients undergoing ECPR found that combined VA-ECLS and mAFP support was associated with improved survival compared to VA-ECLS alone.⁴³⁷ In certain clinical scenarios with unique haemodynamic requirements, a synergistic device strategy, such as mAFP with VA-ECLS or BVAD, may be warranted.²⁷⁴ Various LV unloading strategies have been summarized in a review article.¹⁵¹ Overall, early and effective LV unloading is associated with better survival and increased weaning success.¹⁵¹⁻¹⁵⁴ Thus, LV unloading should be promptly considered in all patients supported by VA-ECLS to further improve outcomes.^{274,438-440} In patients with profound CS, full-support mAFP appears superior to partial support and should be prioritized when feasible. Device selection and unloading strategy should be guided by haemodynamic assessments, preferably with a Swan-Ganz catheter.

Recommendation Table 10: Recommendations for the Institution of tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
Device selection			
It is recommended that device selection should be made in the context of the clinical situation and the team's experience.	I	C	-
It is recommended that evaluation for de-escalation of tMCS support occur on a daily basis.	I	C	-
Serial assessments of intracardiac pressures should be considered to guide tMCS management.	Ila	C	-
Isolated LV support			
In patients supported with tMCS, cannulation strategies to promote patient mobilization should be considered.	Ila	C	-

It is recommended to use echocardiographic or fluoroscopic guidance for the insertion of a peripheral cannula.	I	C	-
Biventricular support			
In patients with biventricular failure, it is recommended to support both ventricles either with VA-ECLS in combination with mAFP, or a combination of 2 mAFP, or 2 tVAD.	I	C	-
Cardiopulmonary support			
In patients requiring cardiac and pulmonary support, it is recommended to utilize VA-ECLS or an oxyBVAD.	I	C	-
In patients with femoral artery cannulation, it is recommended to utilize a distal leg perfusion catheter.	I	A	(423-428)
LV unloading			
In patients supported with VA-ECLS, timely LV unloading should be considered.	IIa	B	(12, 274, 430, 434, 435, 438, 439)

Abbreviations: BVAD, biventricular assist device; ECLS, extracorporeal life support; LV, left ventricle; mAFP, microaxial flow pump; oxy, oxygenator; tMCS, temporary mechanical circulatory support; tVAD, temporary ventricular assist device; VA, veno-arterial.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

14. MONITORING AND ANAESTHETIC MANAGEMENT IN PATIENTS WITH TMCS

All patients in shock should receive invasive arterial and central venous pressure (CVP) monitoring, as well as pulmonary artery catheter (PAC) monitoring, in addition to standard monitoring. Peripheral oxygen saturation monitoring is essential and should be performed at sites that reflect central arterial saturation (eg, right upper limb, nose, or right ear).⁴⁴¹⁻⁴⁵⁶

14.1 Monitoring

14.1.1 Ultrasound and echocardiography monitoring

Real-time imaging of the heart and vascular system is crucial for evaluating anatomy, valvular function, LV and RV contractility, wall motion abnormalities, and signs of vascular congestion. Multiple terms describe imaging techniques and the providers who perform them. POCUS refers to bedside ultrasonography used to assist clinical assessment and diagnosis.⁴⁵⁷ In CS, POCUS enables rapid evaluation of cardiac function, pericardial effusion, pulmonary oedema, pneumothorax, and venous congestion by assessing the IVC and hepatic and portal veins, commonly referred to as venous excess ultrasound scoring (VExUS).⁴⁵⁸ For a more comprehensive cardiac evaluation, formal transthoracic echocardiography (TTE) or transoesophageal echocardiography (TOE) should be performed. TOE is especially useful when TTE imaging is suboptimal, which is often the case in patients on tMCS. TOE can also aid in the positioning of guidewires, IABPs, cannulas, and transvalvular support devices such as mAFP or dual-lumen cannulas. Furthermore, TOE is valuable for assessing RV function using RV fractional area change and RV free wall longitudinal strain.^{447,459,460} In addition, TTE or TOE can also help evaluate LV distension, failure of aortic valve opening, and blood stasis, indicating the need for LV unloading.

14.1.2 Monitoring of extremity perfusion

Regional Oxygen Saturation Monitoring and Peripheral Temperature: Regional oxygen saturation monitoring using near-infrared spectroscopy (NIRS) is a valuable tool for assessing tissue perfusion and should be considered in all tMCS patients. Cerebral rSO₂ monitoring, particularly bilateral, is essential in patients at risk of differential oxygenation (eg, upper-body hypoxia) to help prevent cerebral hypoxia.⁴⁶¹ In addition, bilateral lower-limb NIRS is recommended during femoral VA-ECLS, especially for monitoring perfusion in the cannulated limb.⁴⁶² Signs of limb ischaemia include asymmetrical muscle oxygen saturation (StO₂), lower nadir StO₂ in the cannulated leg, reduced skin temperature, or low flow measured with bedside Doppler ultrasound.^{462,463}

Several small studies have shown the value of NIRS in detecting ischaemia, particularly when asymmetry in limb saturation or low absolute values (<40%) are present.^{458,462,464,465} Two studies identified interlimb oxygen saturation differences of >35% and >15% as indicative of ischaemia.^{462,466}

Antegrade distal perfusion via a separate cannula significantly reduces the risk of limb ischaemia, compartment syndrome, and amputation.^{425,428,467,468} The flow in the distal perfusion cannula may also be monitored using an additional sensor.

Ideally, in peripheral VA-ECLS, extremity perfusion should be repeatedly assessed using a multimodal approach: Clinical examination, continuous rSO₂, Doppler ultrasound, and temperature monitoring. The establishment of institutional standardized protocols for the detection and treatment of vascular complications is recommended.

Ankle–Brachial Index: The ABI is the ratio of systolic blood pressure in the ankles to that in the arms and is used to estimate leg perfusion.⁴⁶⁹ An ABI between 1.0 and 1.4 suggests normal perfusion; an ABI between 0.9 and 0.99 is considered borderline for PVD; an ABI < 0.9 indicates PVD. During VA-ECLS, ABI can be used to evaluate limb perfusion ipsilateral to the femoral artery cannula.⁴⁶⁹ However, ABI may underestimate ischaemia in patients with non-compressible arteries (eg, due to DM and/or calcification). Therefore, ABI should be used in combination with other tools, as indicated above.

Recommendation Table 11: Recommendations for Monitoring Patients on tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
In patients with tMCS, arterial and central venous line placement is recommended.	I	C	(441)
The use of PAC should be considered to guide ongoing haemodynamic management.	IIa	B	(445)
Serial echocardiographic evaluations should be considered in the assessment of biventricular function.	IIa	C	(447, 459, 460)
Near-infrared spectroscopy should be considered to monitor cerebral oxygenation.	IIa	C	(461)
In patients supported with peripheral VA-ECLS, ongoing clinical evaluation and monitoring of limb perfusion, regional near-infrared spectroscopy, Doppler ultrasound, and temperature monitoring are recommended.	I	B	(425, 427, 462, 467, 468)

Abbreviations: PAC, pulmonary artery catheter; tMCS, temporary mechanical circulatory support; VA-ECLS, veno-arterial extracorporeal life support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

14.1.3 Validity of monitoring parameters during various types of tMCS

The use of different tMCS systems may alter normal blood flow by changing its direction and/or characteristics. Because most monitoring techniques were developed for physiologic (native) circulation, it is essential to understand their validity in the context of tMCS. Table 6 provides an overview of the validity and limitations of commonly used monitoring methods. At this moment, we can only recommend using multiple monitoring modalities in patients on tMCS, taking into account their respective strengths and limitations.

Table 6. Validity and Limitation of Common Monitoring Parameters^a

	IABP	mAFP	VV-ECLS	VA-ECLS	Paracorporeal LVAD	Paracorporeal RVAD
ScvO ₂	Valid	Valid	Unreliable	Unreliable	Valid	Valid
ECLS ScvO ₂	-	-	Unreliable	Unreliable	Unreliable	Unreliable
PAC CO	Valid	Valid	Unreliable	Unreliable	Valid	Unreliable
CVP	Valid	Valid	Unreliable	Unreliable	Valid	Unreliable
PAP	Valid	Valid	Valid	Unreliable	Valid	Unreliable
PCWP	Valid	Valid	Valid	Unreliable	Unreliable	Valid
LVOT VTI	Valid	Unreliable	Valid	Unreliable	Unreliable	Valid
Uncalibrated AWA CO	Unreliable	Unreliable	Valid ⁴⁷⁰	Unreliable	Unreliable	Valid
TPCO	Unreliable	Unreliable	Unreliable ^{471,472}	Unreliable	Unreliable	Unreliable
SpO ₂	Valid	Valid	Valid	Valid	Valid	Valid
Cerebral rSO ₂	Valid	Valid	Valid	Valid	Valid	Valid

PPV-SVV	Unreliable	Unreliable	Unreliable	Unreliable	Unreliable	Unreliable
PLR	Valid	Valid	Valid	Valid	Valid	Valid
Peripheral rSO₂	Valid	Valid	Valid	Valid	Valid	Valid
POCUS	Valid	Valid	Valid	Valid	Valid	Valid

Abbreviations: AWA, arterial waveform analysis; CO, cardiac output; CVP, central venous pressure; IABP, intra-aortic balloon pump; LVAD, left ventricular assist device; LVOT, left ventricular outflow tract; mAFP, microaxial flow pump; PAC, pulmonary artery catheter; PAP, pulmonary artery pressure; PCWP, pulmonary capillary wedge pressure; PLR, platelet-lymphocyte ratio; POCUS, point-of-care ultrasound; PPV, pulse pressure variations; rSO₂, regional oxygen saturation; RVAD, right ventricular assist device; ScvO₂, central venous oxygen saturation; SpO₂, oxygen saturation; SVV, stroke volume variations; TPCO, transpulmonary thermodilution cardiac output; VA-ECLS, veno-arterial extracorporeal life support; VTI, velocity time integral; VV-ECLS, veno-venous extracorporeal life support.

^aAdapted from Krishnan et al.⁴⁷³

14.1.4. Monitoring laboratory parameters in relation to adequate tissue perfusion

Patients with CS are typically characterized by elevated lactate levels, impaired renal perfusion, and increased tissue oxygen consumption. Following the initiation of tMCS, timely monitoring of lactate clearance, urine output, and central/mixed venous oxygen saturation (SvO₂) are encouraged as key indicators of adequate tissue perfusion.

14.2 Anaesthetic management

14.2.1 Preoperative assessment

The initial preoperative assessment of a patient in CS should include a rapid but thorough evaluation of allergies, current medications, medical history (including baseline functional status), recent oral intake, and the clinical events leading to the current presentation.

14.2.2 Anaesthetic agents

The choice of anaesthetic medications, such as sedatives, hypnotics, analgesics, and muscle relaxants, should be individualized based on the clinical scenario. In awake patients with CS, minimal sedation may help avoid intubation.⁴⁷⁴⁻⁴⁷⁸ Notably, patients who remain awake on ECLS have demonstrated higher hospital discharge rates and fewer complications.⁴⁷⁵ Similarly, in cases of cardiac arrest with ROSC and preserved consciousness, light sedation and

local anaesthesia may be sufficient for tMCS insertion. However, in patients in extremis, endotracheal intubation is often necessary.

Total intravenous anaesthesia, using combinations of sedatives, analgesics, and neuromuscular blockers, is necessary for the intubation. Agents such as propofol, etomidate, sufentanil, fentanyl, and remifentanyl are frequently used for sedation, depending on patient-specific needs and institutional practice. In an observational study comparing propofol and midazolam in patients with cardiogenic shock, propofol was associated with reduced catecholamine requirements, less bleeding, and lower 30-day mortality.⁴⁷⁹ Volatile anaesthesia with isoflurane resulted in higher costs without improved outcomes.⁴⁸⁰

Sedation depth should ensure patient comfort while remaining light enough to enable early neurologic assessment and potential extubation.^{481,482} Moreover, some studies suggest that routine sedation interruptions in patients with VA-ECLS may facilitate earlier extubation and reduce ICU length of stay.⁴⁸³ Well-designed RCTs are needed to better define optimal sedation strategies in CS. A validated sedation scale is recommended to titrate sedation appropriately.^{481,484}

In all instances, pharmacologic therapies should be tailored to the patient's physiologic state, taking into account the distribution volume (VA-ECLS circuit, capillary leakage, vasodilation, and hepatic and renal function [metabolism and clearance]).^{485,486} The distribution volume of many anaesthetic agents is increased, particularly during ECLS, necessitating higher dosages, especially of propofol, benzodiazepines, dexmedetomidine, ketamine, and opioids.⁴⁸⁷⁻⁴⁹¹

Recommendation Table 12: Recommendations for Anaesthetic Management of Patients Requiring tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
It is recommended to adjust the level of sedation to the clinical situation and the patient's needs.	I	C	-
Implanting tMCS employing local anaesthetics while the patient is under light sedation should be considered.	Ila	B	(474-477)
The use of a validated sedation scale to facilitate titration of sedation agents is recommended.	I	C	(481, 484)

Abbreviation: tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

14.2.3 Inotropic agents and inotropic therapy

Vasoactive inotropic agents are frequently required in the initial management of CS. Commonly used agents include dobutamine, epinephrine, norepinephrine, levosimendan, and milrinone or enoximone.⁴⁹² Recent reports suggest that combining dobutamine or levosimendan with norepinephrine appears effective and offers a favourable safety profile.⁴⁹³

14.2.4 Mechanical ventilation in patients with CS

Patients in CS often present with acute respiratory failure characterized by increased work of breathing, hypoxaemia, and hypercapnia. The decision to proceed with intubation is multifactorial, depending on neurologic status, haemodynamic stability, degree of hypoxaemia—often assessed via the PaO₂/FiO₂ (P/F) ratio—presence of hypercapnia, airway protection, and RV function. Non-invasive ventilation, which has minimal haemodynamic impact, may be an appropriate first-line therapy in carefully selected patients with CS.⁴⁹⁴

Mechanical ventilation strategies should differ based on whether LV or RV failure predominates. In patients with predominantly LV failure, reduced CO, reflex systemic vasoconstriction, and elevated LV filling pressures may lead to pulmonary oedema and hypoxia. For these patients, ventilation with positive end-expiratory pressure (PEEP) of 5–10 cm H₂O—or higher, if tolerated, is recommended to improve gas exchange and reduce pulmonary congestion.

Conversely, in patients with predominantly RV failure, positive pressure ventilation increases RV afterload. Therefore, minimizing intrathoracic pressure and avoiding hypoxia and hypercapnia, which increase PVR, are key principles of ventilatory support.⁴⁹⁵

14.2.5 Combined use of inotropic agents and tMCS

Left ventricular support: In certain clinical scenarios, inotropic support must be combined with tMCS to maintain adequate systemic perfusion. When VA-ECLS is initiated, its retrograde aortic flow leads to increased LV afterload and may additionally increase aortic valve insufficiency, aggravating LV distension as measured by a PAC or displayed on the monitor of an mAFP.⁴⁹⁶ In such cases, insertion of or increasing the speed of the already implanted mAFP, can help unload the LV. Concurrent afterload reduction using vasodilators improves forward

flow in all afterload-sensitive tMCS devices. This strategy is supported by the DANGER-Shock trial, which demonstrated a survival benefit from early mAFP use in ST segment elevation myocardial infarction-related CS.¹⁸ For detailed algorithms on device selection, timing, and weaning, see [Sections 6 and 8](#).

Right ventricular support: In the setting of RV failure, pulmonary hypertension should be avoided, and inotropes, inodilators, and pulmonary vasodilators (though not routinely used in patients in shock) should be considered to support RV contractility and reduce RV afterload.⁴⁹⁷ Treatment of RV failure includes managing the underlying cause (eg, AMI with right coronary artery involvement, LV dysfunction, an acute PE), optimizing preload, and minimizing RV afterload by addressing hypoxaemia, acidosis, hypercapnia, and elevated intrathoracic pressures.

Inotropes without vasoconstrictive effects such as dobutamine and milrinone are preferable, because they support contractility and have little to no effect on the pulmonary vasculature. Additional agents to reduce PVR include inhaled nitric oxide or iloprost, intravenous prostacyclin analogues (eg, epoprostenol), and oral pulmonary vasodilators (eg, sildenafil, bosentan).⁴⁹⁸

tMCS may be required in patients with persistent RV failure despite optimal medical therapy. Percutaneous or surgically implanted RVADs can significantly reduce the need for inotropic support while maintaining adequate LV filling and tissue oxygen delivery.⁴⁹⁷ Importantly, RV failure secondary to LV failure mandates LV support⁴⁹⁹; addressing RV failure alone in this context may worsen LV volume overload and cause lung oedema. In such cases, biventricular support, using VA-ECLS with mAFP or a temporary BVAD, should be considered.

Biventricular support: Several strategies exist for biventricular support.^{500,501} A temporary extracorporeal BVAD is generally implanted via a median sternotomy.⁵⁰²⁻⁵⁰⁴ However, percutaneous implants of both right- and left-sided mAFP devices may also be considered.

Recommendation Table 13: Recommendations for Haemodynamic Management

Recommendations	Class ^a	Level ^b	Ref ^c
It is recommended to maintain an MAP $\geq$ 80 mmHg ($\geq$ 8.66 kPA).	I	B	(442)

In patients with RV failure due to post-capillary pulmonary hypertension secondary to LV failure, it is recommended to initiate primary LV support prior to specific support of the RV.	I	C	(499)
In patients on a tLVAD, it is recommended that RV failure be managed with optimization of the heart rate and rhythm, volume status, and RV afterload.	I	C	(497)

2072 Abbreviations: kPA, LV: left ventricle; MAP: mean arterial pressure; RV: right ventricle; tLVAD:
2073 temporary left ventricular device.
2074 ^aClass of recommendation.
2075 ^bLevel of evidence.
2076 ^cReferences.

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15. POSTOPERATIVE MANAGEMENT IN THE ICU

Patients in CS present with varying degrees of cardiopulmonary compromise and often require individualized medical and/or haemodynamic support to optimize tissue perfusion and oxygenation. As outlined in the previous chapter, effective therapy must be guided by extensive haemodynamic monitoring, including continuous parameters and daily imaging, such as echocardiography, to assess cardiac function.⁵⁰⁵ Additionally, the use of tMCS devices may introduce device-related complications, including infections (bacteraemia, sepsis), local bleeding, pump thrombosis, vascular complications, intracranial bleeding, cerebrovascular events, or global ischaemia, AKI, and haematologic complications, such as haemolysis and coagulation disorders. These patients also face logistical challenges related to appropriate care, such as the need for intrahospital transport for additional investigations and procedures.

15.1 The use of ultrasound

In addition to standard haemodynamic monitoring, echocardiography is indispensable in patients supported with tMCS. It enables assessment of device positioning and functionality, using Doppler techniques, biplane, or 3-dimensional imaging. Echocardiography is essential for diagnosing mechanical complications and identifying anatomical or valvular lesions (eg, aortic or mitral), evaluating ventricular size and function, detecting pericardial tamponade, and determining optimal preload management.⁵⁰⁵ After a VA-ECLS is implanted, it is crucial to confirm adequate LV unloading to reduce the risk of blood stasis and thrombus formation, especially in cases of suboptimal anticoagulation. Echocardiography also plays a key role in evaluating readiness for weaning from tMCS. In patients with transseptal cannulas or mAFP, daily imaging is vital to confirm optimal cannula position.

15.2 Haemodynamic management

Haemodynamic management should be guided by the CS stage (SCAI A–E),³² the predominant pathophysiology (left-, right-, or biventricular failure),⁵⁰⁶ and the underlying aetiology (eg, ischaemic, infectious, toxic, autoimmune, stress-induced, or acute-on-chronic HF).¹ Early use of a PAC should be considered to guide resuscitation, ongoing support, and weaning.⁴⁵⁵ Post-tMCS haemodynamic management focuses on optimizing tissue oxygen delivery, particularly ensuring adequate performance of the unsupported ventricle in univentricular support scenarios, to prevent or reverse organ dysfunction.

VA-ECLS support must be accompanied by monitoring to optimize LV decompression, including ensuring at least intermittent aortic valve openings.⁴⁵⁵ Inotropic agents may facilitate LV ejection, reduce RV afterload, either alone or in combination with mechanical unloading strategies.⁵⁰⁷

In patients with RV failure, treatable mechanisms should be addressed, such as optimizing preload; reducing RV afterload; avoiding hypoxia, hypercapnia, and elevated intrathoracic pressures; and using direct pulmonary vasodilators (inhaled or oral). Echocardiography can help assess LV unloading and its effect on septal motion. Across all cases, RV support requires close attention.^{506,508-510}

For CS due to a PE, VA-ECLS should be implemented promptly. In patients with primary pulmonary hypertension, tRVAD may increase the risk of pulmonary haemorrhage.⁵¹¹⁻⁵¹³

Additional strategies to reduce RV afterload include anticoagulation in the setting of PE, surgical embolectomy,⁵¹⁴ pulmonary vasodilators (inhaled nitric oxide, prostacyclin), phosphodiesterase inhibitors (milrinone), or levosimendan. Efforts to reduce intrathoracic pressure and to correct acidosis are also key.

Haemodynamic targets post-tMCS include maintaining MAP, SvO₂, lactate levels, and CO. MAP should be >80 mmHg with optimized flow rates depending on the type of tMCS.^{442,455}

Assessing arteriovenous oxygen difference (ideally, AV-O₂<5 cc) may better reflect adequate tissue oxygen delivery than utilizing SvO₂ alone. All medical or mechanical support strategies should be tailored to the underlying cause of CS. For example, patients with isolated LV support require preserved RV function. In cases of AMI-related shock, increasing afterload may be detrimental due to elevated myocardial oxygen demand; in such cases, milrinone or levosimendan may be preferable support. However, ongoing inotropic use may be necessary when only temporary univentricular support is in place as the unsupported ventricle remains at risk of volume overload and failure. In patients on VA-ECMO (without venting), assuring ejection through the aortic valve often requiring inotropes is essential to prevent LV stasis.

A central venous line is generally required for CVP monitoring and vasoactive drug administration. It also allows for SvO₂ measurement via blood sampling, which, together with haemoglobin concentration, helps assess AVO₂ difference i.e. the adequacy of tissue oxygen delivery.

Use of structured decision algorithms, such as the one proposed by Kim et al., may assist in post-tMCS management.⁵¹⁵

15.2.1 Special cases

RV failure secondary to LV failure: Patients with LV failure often develop elevated LVEDP, leading to venous pulmonary hypertension. As a consequence, secondary RV failure may develop. These patients are typically on inotropic support before the initiation of left-sided tMCS. It is important to recognize that supporting the RV in isolation, eg, by reducing PVR, can inadvertently increase venous return to the failing LV, exacerbating LV overload. Therefore, early initiation of left-sided tMCS is crucial in such cases to achieve LV unloading. Once the LV is mechanically supported, RV function can be optimized, increasing LV filling, effectively managed by the left-sided tMCS.

Atrial septal defects: Intracardiac shunting, particularly across a patent foramen ovale (PFO), may not be immediately evident in patients with elevated left-sided filling pressures due to LV failure. In such scenarios, when LV unloading is achieved (eg, with tLVAD) and RV pressures remain elevated, the shunt direction can reverse, causing a massive right-to-left flow through the PFO and leading to systemic hypoxaemia. TTE is essential to identify this condition. Prompt reduction of LV unloading should be considered to restore balance, followed by transcatheter PFO closure using interventional devices.

Valvular lesions: Severe aortic regurgitation or MR may occur after an mAFP is implanted due to mechanical interference with native valvular closure. These complications are typically identified via echocardiographic imaging. Persistent regurgitation can lead to progressive LV distension and can compromise myocardial recovery, thereby impeding successful weaning from tMCS. Prompt recognition and appropriate intervention are essential to optimize outcomes in these patients.

Differential oxygenation: tMCS effectiveness in the setting of LV ejection is highly dependent on pulmonary function for adequate oxygenation. When pulmonary function is compromised, systemic arterial oxygen saturations may be critically low. Differentiating between primary lung pathology and cardiogenic pulmonary oedema is essential and can be supported by PAC data, particularly wedge pressure measurements.⁵¹⁶

If pulmonary compromise is secondary to elevated LV pressures, LV unloading with an mAFP may sufficiently improve pulmonary oxygen exchange and restore systemic oxygenation. However, in some cases, mAFP is ineffective, because it may augment aortic output but with inadequately oxygenated blood. In scenarios of primary or secondary pulmonary failure,

right-sided venous oxygenation support is required. It can be achieved via central VA-ECLS, the addition of VV-ECLS to an existing tMCS set-up, or through V-AV ECLS configurations.^{516,517} In patients on femoral VA-ECLS with preserved LV ejection, 2 circulatory flows coexist: native CO and circuit-driven retrograde flow. In the context of pulmonary failure, the LV may eject poorly oxygenated blood into the aortic root, resulting in a phenomenon termed differential oxygenation, characterized by desaturation in the upper body compared to the lower. Continuous monitoring of peripheral and cerebral oxygenation is mandatory in all patients on ECLS and is especially important in peripheral VA-ECLS to detect regional perfusion disparities. Arterial oxygen saturation should be measured from the innominate artery branches. This step includes bilateral forehead NIRS to detect asymmetric cerebral perfusion.⁵¹⁸⁻⁵²⁰ Importantly, even with preserved saturation in the innominate distribution, coronary arteries may still receive desaturated blood, which predisposes the patient to silent coronary ischaemia, an entity that is difficult to detect clinically. In all cases where differential oxygenation is suspected or confirmed during peripheral VA-ECLS support, 1 or more of the following strategies should be promptly implemented⁵²¹:

1. Reconfigure to central aortic cannulation or right axillary cannulation
2. Insert an additional ECLS return cannula into the RA (eg, via the right internal jugular vein), thereby converting the circuit into VA-V (veno-arterial-venous)-ECLS and improving oxygenation of LV inflow
3. Switch to a combination of tLVAD and oxyRVAD
4. In case of adequate RV function, switch to VV-ECMO+tLVAD
5. Vent the left side of the heart into the VA-ECLS circuit to prevent ejection of desaturated blood (see [Section 6.4](#)).

All options above except number 1 would also ensure coronary perfusion with oxygenated blood.

Recommendation Table 14: Recommendations for Patients on tMCS with Inadequate Pulmonary Function

Recommendations	Class ^a	Level ^b	Ref ^c
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In patients with differential oxygenation, ^d the insertion of an additional cannula (VA-V-ECLS) to return oxygenated blood to the right atrium is recommended.	I	C	-
In patients with differential oxygenation, right axillary or aortic cannulation should be considered.	IIa	C	-
In patients with differential oxygenation, ^d venting the left side of the heart into the VA-ECLS circuit should be considered.	IIa	C	-

Abbreviations: tMCS, temporary mechanical circulatory support; VA-ECLS, veno-arterial extracorporeal life support; VA-V-ECLS, veno-arterial-venous extracorporeal life support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

^dHarlequin syndrome.

15.3 Ventilatory management

The most appropriate ventilation strategy for patients on tMCS depends on the mode of circulatory support.

15.3.1 Temporary LVAD and/or RVAD

PEEP is almost universally applied in mechanically ventilated patients due to its benefits in gas exchange, recruitment of alveolar units, counterbalancing of hydrostatic forces leading to pulmonary oedema, and maintenance of airway patency.⁵²² Authors of a small study of 18 patients with CS requiring IABP support reported that those randomized to receive elective mechanical ventilation with 10 cm H₂O of PEEP were more likely to be weaned off the IABP and to survive to discharge than those who received oxygen supplementation alone.⁵²³ However, in LVAD-supported patients with borderline RV function, PEEP may increase RV afterload, potentially exacerbating RV failure.⁵²⁴ In cases of pulmonary oedema, if the LA is adequately decompressed with a tLVAD, the oedema would typically resolve rapidly, allowing for timely weaning from invasive mechanical ventilation.

When primary ventricular failure involves the right side, a membrane oxygenator spliced into a tRVAD circuit or VV-ECLS can provide gas exchange and reduce ventilatory demands,

thereby facilitating extubation.⁵²⁵ Some centres even advocate using oxyRVAD in the setting of LVAD implant for acute RV failure causing CS, which may offer a higher 30-day survival compared with RVAD support alone.⁵²⁶

15.3.2 Veno-arterio-venous extracorporeal life support

V-AV-ECLS is appealing for respiratory support, because it can maintain adequate gas exchange while minimizing the injurious effects of positive pressure ventilation. If extubation is not feasible, ventilator settings can be adjusted to limit ventilator-induced lung injury.

In acute respiratory distress syndrome (ARDS), a low-volume–low-pressure ventilation strategy has been shown to improve outcomes: Mechanical ventilation using a tidal volume of 6.2 ml/kg predicted body weight and a mean plateau pressure of 25 cm H₂O was associated with reduced mortality and fewer days requiring invasive mechanical ventilation, compared to a higher tidal volume of 11.8 ml/kg and a mean plateau pressure of 33 cm H₂O.⁵²⁷ However, the optimal ventilation strategy for patients with severe ARDS undergoing ECLS remains uncertain. Some experts advocate for higher PEEP (>10 cm H₂O) to maintain alveolar recruitment and prevent atelectasis,⁵²⁸ whereas others prefer no external PEEP, achieved by extubating the patient entirely.^{529,530} Although selected patients may tolerate extubation, others may experience profound tachypnoea, which can itself be injurious.

Ventilator settings in patients on V-AV ECLS should be guided by the strategies used in the CESAR and EOLIA trials to minimize lung injury.⁵³¹⁻⁵³³

Recommendation Table 15: Recommendations for Protected Ventilation

Recommendations	Class ^a	Level ^b	Ref ^c
In patients with respiratory insufficiency on VV- or V-AV-ECLS, goal ventilator settings should be considered: • Respiratory frequency of 6 to 10 breaths/min • FiO₂ adjusted to an appropriate saturation level • Tidal volume ≤ 6 ml/kg of predicted body weight • Plateau pressure ≤ 25 cm H₂O • Inspiratory pressure ≤ 15 cm H₂O • PEEP between 10 and 15 cm H₂O • Inspiratory to expiratory ratio of 1:1.	IIa	C	(533)

Abbreviations: ECLS, extracorporeal life support; FiO₂, fraction of inspired oxygen; PEEP, positive end-expiratory pressure; VA-V-, venoarterial/venous; VV-, veno-venous.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

15.4 Renal management

The incidence of tMCS-associated AKI has been best described in the ECLS populations and appears to be in the range of 60% to 90%, with more than half of the patients requiring RRT.⁵³⁴

Associated early mortality ranges between 65% and 90% among those requiring dialysis.^{534,535}

In a meta-analysis of more than 4,150 patients, 50% of a heterogeneous group of patients with CS supported by VA-ECLS developed AKI requiring dialysis.⁵³⁶ Risk factors for AKI included APACHE II and SOFA scores at the time of ECLS initiation, as well as age, DM, reduced LVEF, elevated pre-ECLS lactate, and the duration of circulatory support. In post-cardiotomy CS specifically, LCOS with LV failure and RV failure with elevated CVP were major causes of severe AKI. Additional risk factors included PRBC transfusions, hyperbilirubinaemia, and elevated pre-ECLS creatinine levels. Initiation of ECLS postoperatively (as opposed to intraoperatively) increased the risk of AKI requiring dialysis.

Based on the results of several studies, earlier initiation of tMCS likely reduces the severity of AKI and the need for RRT.⁵³⁷ However, once tMCS is initiated, continued use of high-dose vasoactive inotropes, unfavourable device configurations and settings (eg, high RPMs or high negative drainage pressures that damage red blood cells), and prolonged support may all contribute to AKI.^{538,539}

15.4.1 Timing of RRT

Although early initiation of dialysis or haemofiltration may relieve fluid overload more rapidly than diuretics in general ICU populations, RCTs have not demonstrated a survival advantage with early RRT. In fact, early initiation has been associated with a higher risk of dialysis dependence at 90 days, along with an increased risk of adverse events, particularly hypotension. In this patient population, RRT should be initiated based on standard absolute and relative indications, with no justification for pre-emptive initiation.⁵⁴⁰⁻⁵⁴³

15.4.2 Modality

Given the haemodynamic instability of many patients on tMCS, continuous VV ultrafiltration and/or dialysis is generally preferred, because the rapid fluid shifts required by intermittent haemodialysis or prolonged intermittent RRT (PIRT) are often poorly tolerated. In the ECLS setting, RRT can be delivered by integrating the dialysis circuit into the ECLS circuit (via various configurations) or via a separate percutaneously placed dialysis catheter. However, using the ECLS circuit for dialysis increases the risks of stroke and infection, and high ECMO flows and pressures may preclude this approach.

There are no comparative studies, but the infectious and procedural risks associated with separate percutaneous access should be considered when determining the most appropriate RRT strategy.⁵⁴⁴

15.4.3 Drug dosing

Temporary mechanical circulatory support with extracorporeal pumps and tubing increases the volume of distribution of most drugs and decreases drug clearance. Additionally, components of the circuit can adsorb medications, particularly lipophilic compounds, which are known to be sequestered by ECLS circuits. RRT adds further complexity to drug dosing beyond the ECLS circuit alone. Therefore, individualized therapy guided by drug level monitoring should be standard practice in this setting.⁵⁴⁵

15.4.4 Prognosis

Although long-term prognosis for patients with ECLS-associated AKI is not fully understood, study results indicate that patients who survive to hospital discharge often experience early recovery from dialysis dependence.^{546,547} Nevertheless, in adults, renal injury during ECLS is associated with a high likelihood of major adverse kidney events at 1 year, including death, significantly reduced glomerular filtration rate (GFR), and ongoing need for RRT.⁵⁴⁸

Recommendation Table 16: Recommendations for RRT for Patients on tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
RRT is recommended when conventional indications develop.	I	A	(541, 543)
The utilization of the ECLS circuit for RRT may be considered if no other vascular access is available.	IIb	C	(516)

Abbreviations: ECLS, extracorporeal life support; RRT, renal replacement therapy; tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

15.5 Anticoagulation, bleeding, and thrombosis

15.5.1 Anticoagulation agents

Heparin binds to and increases the activity of antithrombin, which inhibits both factor Xa and thrombin. Most tMCS programs use unfractionated heparin (UFH) as the first-line anticoagulant, owing to its availability and relatively low cost.^{549,550} However, UFH pharmacokinetics are highly variable due to nonspecific binding to circulating and tissue proteins as well as varying antithrombin levels.

Three tests are commonly used to monitor heparin therapy: Activated clotting time (ACT), activated partial thromboplastin time (aPTT), and anti-factor Xa (anti-Xa). ACT, initially designed for point-of-care use, measures clot formation in whole blood and is influenced by factors such as haemodilution, hypothermia, platelet count, and concomitant antiplatelet therapy.^{551,552} aPTT measures clot formation in platelet-poor plasma. Anti-Xa, by contrast, directly quantifies heparin's inhibition of factor Xa.^{553,554}

Several studies have suggested a poor correlation between ACT and anti-Xa levels, with anti-Xa providing a more accurate reflection of heparin's anticoagulant effect.⁵⁵¹ In a retrospective cohort of 103 tMCS-supported patients (VA-ECLS with or without mAFP), Feih et al. reported significantly lower bleeding rates when anticoagulation was guided by anti-Xa rather than by ACT.⁵⁵⁵ Similarly, a meta-analysis of 26 studies involving 2,086 patients found that anti-Xa-guided anticoagulation was associated with reduced bleeding and mortality.⁵⁵⁶ However, a single-centre study found no significant outcome differences between aPTT/anti-Xa- and ACT-guided protocols.⁵⁵⁷ Other studies have similarly shown no significant differences in bleeding or thrombosis rates between ACT- and aPTT-guided anticoagulation in ECLS-supported patients.⁵⁵⁸

Two direct thrombin inhibitors (DTIs), bivalirudin and argatroban, have been primarily studied in tMCS.⁵⁵⁹ These agents inhibit thrombin directly and independently of antithrombin, making them preferable in patients with HIT. Some centres favour DTI use in tMCS due to their more

predictable pharmacodynamics and short half-lives. Bivalirudin binds both circulating and clot-bound thrombin and is cleared via renal excretion and enzymatic degradation. Argatroban is a univalent DTI that binds reversibly to thrombin's active site and is primarily metabolized in the liver.⁵⁶⁰

15.5.2 Bivalirudin versus UFH in ECLS

A retrospective study of 52 adult patients on ECLS, 33 on UFH, and 19 on bivalirudin, found no significant differences in thrombosis, bleeding, or mortality, although bivalirudin more consistently achieved therapeutic targets.⁵⁶¹ In a subsequent study by Giuliano et al. involving 144 patients on VA-ECLS (13 of whom were converted to bivalirudin), those treated with bivalirudin required fewer PRBC transfusions.⁵⁶² Several meta-analyses have shown reduced thromboembolic events with bivalirudin compared to UFH,⁵⁶³⁻⁵⁶⁶ although differences in patient populations and ECLS configurations warrant cautious interpretation.

These findings suggest that both UFH and DTIs are viable anticoagulation options during tMCS, each with unique advantages and limitations. DTIs like bivalirudin offer predictable anticoagulant effects and favourable pharmacokinetics, but their use may be limited in low- or no-flow states (eg, during cannulation), where thrombus formation can occur despite systemic anticoagulation. Additionally, institutional familiarity with DTI use remains limited in many centres. Therefore, anticoagulant selection should be guided by patient-specific considerations, centre expertise, and continual reassessment of bleeding and thrombotic risks, pending further high-quality comparative data.

15.5.3 ECLS with low-dose or no anticoagulation

Anticoagulation increases bleeding risk, and advances in ECLS circuit coatings have initiated evaluation of low- or no-anticoagulation strategies.⁵⁶⁷ In a cohort of 32 patients undergoing ECPR with ACT <160 seconds, no increase in thromboembolic events was observed.⁵⁶⁸ Conversely, an aPTT >60 seconds was linked to increased bleeding risk.⁵⁶⁹ In a retrospective study of 121 VA- or VV-ECLS patients, survival without bleeding complications was higher when anti-Xa levels were maintained below 0.46 IU/mL.⁵⁷⁰ A recent analysis of 41 patients on ECLS (mostly on VA-ECLS) found lower bleeding rates with a "restrictive Xa" target (0.2–0.5 IU/mL) compared to a "traditional" target (0.3–0.7 IU/mL).⁵⁷¹ A meta-analysis of 17 studies with 3,249 patients on VA- or VV-ECLS found no consistent aPTT threshold associated with bleeding risk.⁵⁷²

15.5.4 Microaxial flow pump

Anticoagulation strategy with an mAFP must balance thrombotic risk, systemic haemostasis, and device-specific considerations. Modern mAFP systems utilize purge solutions that prevent blood entry into the motor and maintain an alkaline environment in the bearing area, thereby reducing protein denaturation and thrombosis.^{573,574}

15.5.5 Direct thrombin inhibitor-based purge solution

UFH has been successfully substituted with bivalirudin⁵⁷⁵⁻⁵⁷⁷ or argatroban^{578,579} in purge solutions, including in patients with HIT. Additionally, bicarbonate-based purge solutions have proven effective in reducing device-associated thrombus formation^{580,581} and are approved for use with an mAFP. If a purge solution lacks systemic anticoagulants, systemic anticoagulation must be provided independently.^{582,583}

Recommendation Table 17: Recommendations for Anticoagulation Management with tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
It is recommended that institutions establish protocols for anticoagulation management.	I	C	-
It is recommended to maintain anticoagulation after tMCS initiation using either UFH or DTI, depending on patient-specific factors and institutional protocols.	I	B	(563-566)
In patients with bleeding complications, discontinuation or reduction of anticoagulation should be considered based on bleeding severity, location and controllability, tMCS flow, and patient-specific thrombotic risk factors.	IIa	B	(568-570, 572)

Abbreviations: DTI, direct thrombin inhibitor; tMCS, temporary mechanical circulatory support, UFH: unfractionated heparin:

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

15.5.6 Bleeding

tMCS inherently introduces blood–foreign surface interactions, altered flow dynamics, shear stress, and endothelial disruption. These factors contribute to dysregulation of the inflammatory and coagulation cascades.^{584–586} Foreign surfaces and disturbed flow regions act as a nidus for thrombus formation, necessitating systemic anticoagulation. Given the impaired innate haemostatic balance in these patients, meticulous anticoagulation management is required to prevent both thrombosis and bleeding.

Definitive guidelines on bleeding prevention and management in tMCS remain limited due to heterogeneity in study designs and outcomes across the published literature.⁵⁴⁹ Consequently, institutional variability exists in anticoagulation regimens and monitoring strategies.⁵⁵⁰ It is essential for individual centres to implement standardized protocols tailored to each tMCS modality in regular use. Systemic anticoagulation, most often with UFH or DTI, remains the cornerstone of thromboprophylaxis in this context.

15.5.7 Thrombosis

Thrombosis remains a critical complication of tMCS, potentially leading to haemolysis, mechanical failure, or thromboembolic events. Among patients supported with VA-ECLS, thrombotic complications occur in 8% to 15% of cases.^{549,587} Risk factors include elevated fibrinogen-to-albumin ratio, use of a surgical LV vent, increased haemoglobin concentration, and central cannulation configurations.^{569,588}

Intracardiac thrombosis is a particular concern in patients with diminished myocardial contractility, which predisposes to intracardiac stasis. Echocardiography should be performed before an mAFP is implanted, because mobile or floating intracardiac thrombi are contraindications for its use. Although the incidence is not well defined, intraventricular thrombus formation may occur even after mAFP placement.⁵⁸⁹ Active LV unloading should be pursued to mitigate stasis in the LV and LA, with serial echocardiographic reassessments recommended (see [Section 15.5.1, Monitoring](#)).

Haematologic markers for thrombus formation, including plasma-free haemoglobin and D-dimer levels, should be monitored daily in patients on tMCS. Clinicians should vigilantly observe for signs of haemolysis, such as dark or reddish urine, particularly by positioning the urine collection bag in the visual field.^{590,591} The urine bag should be placed on the side of the bed facing the door of the room to keep the changes constant in the vision field. Serum D-dimer level has demonstrated utility as a marker for thrombus formation.⁵⁹² Continuous or

near-continuous free haemoglobin monitoring, especially useful in anuric patients, may help detect early thrombus formation.

In patients supported with ECLS, thrombus formation frequently occurs at the oxygenator, circuit connections, and reinfusion cannulas.^{593,594} The ECLS circuit should be visually inspected at least twice daily, with special attention to these high-risk sites. If thrombus is identified in external components (eg, oxygenator or pump), component exchange is generally indicated. For mAFP, routine assessment of laboratory markers and purge pressure trends is essential for early detection of device thrombosis. Researchers conducting small studies have reported that adding tissue plasminogen activator (tPA) at low concentrations (0.04–0.08 mg/mL) to the purge solution may facilitate resolution of thrombus within the mAFP system.⁵⁹⁵⁻⁵⁹⁷

15.6 Infections

Nosocomial infections remain a major source of complications in patients receiving any type of mechanical circulatory support (MCS), contributing significantly to morbidity and mortality. A recent meta-analysis reported a 32% increase in the relative risk of death among adult patients on ECLS who developed nosocomial infections.⁵⁹⁸

Infection prevention strategies initiated prior to MCS should be maintained post-implantation. The development of institutional antibiotic protocols is recommended.

15.6.1 General infection preventative measures

The ELSO Infectious Disease Task Force has issued recommendations aimed at preventing nosocomial infections. Standard measures include regular hand hygiene by care givers and the use of chlorhexidine wipes for skin antisepsis. Line interruptions and the use of needleless hubs should be minimized. Unnecessary vascular access and long-term access lines should be avoided.⁵⁹⁹

The ISHLT has published specific guidance on the care of invasive lines. Key recommendations include avoiding femoral access, using tunnelled catheters for long-term therapy, and avoiding line exchanges over a guidewire. Lines should be assessed daily and promptly removed if signs of infection are present.⁶⁰⁰

15.6.2 Antibiotic prophylaxis

The use of antibiotic prophylaxis in tMCS remains controversial due to concerns over the emergence of multidrug-resistant organisms and the risk of *Clostridium difficile*-associated

colitis. Current evidence supporting the routine use of prophylactic antibiotics during ECLS is limited.^{601,602} For surgically implanted tMCS devices, antibiotic prophylaxis should target *Staphylococcus* species. In patients with unknown Methicillin-resistant *Staphylococcus aureus* (MRSA) status, empirical coverage against MRSA is advisable. Some centres continue antibiotic prophylaxis until chest closure in patients with an open sternum; however, no clinical data support this routine practice.⁶⁰³ Routine inclusion of broad-spectrum gram-negative or antifungal prophylaxis is not recommended.⁶⁰³⁻⁶⁰⁵

15.6.3 Percutaneous cannula and driveline management

Following operations, percutaneous cannulas and drivelines should be promptly secured using anchoring devices to prevent movement and minimize trauma at the exit site⁶⁰⁶⁻⁶¹⁰ and reduce the risk of dislodgement and local infection. The exit site should be protected with a sterile dressing, applied under aseptic technique by trained staff in accordance with institutional policy. Initially, dressing changes should be performed daily; the frequency may be reduced once the site has healed adequately.

Recommendation Table 18: Recommendations for Preventing Infection

Recommendations	Class ^a	Level ^b	Ref ^c
The development of institutional antibiotic protocols is recommended.	I	C	-
It is recommended that percutaneous cannulas and drivelines be secured with an anchoring device(s) to minimize movement and exit site trauma.	I	C	(606-610)

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

15.7 Vascular complications including hypo- or hyper-perfusion

Vascular complications occur in 9% to 44% of patients supported with ECLS and are associated with increased mortality.³⁶⁷⁻³⁶⁹ Given the potentially devastating consequences of acute limb

ischaemia in patients receiving tMCS, a limb protection strategy is essential whenever peripheral arterial cannulation is used.

Peripheral VA-ECLS frequently requires femoral artery cannulation to provide retrograde systemic perfusion. When employing percutaneous techniques, ultrasound guidance is recommended.³⁶⁸ However, studies comparing complication rates between percutaneous and surgical cutdown approaches have reported inconsistent findings.⁶¹¹⁻⁶¹³

Cannula size has been implicated in the risk of vascular complications during femoral cannulation. Smaller cannulas may reduce this risk but may also limit achievable flow rates.⁶¹⁴

The authors of a meta-analysis of 22 studies reported that cannulas <17 Fr were associated with fewer arterial complications,⁴²³ although a subsequent study found no significant association between cannula size and complication rates.³⁷²

Patients undergoing femoral VA-ECLS cannulation are at risk of distal limb ischaemia due to obstruction of ipsilateral arterial flow. This cannula may be positioned antegrade distal to the arterial cannula or retrograde via the posterior tibial artery.⁶¹⁵

Multiple single-centre studies have shown that distal perfusion cannulation reduces vascular complications.⁴²⁴⁻⁴²⁷ Although some studies did not find a significant protective effect,^{369,372} meta-analyses demonstrated a decreased incidence of vascular complications when a prophylactic distal perfusion cannula was used.^{423,428} Given the preponderance of supporting evidence, prophylactic placement of a distal perfusion cannula is recommended in all cases. Novel bi-directional cannulas may eventually eliminate the need for a separate distal perfusion cannula, but further validation is needed.⁶¹⁶

After peripheral cannulation, the cannulated limb should be closely monitored for signs of impaired perfusion (see [Section 14.1.2 Monitoring of extremity perfusion](#)). Clinical examination should be complemented by continuous monitoring using NIRS.

Importantly, any signs of limb ischaemia should be treated as a vascular emergency. An interdisciplinary ECLS team, including vascular surgery specialists, should be involved promptly. ECLS is not a contraindication to vascular intervention; a full range of endovascular and surgical options can and should be considered.^{617,618}

Recommendation Table 19: Recommendations for the Prevention of Leg Ischaemia

Recommendations	Class ^a	Level ^b	Ref ^c
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A distal perfusion cannula is recommended for patients with femoral arterial cannulation.	I	B	(423, 425-428)
Monitoring of distal perfusion is recommended in patients with peripheral cannulation.	I	B	(462, 464, 465, 518)
In patients with peripheral cannulation, differences in oxygen saturation between the extremities > 15% may be considered an indication for prompt evaluation of extremity perfusion.	IIb	C	(462, 466)

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

15.8 Neurologic complications

Acute brain injury is among the most devastating complications during tMCS⁶¹⁹⁻⁶²¹ and may manifest as ischaemic stroke, intracranial haemorrhage, hypoxic-ischaemic brain injury, or seizures.⁶²² The reported incidence of stroke ranges from 5% to 10% in this population⁶²³⁻⁶²⁶ and may double in patients who experienced cardiac arrest prior to tMCS initiation.⁶²⁷⁻⁶²⁹ Across virtually all studies, any form of acute brain injury is associated with a 2-fold increase in mortality.

15.8.1 Management

The management of modifiable risk factors is essential to reducing neurologic complications. Infections, arrhythmias, hypertension, and poorly controlled anticoagulation can all predispose patients to thromboembolism.^{630,631} ECLS-specific factors include management of arterial blood gases: During the first 24 hours after cannulation, hyperoxia ($\text{PaO}_2 > 200$ mmHg) and rapid decreases in PaCO_2 should be avoided, as both have been linked to increased risk of brain injury.^{632,633}

In cases of ischaemic stroke, thrombolytic therapy is generally contraindicated due to systemic anticoagulation. However, mechanical thrombectomy remains an option and should be pursued promptly if ischaemic stroke is suspected; CT angiography should be performed without delay.⁶³⁴⁻⁶³⁷ The interruption of early sedation facilitates timely neurologic

assessment. In the setting of ischaemic stroke, the AHA recommends permissive hypertension, but this may be counterproductive in patients with LV distension on tMCS. Antiplatelet therapy is advised due to the high risk of recurrent embolism. Although data are limited, restarting anticoagulation within 1 week may be appropriate for small infarcts.⁶³⁸ Intracranial haemorrhage, most often intraparenchymal, is a catastrophic event.^{584,639-642} The AHA guidelines recommend rapid lowering of systolic blood pressure to <140 mmHg. In patients without pulsatile flow, targeting an MAP <85 mmHg is considered reasonable.⁶³⁰ Although surgical evacuation of haematomas may be considered, it carries a high risk of rebleeding and death in anticoagulated, critically ill patients.^{643,644} Hypoxic-ischaemic brain injury frequently arises due to cardiac arrest or prolonged hypoperfusion in refractory CS.^{621,645} Additionally, peripheral VA-ECLS may lead to unrecognized differential oxygenation, contributing to cerebral hypoxia.⁶⁴⁶ Seizure incidence in tMCS is likely underestimated. Although registry data suggest a rate of 1% to 6%,^{631,647} standardized monitoring has revealed rates as high as 10%, highlighting the prevalence of subclinical seizures. Antiepileptic treatment is complicated by altered pharmacokinetics in ECLS circuits. Midazolam, propofol, phenobarbital, and carbamazepine are sequestered within the circuit and require frequent monitoring of drug levels. Levetiracetam, with low protein binding and limited volume of distribution, may be preferred.⁶⁴⁸ In cases of suspected brain death, traditional apnoea testing may be unreliable, particularly in patients with peripheral ECLS with preserved LV ejection. Alternative approaches include reducing sweep gas or adding CO₂ to the circuit. For patients without ejection, results may be more predictable. Ancillary assessments (electroencephalography [EEG], transcranial Doppler ultrasound, cerebral angiography) may be needed to confirm brain death.⁶⁴⁹

15.8.2 Neuromuscular complications

Peripheral neuropathy has been reported in patients on ECLS, though data are limited for other tMCS modalities. In lung transplant candidates bridged with ECLS, symmetrical or asymmetrical leg weakness occurred in 14% and 8% of patients, respectively, and was associated with DM, myositis, and RRT.⁶⁵⁰ Although ischaemic complications from arterial lines used for haemodynamic monitoring are rare according to ELSO registry data, some single-centre studies report rates as high as 7%.⁶⁵¹ No specific prophylactic strategies have been studied; however, careful monitoring of perfusion distal to all arterial lines (radial or

femoral) is advised. Bilateral NIRS should be employed for both absolute values and inter-limb comparison.

15.8.3 Neuromonitoring of the patient on tMCS

Despite the high incidence of neurologic complications, systematic neurologic monitoring is rarely implemented in tMCS programs. Neurologic examination reliability is limited by frequent use of sedatives, anaesthetics, and neuromuscular blockers. However, when utilized, standardized multimodal neuromonitoring protocols have been associated with earlier detection of brain injury and more timely interventions.⁶⁵²

Early reduction of sedation and extubation, when feasible, or daily sedation interruption ("sedation vacations"), is recommended. In unresponsive patients, routine multimodal non-invasive neurologic monitoring should be considered.⁶⁵³ Reflex pupillometry, Glasgow Coma Scale, EEG, and transcranial Doppler ultrasound can all provide valuable insight into cerebral status. NIRS should be used in all patients to monitor bilateral cerebral oxygen saturation. A significant drop from baseline or interhemispheric asymmetry may indicate evolving acute brain injury.⁶⁵⁴⁻⁶⁵⁷

Recommendation Table 20: Recommendations for the Prevention of Brain Injury in Patients on tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
At the initiation of VA-ECLS, it should be considered to avoid hyperoxia (eg, pO ₂ > 200 mmHg [> 26.66 kPa]) or rapid decreases in pCO ₂ .	IIa	B	(632, 633)
If a stroke is suspected, emergency computed tomography angiography should be considered.	IIa	A	(634-637)
In comatose patients unable to undergo a sensitive neurologic examination, an EEG should be considered to rule out a seizure.	IIa	B	(647, 654)
Protocolized neurologic monitoring may be considered to mitigate acute brain injury.	IIb	B	(654)

Abbreviations: EEG, electroencephalogram; pCO₂, partial pressure of carbon dioxide; pO₂, partial pressure of oxygen; tMCS, temporary mechanical circulatory support; VA-ECLS, veno-arterial extracorporeal life support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

15.9 Nutrition

Despite growing awareness of the adverse effects of malnutrition, nutritional support in critically ill cardiac surgery patients remains underemphasized. To date, only broad and non-specific guidelines exist for nutritional support in this population.⁶⁵⁸⁻⁶⁶⁰ During tMCS, altered haemodynamics, systemic inflammation, multiorgan dysfunction, deep sedation, neuromuscular blockade, and the use of vasoactive-inotropic agents all compound the inaccuracy of predictive nutritional equations. Furthermore, recent prospective international^{661,662} and single-centre studies^{663,664} have consistently demonstrated that critically ill cardiac and ICU patients receive less than 50% of their estimated caloric and protein requirements.

Predictive equations used to estimate energy expenditure often result in underfeeding or overfeeding.⁶⁶⁵ Although European and American societies recommend indirect calorimetry to guide nutritional support in patients in the ICU, this method can be challenging in patients on ECLS or who are non-ventilated.⁶⁶⁶ If indirect calorimetry is not available, a simple weight-based formula is recommended for estimating caloric needs.⁶⁶⁷

Several factors contribute to inadequate nutrition delivery in this population, including (1) delayed initiation of nutrition support; (2) reluctance to feed due to concerns over bowel ischaemia in patients on high-dose vasopressors; (3) frequent interruption of enteral nutrition (EN) for procedures and daily care; and (4) low utilization of supplemental parenteral nutrition (PN).

With tMCS now often required for weeks or longer, meeting the nutritional needs of these patients has become increasingly important. In patients on ECLS, achieving caloric intake within 10% of the goal by the second week of ICU care has been associated with improved 90-day survival.^{664,668}

15.9.1 Timing of nutrition initiation

EN should be initiated within 24–48 hours of ICU admission, depending on vasopressor dosage.^{664,668-670} Delays in EN initiation are associated with increased risks of infection, prolonged ICU length of stay and higher mortality.^{662,670} Although gastrointestinal intolerance is common, true intestinal ischaemia remains rare.^{671,672}

15.9.2 Feeding strategy considerations

Initiation and advancement of EN should be guided by the degree of vasoactive-inotropic support.⁶⁷³ In patients on high-dose vasopressors, delaying full EN may be appropriate. Use of the VIS is recommended to guide these decisions.^{669,674} Post-pyloric tube placement may help maintain EN delivery in patients with delayed gastric emptying. However, gastric feeding is generally better tolerated and preferred when possible.^{669,674,675}

15.9.3 Parenteral nutrition and glycaemic control

Enteral nutrition remains the preferred route of feeding. If PN is necessary, it should be administered through a dedicated access port. Although supplemental PN can help meet nutritional targets, it is not recommended during the first week of ICU care. Afterwards, it should be considered in patients who fail to meet nutritional goals with EN alone.⁶⁷⁶⁻⁶⁸⁰ Patients should be managed with relatively tight glycaemic control (<180 mg/dL).⁶⁷⁷ Due to unreliable subcutaneous absorption, insulin infusion protocols should be implemented for optimal glucose management.

Recommendation Table 21: Recommendations for Enteral Nutrition in Patients on tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
Early enteral nutrition should be considered within 24–48 hours of admission to the ICU.	Ila	B	(668, 670, 674, 675)
In patients requiring high-dose vasoactive support, it may be considered to postpone the introduction of full enteral nutrition.	Ilb	C	(669, 673, 674)

Abbreviations: ICU, intensive care unit; tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

15.10 Mobilization

Historically, the management of patients on tMCS involved prolonged bed rest to minimize the risk of haemodynamic instability, cannula displacement, and patient distress. However, many patients receive prolonged tMCS treatment, often lasting weeks or months. Prolonged immobilization in critically ill patients is known to negatively impact functional recovery and QoL among survivors. There is growing interest in promoting early ambulation in this population,⁶⁸¹ with the goals of reducing muscle wasting and peripheral neuropathy associated with immobility, preserving skin integrity, and enhancing psychological well-being. A recent systematic review of "awake ECLS" identified 30 studies published over the past decade, encompassing a total of 1,157 adult patients. Among them, 39.6% were on VA-ECLS and 56.8% on VV-ECLS. Femoral cannulation was used in 69% of cases, whereas 31% had upper-body cannulation. Of the 676 patients with reported physiotherapy data, 47.9% performed in-bed exercises, 9.3% could stand, and 42.7% were ambulatory. Notably, patients on VA-ECLS had a higher rate of ambulation (58.5%) than those on VV-ECLS (24.9%).⁶⁸² Early mobilization is therefore recommended to be initiated as soon as feasible in appropriate patients on tMCS. In those with femoral cannulation, conversion to axillary artery cannulation may be considered to facilitate ambulation when clinically appropriate.

The main challenges to mobilization are linked to risk of cannula dislodgement or circuit malfunction, agitation in awake patients, local trauma or bleeding at the cannulation site, respiratory compromise requiring re-intubation, and the need for a well-coordinated interdisciplinary team. Despite these concerns, it is noteworthy to emphasize that mechanical complications were rare, occurring in only 3.2% of femoral cannulation cases and none in cases with cervical cannulation.⁶⁸²

Study results suggest that early mobilization in patients on VA-ECLS is associated with higher ICU Mobility Scale scores, shorter durations of VA-ECLS support, reduced mechanical ventilation time, and decreased hospital length of stay.^{683,684} Similarly, in patients with extracorporeal VADs as a BTT, high mobility levels were associated with better nutritional status and shorter durations of mechanical ventilation.^{422,685-688} Although upper-body cannulation theoretically facilitates ambulation in patients on VA-ECMO, it is not without complications. These include bleeding at the axillary cannulation site, sometimes requiring surgical re-intervention and transfusions, if vascular graft was used for arterial access. Some

centres have successfully implemented ambulation protocols even for patients with femoral cannulation. A variety of assessment tools have been used to evaluate patient suitability and safety for mobilization. However, there remains no consensus on the optimal timing or rate of progression for mobilization in patients on tMCS.

Recommendation Table 22: Recommendations for Early Mobilization in Patients on tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
In appropriate patients, it is recommended that mobilization occur as soon as possible.	I	C	(683, 686-688)
In patients with femoral cannulation, relocation of the femoral cannula(s) should be considered to facilitate mobilization when feasible.	IIa	C	-

Abbreviation: tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

16. ORGAN AND MYOCARDIAL RECOVERY AND WEANING

The decision to wean from tMCS depends on achieving a balance between the degree of myocardial and end-organ recovery and the risks associated with ongoing mechanical support. During support, the overarching goal should be to transition the patient to the least invasive form of support that ensures adequate perfusion while enabling mobility.⁶⁸⁹

When considering weaning, it is important to recognize that VA-ECMO should be used for the shortest duration possible.^{29,690,691} In contrast, extracorporeal VADs or mAFPs are approved for longer durations, allowing a more gradual weaning process. The primary purpose of VA-ECMO is to stabilize the patient until a more durable and less invasive support strategy can be employed. Ideally, the duration of VA-ECMO should not exceed 10–14 days. An individualized plan should be developed early, defining the bridge strategy, whether it be to recovery, to a dLVAD, to an HTx, or to palliation.^{172,692} During tMCS support, myocardial

recovery must be continually evaluated, with the configuration adjusted accordingly, as discussed elsewhere in this document.

16.1 Evaluation and facilitation of organ recovery on tMCS

16.1.1 Optimal flow

Titration of systemic flow is essential to ensure adequate end-organ perfusion. Initial flow should target a cardiac index > 2.5 L/min/m². Although many tMCS modalities allow for CO measurement via a PAC, this is not feasible with VA-ECLS or RVADs, limiting the ability to determine total systemic flow. In such cases, flow should be titrated to maintain an optimal MAP while minimizing the need for vasoactive agents.⁶⁹³

The primary goal is to restore perfusion and oxygen delivery, as indicated by improving lactate clearance, pH normalization, and resolution of end-organ injury markers.⁶⁹⁴ In a series by Ohira et al., 7 patients supported with combined ECLS and mAFP demonstrated rapid lactate clearance and haemodynamic improvement due to high flows delivered via a combination of mAFP and VA-ECMO.¹²⁹ Similarly, Montero et al. retrospectively reviewed 209 patients with CS supported with VA-ECMO and found that patients with more severe illness required higher flows.⁶⁹⁵ Ultimately, flow targets should be individualized to achieve adequate end-organ perfusion and prevent further ischaemic injury.

16.1.2 Lactate and systemic markers of generalized recovery

End-organ injury is a key concern in CS.¹¹³ Serum lactate is a well-established prognostic marker in patients requiring ECLS. Rapid normalization of lactate levels following tMCS initiation is associated with better outcomes.⁶⁹⁶⁻⁷⁰⁰ A lactate level < 2 mmol/L at 72 hours is predictive of improved long-term prognosis.^{696,698} It is recommended that serial lactate measurements be performed throughout the duration of tMCS support.^{346,701}

16.1.3 Recovery of hepatic function

Hepatic injury, reflected by elevated levels of transaminases and bilirubin, occurs in approximately 25% of patients with CS^{702, 703} and in 40% of those with acute HF.^{403,702} Among patients not supported with tMCS, hepatic ischaemia is associated with a doubling of mortality risk.^{703,704} Fortunately, liver function typically normalizes with adequate circulatory support.^{413,692,705,706}

Most weaning protocols recommend waiting for normalization of LFT results before weaning is attempted. Persistently elevated LFT results should prompt further investigation, including

assessment of MCS cannula positioning and evaluation for other hepatic pathologies such as portal or hepatic vein thrombosis. Irreversible hepatic injury has a poor prognosis. Unfortunately, advanced cardiac therapies with a concomitant liver transplant remains limited to a few highly specialized centres.

16.1.4 Renal recovery

Acute kidney injury is common in CS shock, occurring in approximately one-third of patients, and it is associated with increased mortality.¹⁶⁵ Monitoring of continuous renal function is recommended for early detection and management of AKI. tMCS may facilitate renal recovery.^{413,692} A study of 124 patients on ECLS identified renal function recovery as a strong predictor of successful weaning from VA-ECLS.⁷⁰⁷ Although the need for continuous RRT is associated with higher mortality, its initiation can help prevent fluid overload and venous congestion.^{306,708-711} Unlike hepatic failure, long-term dialysis or a combined heart-kidney transplant is a viable pathway for some patients.^{712,713}

Recommendation Table 23: Recommendations for Organ Recovery

Recommendations	Class ^a	Level ^b	Ref ^c
In patients on tMCS, an initial combined output ≥ 2.5 L/m ² , depending on the patient's metabolic demands, is recommended.	I	C	-
In patients on tMCS, serial monitoring of lactate levels is recommended.	I	B	(346, 697, 698, 700, 701)
In patients on tMCS, serial monitoring of liver and renal functions is recommended.	I	C	(413, 692, 705, 706, 714)
In patients on univentricular support, serial monitoring of the function of the unsupported ventricle is recommended.	I	C	(715, 716)

Abbreviation: tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

16.2 Facilitating myocardial recovery on tMCS

Facilitating myocardial recovery is a central goal of tMCS. However, VA-ECLS increases afterload, which can lead to impaired aortic valve opening and reduced ejection in a compromised heart. This, in turn, elevates LVEDP, causes LV distension, increases LV end-diastolic wall stress, and may result in pulmonary congestion⁷¹⁷⁻⁷¹⁹ with a detrimental effect on the RV. To reduce these adverse effects, early LV decompression is recommended. This strategy has been shown to improve clinical outcomes, including reduced mortality and increased success in weaning from mechanical ventilation.^{435,720-722} In addition to mechanical unloading, addressing underlying or new emergency pathologies is critical to optimizing myocardial perfusion and function. These may include obstructive CAD, VSD, significant valvular disease, and haemodynamically relevant pericardial effusions.⁷²³ Maintaining an appropriate cardiac rhythm, ideally sinus, is also essential to optimize myocardial recovery. In patients with non-postoperative CS supported by peripheral VA-ECLS, the use of calcium sensitizers was not associated with improved rates of successful weaning or enhanced 30-day or 6-month bridge-free survival.⁷²⁴⁻⁷²⁸ At present, calcium sensitizers are not FDA-approved.

Recommendation Table 24: Recommendations for Myocardial Recovery

Recommendations	Class ^a	Level ^b	Ref ^c
Device selection			
In patients on tMCS, continuous assessment of myocardial recovery is recommended.	I	B	(691, 693, 729-738)
In patients with persistent end organ dysfunction, the initiation of weaning circulatory support is not recommended.	III	C	(692, 705-707)
In patients with respiratory insufficiency who are considered for weaning from VA-ECLS, initial conversion to V-AV- or VV-ECLS should be considered.	IIa	C	(693, 739)

Abbreviations: ECLS, extracorporeal life support; tMCS, temporary mechanical circulatory support; VA, veno-arterial; V-AV, veno-arterial-venous; VV, veno-venous.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

16.3 Post-tMCS considerations

As part of the ongoing assessment of patients supported with tMCS, long-term treatment strategies should be regularly re-evaluated and revised. Explanting tMCS should be performed as soon as technically and clinically feasible in patients who have achieved complete myocardial recovery. In cases of partial myocardial recovery, tMCS may be discontinued if the remaining myocardial function is sufficient to maintain adequate systemic perfusion and acceptable QoL. Alternatively, transition to dMCS or HTx should be considered. For patients not eligible for either option, withdrawal of tMCS should be approached within a palliative care framework.

Understanding the natural history of the underlying cardiac pathology is critical when considering weaning. In the presence of irreversible myocardial injury, such as ischaemic cardiomyopathy, chronic dilated cardiomyopathy, or untreated structural abnormalities, weaning trials may be detrimental and should be avoided. Conversely, other conditions may allow for delayed myocardial recovery, requiring prolonged support beyond the approved tMCS duration; in such scenarios, transition to dMCS should be discussed.

For patients supported with multiple tMCS modalities, such as VA-ECLS and mAFP, ECLS should be weaned first, if feasible, as recovery progresses. In the setting of isolated LV or RV failure, VA-ECLS may be transitioned to univentricular support.

Serial echocardiography is essential for monitoring myocardial recovery and detecting complications such as pericardial effusion, post-infarct VSD, acute MR, or severe AS. These findings should be addressed when possible to support weaning success and improve outcomes.

16.4 Weaning from tMCS

16.4.1 Determining patient suitability for weaning

Patients should be haemodynamically stable prior to initiating a weaning trial. Treatable, acute intercurrent conditions, such as infections, should be addressed beforehand to avoid

negatively impacting the weaning process.^{740,741} The systemic metabolic profile and markers of end-organ injury should trend towards normalization.^{694,714} Although complete renal recovery is not a strict requirement, fluid balance should be optimized, either through diuresis or RRT.

Besides the clinical signs, assessing the volume status is complex and involves multiple modalities. Although several tools exist, limited data are available comparing their effectiveness in determining optimal fluid status. Ultrasound and echocardiography can be particularly valuable for assessing volume status in patients on tMCS.⁷⁴²⁻⁷⁴⁵ Some weaning protocols incorporate target haemodynamic parameters for euvolaemia,^{732,746} including respiratory variation in IVC diameter, provided that imaging is not impeded by drainage or return cannulas.^{732,733,736,742-747} However, the simplest parameter, but one that is difficult to measure in immobilized patients, is the weight of the patient. The achievement of the target weight should be one of the parameters of the euvolaemia.

Haemodynamic stability should be optimized before attempting weaning from tMCS. Generally, an MAP > 60 mmHg with low-dose vasopressor support is recommended,^{693,739} although definitions of "low-dose vasopressors" vary among studies.^{739,748,749} The opening of the aortic valve is a favourable prognostic sign and is associated with a higher likelihood of myocardial recovery,^{691,738} and, in some protocols, it is considered a prerequisite for initiating weaning.^{693,739}

Pulmonary function should be adequate, with evidence of effective ventilation and oxygenation.^{693,739} Cavarocchi et al. suggested using the presence of clear lung fields on chest radiography and adequate systemic arterial oxygenation on 50% FiO₂ on both the ventilator and ECLS blender as readiness criteria.⁷³⁶

16.4.2 The mechanics of weaning from VA-ECLS and mAFP

Once adequate clinical stability has been achieved, the weaning process may begin (**Figure 8**). To promote consistency, efficiency, quality improvement, and staff education, programs are encouraged to standardize their VA-ECLS weaning protocols. Although randomized data comparing specific weaning protocols are lacking, observational studies have reported a variety of strategies and criteria associated with successful separation from support.^{29,435,690,691,693,694,732,736,738,739,746,749-754}

Weaning from tLVAD is typically performed over 2–3 days by gradually reducing flow. Once the patient is on minimal support, echocardiographic assessment should be performed.

Echocardiography during reduced tMCS flow has proven useful in evaluating myocardial recovery and the likelihood of successful weaning.^{729,731,733,736} Key echocardiographic parameters indicating readiness for weaning include LVEF^{691,693,729-732} and LVOT velocity-time integral (VTI) ≥ 12 cm.^{693,729,733} Qualitative and quantitative assessments of RV ejection have demonstrated utility in evaluating weaning readiness using^{729,734,755} tricuspid annular plane systolic excursion (TAPSE), tricuspid annular systolic velocity (S'), TAPSE/RV systolic pressure (RVSP) > 0.45 , and $S'/RVSP > 0.33$.^{729,734,735,755}

Signs of LV or RV distention during flow reduction should be interpreted as failure to wean.⁷³⁶ Favourable indicators for successful separation include improvement in pulse pressure,^{691,737,738} stable MAP, and non-rising CVP while on minimal vasopressor support. In patients with borderline LV function, a PAC and measurement of PA and wedge pressures may offer additional guidance and should be considered.

Certainly, it is recommended that an interdisciplinary team should guide the weaning process to ensure safety and optimize decision making. For patients on combined VA-ECLS and mAFP support, it is reasonable to wean VA-ECLS first, while continuing mAFP support, with plans for subsequent mAFP weaning (**Figures 9 and 10**).⁴³⁹

In the setting of adequate RV function (pulmonary artery pressure index [PAPi] > 0.9 , TAPSE/RVSP > 0.45 , absent RV distention under reduced flow, minimal increase in CVP) but LV failure (VTI, LV distention, low pulse pressure), patients should be transitioned from ECLS to longer-term tLVAD support. Conversely, if LV function is adequate but RV failure persists, transition to tRVAD support is recommended. During ECLS weaning with flows ≤ 2 L/min, strict anticoagulation management is essential. Bolus anticoagulation may be required if flow is temporarily stopped.

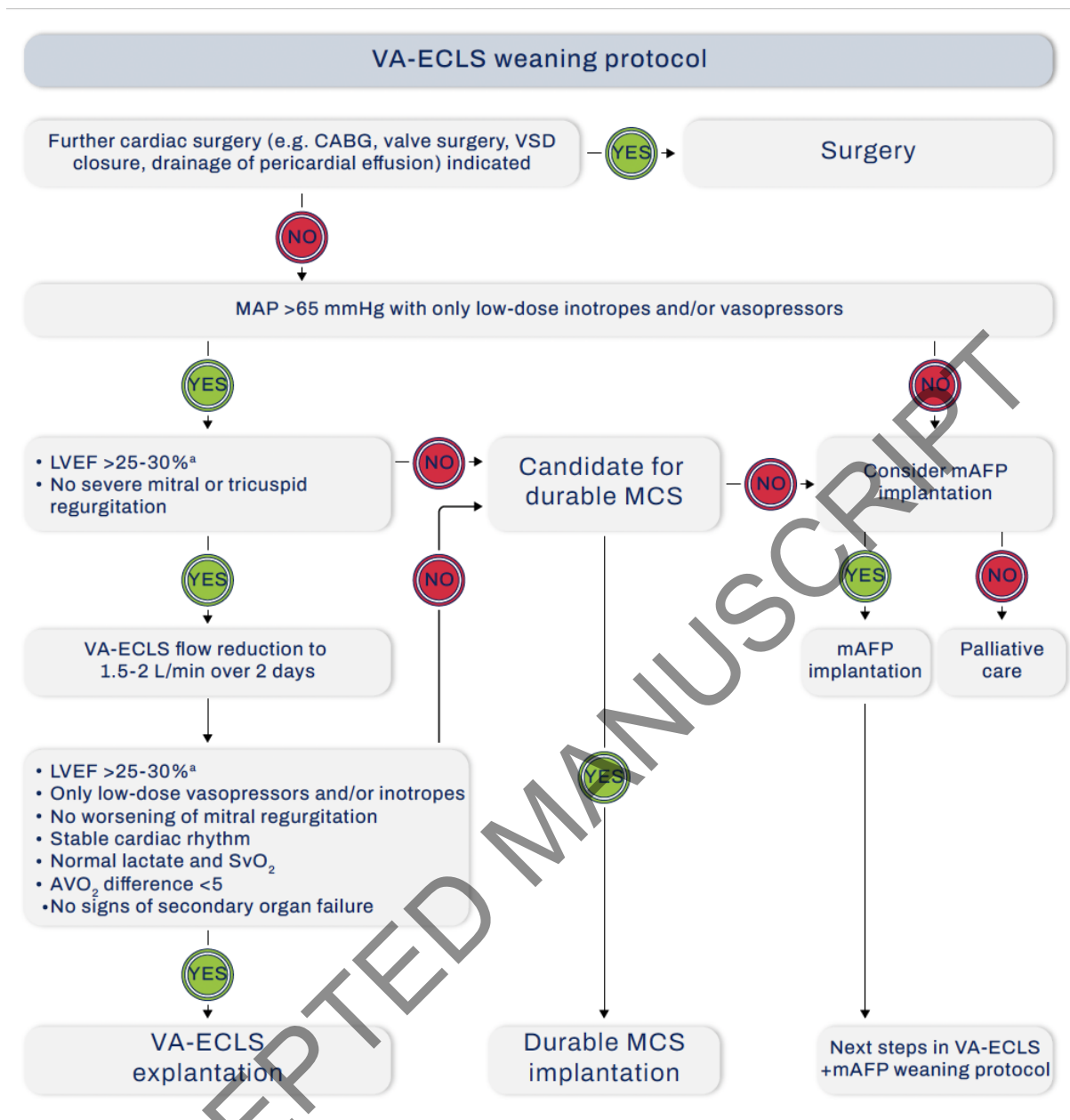


Figure 8. Proposed weaning protocol for VA- ECLS. AVO₂: Arteriovenous oxygen difference; CABG: Coronary artery bypass grafting; CVP: Central venous pressure; LVEF: Left ventricular ejection fraction; mAFP: Microaxial flow pump; MAP: Mean arterial pressure; MCS: Mechanical circulatory support; PCWP: Pulmonary capillary wedge pressure; SV: Stroke volume; SVO₂: Mixed venous oxygen saturation; VA- ECLS: Veno-arterial extracorporeal life support; VSD: Ventricular septal defect. Adapted from Nersesian et al.¹⁵ with permission from Multidisciplinary Digital Publishing Institute. ^aPatients with LVEF 25-30% require further evaluation, including in particular, but not exclusively measurement of SV, PCWP, CVP during pump reduction trial.

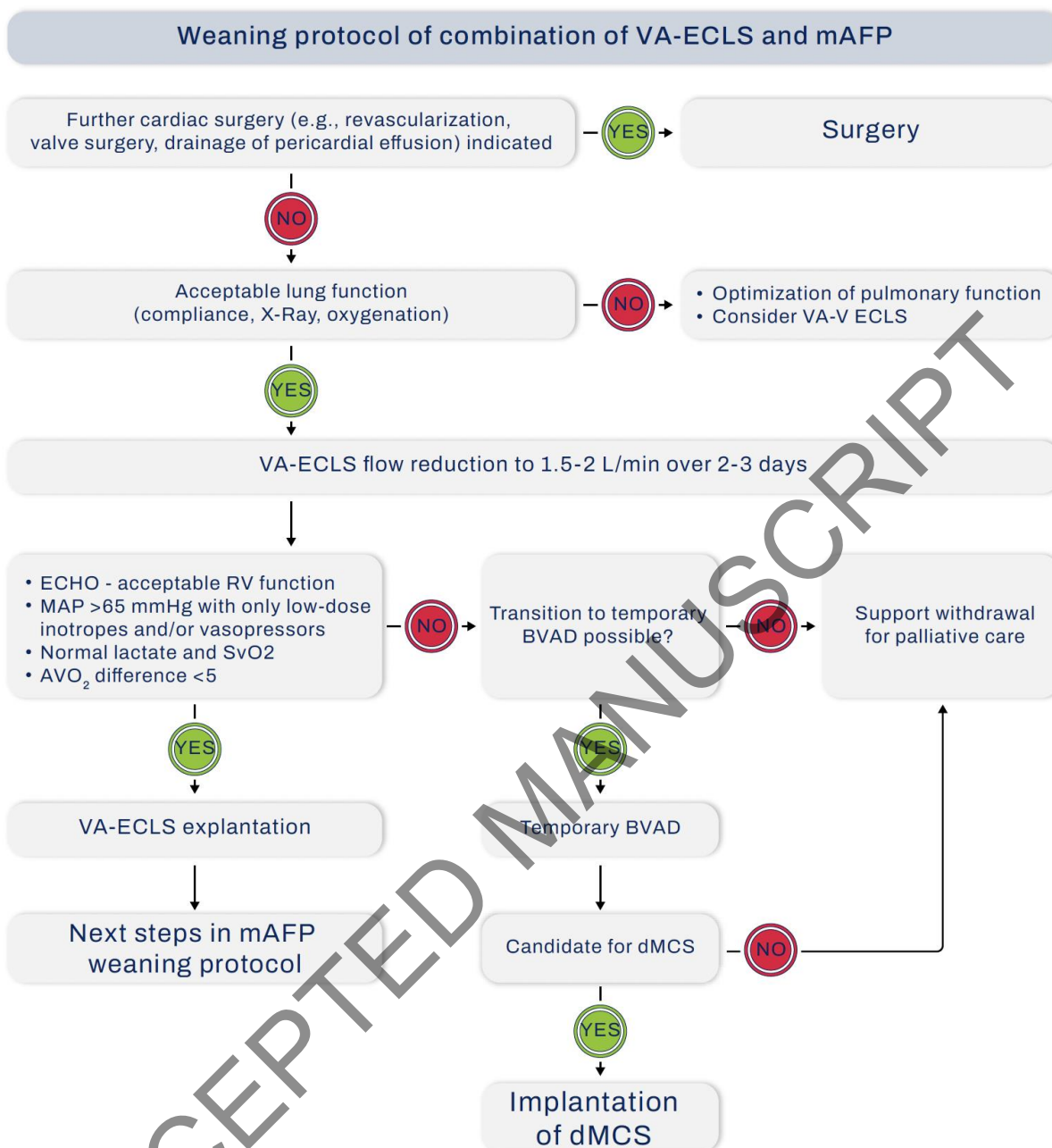


Figure 9. Proposed weaning Protocol of Combination of Veno-arterial Extracorporeal Life Support and Microaxial Flow Pump. AVO₂ indicates arteriovenous oxygen difference; CABG, coronary artery bypass graft; CVP, central venous pressure; LVEF, left ventricular ejection fraction; mAFP, microaxial flow pump; MAP, mean arterial pressure; MCS, mechanical circulatory support; PCWP, pulmonary capillary wedge pressure; SV, stroke volume; SvO₂, venous oxygen saturation; VA-ECLS, venous-arterial extracorporeal life support; VSD, ventricular septal defect. Adapted from Nersesian et al.¹⁵ with permission from Multidisciplinary Digital Publishing Institute.

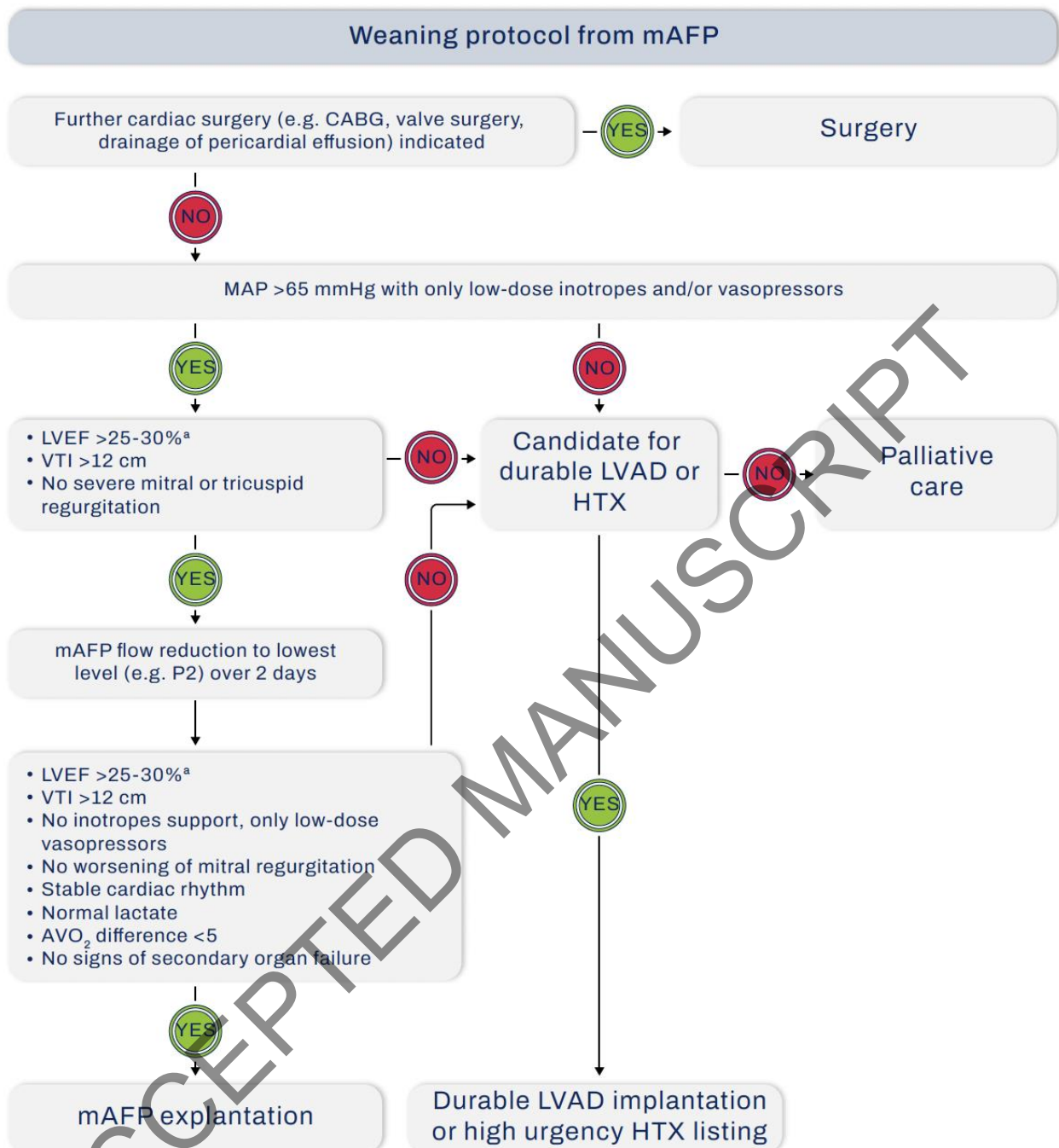


Figure 10. Proposed weaning protocol from mAFP. AVO₂ indicates arteriovenous oxygen difference; CABG, coronary artery bypass graft; CVP, central venous pressure; HTx, heart transplant; LVAD, left ventricular assist device; LVEF, left ventricular ejection fraction; mAFP, microaxial flow pump; MAP, mean arterial pressure; PCWP, pulmonary capillary wedge pressure; SV, stroke volume; VTI, velocity-time integral. Adapted from Nersesian et al.¹⁵ with permission from Multidisciplinary Digital Publishing Institute. ^aPatients with LVEF 25-30% require further evaluation, including in particular, but not exclusively measurement of SV, PCWP, CVP during pump reduction trial.

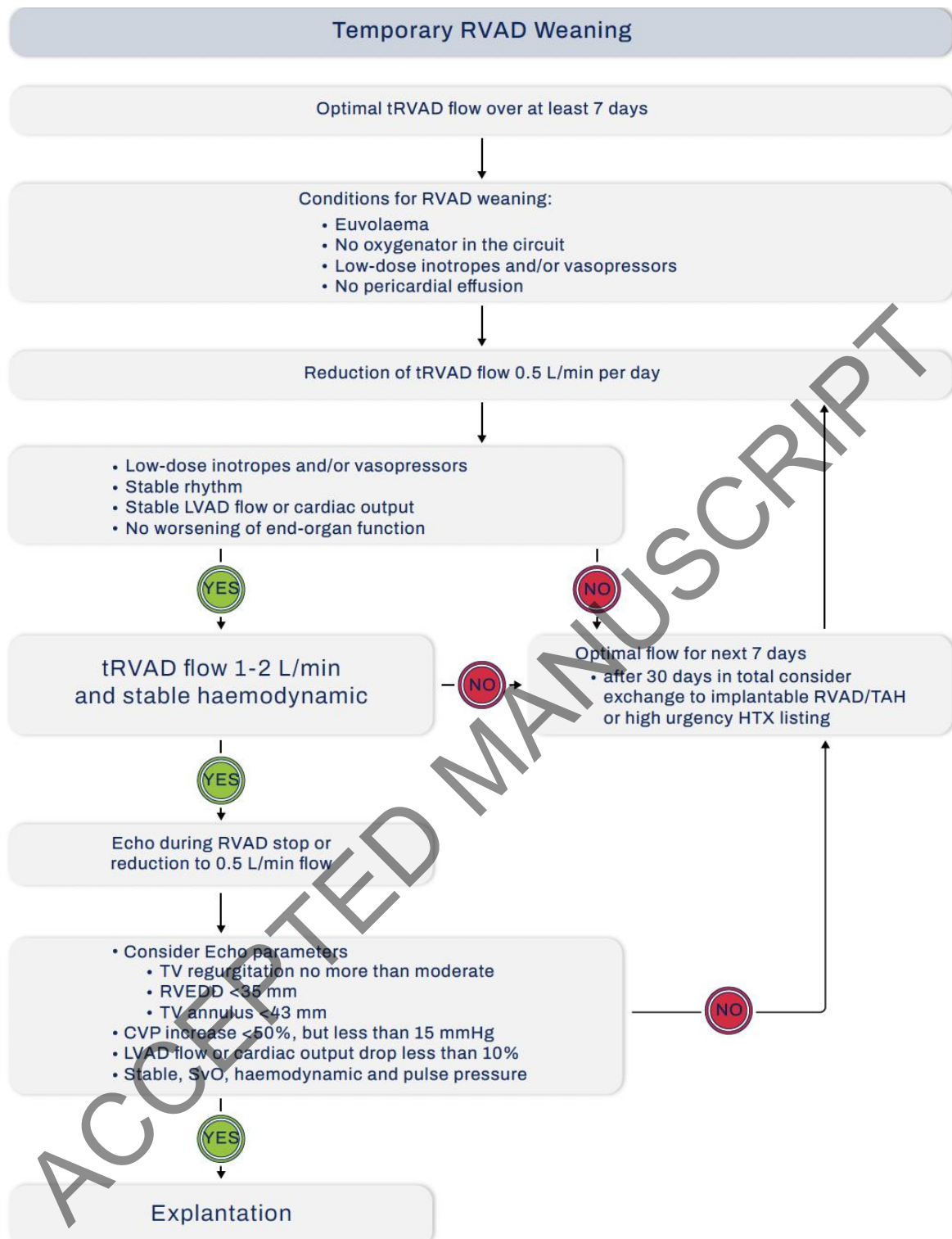


Figure 11. Proposed weaning protocol for tRVAD. CVP indicates central venous pressure; echo, echocardiogram; HTx, heart transplant; LVAD, left ventricular assist device; RVAD, right ventricular assist device; RVEDD, right-ventricular end-diastolic diameter; SvO, mixed venous oxygen blood saturation; TAH, total artificial heart; tRVAD, temporary RVAD; TV, tricuspid valve. Adapted from Nersesian et al.¹⁵ with permission from Multidisciplinary Digital Publishing Institute.

16.5 RVAD weaning

The criteria for weaning from tRVAD are similar to those for tLVAD weaning. However, in most cases, tRVAD is used in patients already supported with a dLVAD.⁷⁵⁶ In such patients, stable haemodynamics are largely dependent on optimal dLVAD function. Notably, severe tricuspid regurgitation or reduced RV contractility does not necessarily preclude successful tRVAD weaning. The key prerequisite for tRVAD weaning is stable haemodynamics with appropriate LV filling and stable dLVAD flow or stroke volume, particularly in patients who still retain native LV function. For patients supported with oxyRVAD, weaning should begin only after removal of the oxygenator, as suggested in [Figure 11](#). Due to differences in anatomical properties, working pressure, and volume response, weaning from the tRVAD usually takes longer than weaning from the tLVAD.

Recommendation Table 25: Recommendations for Weaning from tMCS^d

Recommendations	Class ^a	Level ^b	Ref ^c
Device selection			
In patients on tMCS, weaning may be considered when patients can maintain an MAP ≥ 65 mmHg (≥ 8.66 kPa) with minimal vasoactive support (VIS < 10).	IIb	C	(693, 739, 748, 749, 755)
Echocardiography is recommended to evaluate myocardial function for weaning.	I	C	-
Right heart catheterization should be considered to evaluate myocardial function for weaning.	IIa	C	-
In patients on tMCS, algorithms specific to individual support methods (VA-ECLS with or without mAFP, RVAD) should be considered for weaning. ^d	IIa	C	-
In patients undergoing echocardiographic evaluation for weaning from tMCS, it is not recommended to continue with weaning in the face of acute dilation of the RV or the LV. ^d	III	C	(736)

In patients on ECLS with both respiratory and RV recovery, transition to mAFP may be considered.	IIb	C	(690)
In patients on tMCS, it is recommended that weaning strategies be implemented with the guidance of an interdisciplinary Heart Team.	I	C	-

Abbreviations: ECLS: extracorporeal life support; LV: left ventricle; mAFP: microaxial flow pump; MAP: mean arterial pressure; RV: right ventricle; RVAD: right ventricular assist device; tMCS: temporary mechanical circulatory support; VA: veno-arterial; VIS: vasoactive inotropic score.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

^dSee flow charts in Figures 8-11 as companion guides to the recommendations.

17. TRANSPORTATION

Transitioning care to a higher-level facility is often in the patient's best interest, because access to interdisciplinary care, including HTx and dLVAD, may improve outcomes.⁷⁵⁷⁻⁷⁵⁹

However, interhospital transfers pose significant challenges for both the referring and receiving centres.⁷⁶⁰⁻⁷⁶² Transporting a patient in CS is a high-risk procedure that requires meticulous planning and coordination by both sending and receiving teams.⁷⁶³

The transport crew must be trained to anticipate and avoid complications, recognize adverse events early, and troubleshoot issues as they arise. The medical transport team should be fully self-sufficient for the duration of the journey in terms of clinical expertise, equipment, monitoring, consumables, and diagnostics.⁷⁶⁴ The required level of health-care supervision depends on the type and intensity of tMCS support provided (eg, requirements differ significantly for patients on IABP vs those on VA-ECLS or mAFP). Large case series have reported safe and successful transport of patients supported with different types of tMCS using various team configurations,^{531,761-763 765-770} and there is growing experience with the transport of patients on other tMCS devices.⁷⁷¹⁻⁷⁷⁷

The composition of the medical transport team for patients on tMCS is guided primarily by expert consensus.

For VA-ECLS-supported patients, ELSO recommends a team composed of the following members⁷⁷⁸:

1. An ECLS-experienced physician
2. A perfusionist or ECLS specialist
3. A transport nurse or respiratory therapist

For other patients on tMCS, the medical transport team may include:

1. A perfusionist or a tMCS operator
2. A transport nurse or respiratory therapist

If the designated team members lack expertise in essential tasks (eg, ventilator management), additional personnel with the required skills should accompany the patient to ensure safety during transfer.

Successful transport requires stabilization and continuous monitoring of vital signs, management of complex equipment, and thorough handover of the patient's clinical condition. A pre-transport checklist is strongly recommended to confirm the secure placement of tMCS devices, the readiness of essential equipment, and the availability of qualified personnel. Although tMCS may stabilize patients with refractory CS, it significantly increases the complexity of transport logistics.

17.1 Team Structure

17.1.1 Receiving Hospital

Appropriate management of patients on tMCS requires a dedicated interdisciplinary team. Ideally, this team should include cardiothoracic surgeons, interventional cardiologists, HF specialists, intensivists, cardiac-trained ICU nurses, perfusionists, and social workers. Administrative staff play a key role in managing logistics, such as securing emergency credentialing for procedures at referring hospitals, unless an "umbrella" agreement for emergency procedures is in place. Consultation from other specialties (e.g. nephrology, neurology, gastroenterology, infectious diseases, palliative care) should be readily available. At tertiary care centres, such experts increasingly collaborate as organized Shock Teams. These teams establish facility-specific infrastructure, coordinate communication channels and on-call systems, and implement standardized management algorithms, significantly streamlining the transport and transition processes.

Furthermore, transport capability requires a dedicated intake team available on a 24/7 basis, including an on-call physician to evaluate and troubleshoot the external patient and determine the need for transport, an on-call surgical team for cannulation and transport, and a pool of on-call tMCS specialists (eg, perfusionists).

Establishing interfacility relationships in advance improves the effectiveness of emergency transitions of care. Such collaboration allows a clear understanding of each hospital's capabilities, shared protocols and checklists, and expectations across the care continuum (eg, a "hub-and-spoke" model). This approach enables the referring hospital to act as an extension of the receiving centre. Clear communication and documentation between institutions and teams are essential for safe, coordinated transfer. Simulation-based training involving all stakeholders can enhance preparedness and team dynamics. Checklists should be integrated at every step of the process.

17.1.2 Transport team

Critical care paramedics are key members of the transport team, with essential expertise in ventilator management, vasoactive medications, and clinical documentation. For off-site cannulation, experienced cannulators (eg, surgeons or interventional cardiologists) must be included. Additional personnel, such as perfusionists or VA-ECLS specialists, respiratory therapists, and ICU-trained nurses, may accompany the team.

Prepacked and routinely checked tMCS transport bags are recommended to allow rapid mobilization. These should include back-up cannulas and supplies, extra disposable circuits, and a hand crank or backup motor. Transport teams must be trained in patient assessment, tMCS management, and safe transport procedures. Preplanning and simulation exercises are encouraged to maintain technical readiness and improve efficiency.⁷⁷⁸

Vehicle selection should consider crew and equipment needs, transport distance, and weather conditions. Safe transfer using ambulances, helicopters, or fixed-wing aircraft has been documented. The chosen vehicle must accommodate necessary personnel and equipment, have a functional electrical inverter for continuous AC power and provide 2 oxygen sources (1 for the ventilator and 1 for the oxygenator). Also, at low barometric pressure (eg, at higher altitudes), attention must be paid to potential hyperoxygenation due to reduced barometric pressure, although this is rare in pressurized cabins.⁷⁷⁸ Transport-capable centres should identify which vehicles in their fleet are suitable for tMCS transport.

This decision highlights the critical role of a dedicated logistics coordinator to ensure that the right personnel, equipment, and vehicle are available for safe patient transfer.

17.1.3 Referring hospital

Referring centres often have limited experience and resources for managing patients in CS due to low case volumes. Available local expertise should be fully utilized to stabilize the patient prior to transport. The local team should typically include cardiologists, intensivists, ICU nurses, and respiratory therapists. Cannulating physicians should also be engaged for patients with post-cardiotomy shock or for those planned for tMCS insertion prior to transfer.

17.2 Initiating the transfer process

A standardized referral form may be useful to streamline communication and documentation. The initial call from the referring hospital should ideally be made early in the course of the illness. This timing allows the receiving centre to advise on the evolving management strategy, potentially avoiding unnecessary transfer. Early communication also provides time to assemble the appropriate team members and organize logistics. The transfer process typically requires a series of direct, bidirectional communications and, in some cases, short delays to stabilize the patient. These interactions must be coordinated efficiently to ensure patient safety and optimize outcomes.

17.3 Initiating tMCS

Those making the decision to initiate tMCS at the referring hospital prior to transport must weigh the benefits of greater haemodynamic stability during transfer against the risks of potential delays and troubleshooting during transit. Key considerations include (1) duration of transport, (2) an adequate transport vehicle, (3) availability of an appropriate transport vehicle, (4) proficiency of on-site personnel, (5) whether the equipment can accompany the patient from the referring hospital, (6) device compatibility between referring and receiving centres, and (7) availability of necessary resources at the referring hospital (eg, fluoroscopy to guide cannulation).

17.4 Transporting the patient on tMCS

Upon arrival at the referring hospital, the transport team should perform a final assessment to determine the patient's suitability for transport and optimize overall stability. A device-

specific checklist should be used to verify system integrity and ensure patient safety. Cannulas and catheters must be properly secured at the insertion site, and all tubing connections should be tightened to prevent inadvertent decannulation during transport. Limb ischaemia should be identified and addressed whenever possible to prevent prolonged ischaemia and limb loss.⁷⁷⁹

With proper planning, major complications during transport are uncommon.⁷⁸⁰ However, the transport team must be prepared to address issues such as electrical failure, tubing dislodgement, or unexpected delays. The tMCS device should be firmly secured to prevent movement or malfunction, and approved transport units should be used whenever available. Irrespective of distance or mode of transport, the safety of the patient, staff, and the public must remain paramount, in line with local traffic regulations. Although high-quality evidence is limited, structured risk assessments can help minimize adverse events during tMCS transport.

A standardized equipment checklist before setting off should include (adapted from the ELSO guideline for transporting patients on VA-ECLS)⁷⁷⁸:

1. Portable monitoring equipment:
 - a. ECG and pulse oximetry
 - b. Invasive haemodynamic monitor
 - c. Ultrasound device for vascular imaging
 - d. Infusion pumps
2. Consumables: Intravenous fluid for infusion
3. Insertion kits: arterial and venous line kits, sterile drapes, gowns, gloves, etc.
4. Medications: heparin, inotropes, vasopressors, sedatives, analgesics, etc.
5. Blood products
6. Point-of-care test equipment: Anticoagulation testing, blood gas analysers
7. Transport ventilator, if clinically required
8. Sufficient medical gases, calculated for at least double the expected journey duration

17.5 Receiving the patient

By the time the patient arrives, the receiving team should be fully briefed and prepared. The patient should be delivered directly to the appropriate location, such as the catheterization laboratory, operating room, or intensive care unit. Upon arrival, the stabilization of both the

patient and the tMCS device(s) must be confirmed. The management plan should be immediately reviewed and discussed by the interdisciplinary Shock Team. A post-transport debriefing involving all relevant parties should be conducted to evaluate any adverse events, technical issues, or near-miss complications.

17.6 Transport considerations for specific devices

17.6.1 Veno-arterial extracorporeal life support

Literature on ECLS transport-related complications is limited.^{762,763,781} In a single-centre series involving 908 ECLS transports, complications occurred in 20% of cases but were not associated with increased mortality (2 deaths reported).⁷⁶³ Most complications were patient-related (62%), followed by equipment-related issues (19%), transport vehicle complications (13%), human error (5%), and environmental factors (2%). Another retrospective study of 221 patients transported on VA- or VV-ECLS reported the following: 18% electrical complications, 10% missing items, 9% circuit issues, 9% inadequate flows, 4% delays in travel, and 0.5% patient death (1 patient death due to refractory cardiac arrest on VV-ECLS).⁷⁶²

17.6.2 Microaxial flow pumps

Literature regarding the transport of patients on mAFP is scarce, too.^{764,782} A key concern is the risk of device dislodgement during transport. Because repositioning an mAFP is challenging even under echocardiographic guidance, repositioning en route is virtually impossible. Therefore, verifying optimal positioning and secure fixation of the catheter before departure is essential.

17.6.3 Intra-aortic balloon pump

Compared to other tMCS devices, the IABP has a favourable safety profile and is relatively easy to monitor during transport. Before transfer, proper positioning and securing sutures must be checked. Additionally, it is crucial to carry an extra helium tank to ensure uninterrupted pump function during transit.

17.6.4 Paracorporeal VADs

With the increasing use of mAFP, other forms of short-term external VADs are being used less frequently in CS.⁷⁸³ When transporting patients on paracorporeal VADs, the cannulation approach and insertion sites should be reviewed in advance. Cannula fixation and overall device stability must be confirmed. The transport team should include personnel trained in

the use of the specific device. Case reports from experienced centres have demonstrated the safety of this practice⁷⁸⁴⁻⁷⁸⁶

17.7 Transfer after post-cardiotomy CS

Transferring patients on tMCS due to post-cardiotomy cardiogenic shock requires close and timely communication between the referring and receiving surgical teams, including a detailed review of intraoperative findings. Haemostasis must be ensured prior to transfer, because these patients typically undergo complex procedures involving prolonged CPB and often experience significant bleeding. Appropriate cannula selection and anchoring are essential to prevent complications during transport. For detailed recommendations on the management of post-cardiotomy cardiogenic shock and the use of ECLS in this setting, we refer the reader to the previously published expert consensus document.¹

Recommendation Table 26: Recommendations for Transport on tMCS

Recommendations	Class ^a	Level ^b	Ref ^c
In hospitals without an established tMCS programme, early communication with an experienced tMCS hospital is recommended to discuss management, options, and possible transfer.	I	B	(760, 762, 763, 786, 787)

Abbreviation: tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

18. EXPLANTATION TECHNIQUES

Explantation techniques depend on the type of MCS and the method of insertion. In all cases, efforts should be made to use the least invasive approach possible to aid patient recovery.

18.1 Explantation of percutaneous tMCS

Following percutaneous implanting of VA-ECLS, decannulation can often be accomplished entirely percutaneously using vascular or suture-mediated closure devices.⁷⁸⁸⁻⁷⁹¹ This technique demonstrates a high technical success rate, reaching up to 93%,^{789,791} with recent data indicating a low incidence of bleeding, vascular, and wound-related complications.⁷⁹¹ Nevertheless, surgical decannulation should still be considered in selected cases to mitigate the risk of vascular injury.⁷⁸⁸ Despite advances in percutaneous closure methods, open vascular closure remains the preferred first-line approach. Historically, the use of manual compression as the primary decannulation strategy for peripheral percutaneous tMCS devices larger than 8F has been associated with higher rates of vascular complications requiring surgical intervention and is therefore not recommended as the sole method for achieving haemostasis.^{790,792}

For other tMCS devices inserted percutaneously via the femoral artery, the use of vascular closure devices during explantation should be considered.^{793,794} However, their applicability is typically limited to sheaths up to 18 Fr. When performing extended low-flow assessments during tMCS explantation, systemic anticoagulation should also be considered.⁷⁹⁵ Manual compression remains an appropriate and commonly used technique for IABP removal.⁷⁹⁶

In patients with peripheral arterial cannulation, immediate and serial assessments of distal limb perfusion are recommended following device explantation.⁷⁹⁷ Removal of femoral venous cannulas, such as those used for VA-ECLS or tRVAD, is generally straightforward and can be safely performed by temporarily constraining the tissue around the venous puncture location by employing a figure-of-eight suture combined with temporary manual groin compression.⁷⁹⁸

18.2 Explantation of surgically implanted tMCS

A surgically implanted VA-ECLS (femoro-femoral, axillo-femoral, or aorto-femoral) device is recommended to be explanted using an open surgical approach. When a vascular prosthesis is sutured to the axillary artery or ascending aorta, a minimally invasive explantation can be facilitated by ensuring that the prosthesis is of sufficient length and that fixation sutures are placed externally. A similar technique may be applied for explanting an mAFP inserted via the axillary artery or aorta using a vascular access graft.^{799,800} Once the pump is stopped and removed through the graft, the graft can be divided at the skin level and oversewn. Providing a generous intracorporeal buried graft length helps maintain sterility at the division point. The

closed stump then retracts into the wound, and the skin incision is closed over it. In selected cases, complete graft removal and vascular reconstruction may be necessary, especially in patients receiving immunosuppression after HTx. When both VA-ECLS and mAFP are implanted via the same prosthesis, termination of ECLS involves clamping and removing the arterial return cannula from 1 graft branch, which is subsequently ligated. The mAFP remains in place to allow staged weaning. Blood in the ECLS circuit is returned to the patient via the venous drainage cannula, which is then clamped and removed.⁸⁰¹

Cannulas tunnelled under the xiphoid for RVAD support can be explanted in a similar manner. As previously described, the pump is stopped, the device removed, and the graft oversewn and left within the chest. The closed graft stump retracts, and the skin incision is closed over it.⁸⁰²

Recommendation Table 27: Recommendations for Explanation Techniques

Recommendations	Class ^a	Level ^b	Ref ^c
In patients with peripheral cannulation, after tMCS explantation, it is recommended that distal limb perfusion be assessed immediately and serially.	I	C	(797)
During tMCS explantation, the use of systemic anticoagulation should be considered during extended low-flow assessment.	IIa	C	(795)
During the explantation of peripheral percutaneous tMCS > 8 Fr, relying only on direct compression for haemostasis is not recommended.	III	B	(790, 792, 793)
During explantation of peripheral ECLS, surgical removal of arterial cannula should be considered to mitigate vascular complications.	IIa	C	(797)
During explantation of percutaneously implanted mAFP, a percutaneous vascular closure device should be considered.	IIa	B	(793, 794)

In patients with an mAFP implanted surgically, it may be considered that the graft should be shortened and buried or completely removed after explantation.	IIb	C	(799-801)
In patients on a tRVAD using a PA graft, the explantation may be considered via the skin exit site without opening the primary incision.	IIb	C	(802)
In patients on tRVAD using a PA graft undergoing explantation, complete graft removal may be considered.	IIb	C	-
In patients undergoing removal of the venous cannula in tMCS, it is recommended to use a safety figure-of-eight suture and manual compression.	I	C	-
During explantation of peripheral ECLS, the use of a percutaneous vascular closure device may be considered.	IIb	C	-

Abbreviations: ECLS: extracorporeal life support; Fr, French scale; mAFP: microaxial flow pump; PA: pulmonary artery; tMCS: temporary mechanical circulatory support; tRVAD: temporary right ventricular assist device.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

19. INTERDISCIPLINARY TEAM

As in many other areas of medicine, a systematic and standardized interdisciplinary approach is essential for the effective management of CS and improved patient outcomes.⁸⁰³ The concept of team-based cardiovascular care is well established, with Heart Teams already involved in the decision-making process for various elective cardiac interventions.⁸⁰⁴ Recent retrospective studies have demonstrated lower risk-adjusted mortality in centres that employ dedicated CS teams^{187,188,805} with a consistently positive impact on survival.^{806,807} Additionally, care at high-volume centres offering 24/7 access to neurologic and cardiovascular intensive care specialists following cardiac arrest has been independently associated with improved

survival compared to treatment at lower-volume centres.⁸⁰⁸ Nevertheless, the primary obstacle remains the failure or delay in deploying tMCS. Although interdisciplinary decision making is ideal, it must not come at the cost of urgent patient care.

19.1 Structure of interdisciplinary teams

A dedicated CS team plays a critical role in improving the survival and overall outcomes of patients in acute haemodynamic compromise.^{187,188,804,806,809-811} It is recommended that CS team activation occur via a single-call “shock line,” available 24/7, to ensure immediate involvement of interventional cardiology, cardiovascular surgery, advanced heart failure, and critical care physicians for timely interdisciplinary decision making.⁸¹¹⁻⁸¹³ Additional team members may be engaged later in the process, as needed.

A specific subgroup of the CS team may include ECLS specialists who support care both within their own institution and through consultations with referring centres lacking ECLS capability. Beyond the core CS team, early involvement of other health-care professionals, such as nutritionists, physical and occupational therapists, palliative care specialists, and ethics consultants, is crucial to address long-term recovery goals and EOL considerations.⁸⁰³ In specific clinical contexts, additional specialists may be needed, including obstetricians for peripartum cardiomyopathy, trauma surgeons for trauma-related CS,⁸¹⁴ and neurologists for patients with acute brain injury.⁸¹⁵

Recommendation Table 28: Recommendations for the Interdisciplinary Team

Recommendations	Class ^a	Level ^b	Ref ^c
It is recommended that the Cardiogenic Shock Team include representatives from critical care, advanced heart failure and transplant cardiology, interventional cardiology, tMCS and cardiac surgery specialties, nutritionists, physical and occupational therapists, experts in palliative care, and an ethics committee member.	I	B	(803, 804, 806, 807, 809, 810, 812, 813)

It is recommended that the Cardiogenic Shock Team be activated through a 1-call “shock line” available 24/7 to involve physicians from interventional cardiology, cardiovascular surgery, advanced heart failure, and critical care for interdisciplinary consultation and decision making.	I	C	(803, 807, 810)
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Abbreviation: tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

20. NEED FOR STANDARDS PROTOCOLS AND ECLS EDUCATION AND TRAINING

The use of tMCS, although reliant on advanced technologies, requires comprehensive training for the entire care team.^{1,11,14,806,816-847} Establishing a clear framework outlining the required

knowledge base and organizational structure for tMCS patient management is essential. Core training elements should include in-depth knowledge of the organizational structure of the care team, indications and contraindications for tMCS, step-by-step procedural guidance for cannulation, and device specifications and requirements.^{1,11,816-818,845-847}

tMCS deployment demands significant experience due to the wide range of potential device- and treatment-related complications, many of which are life-threatening.^{1,11,816,817}

Competence in device management and troubleshooting cannot be achieved solely through occasional patient exposure; instead, repetitive and structured training is critical. As with all infrequent but emergency procedures, preparedness involves mastering not only the techniques for cardiopulmonary support, but also the skills needed to manage unexpected device-related issues.^{14,806,819-844,848-853}

Simulation training plays a central role in preparing for real-time emergencies. These exercises should emphasize both individual and team-based behavioural competencies, including leadership, communication, and collaboration among Shock Team members and tMCS caregivers.^{14,806,819-846,848-853} Clinical decisions should be based on interdisciplinary consensus by tMCS-trained clinicians.^{11,816-818,845-847}

Clearly defined roles and responsibilities are vital to effective team performance.^{1,11,816-818,845-}

⁸⁴⁷ Importantly, the team leader, similar to roles in advanced trauma life support, should not

be the proceduralist, whose focus is on cannulation rather than on overseeing overall resuscitation.^{1,11,816-818,845-847} Leadership detached from procedural tasks ensures coordination of the interdisciplinary team and has been associated with improved outcomes in trauma care.

Continuous performance evaluation is essential for quality improvement. Errors, whether major or minor, should be analysed, including the assignment of first responders and early patient preparation for tMCS initiation. Team debriefings following both real events and simulations enhance collective learning and future preparedness.^{1,11,816-818,845-847} Mandatory participation in safety briefings and error reporting systems should be standard practice for all team members.

Structured educational activities, such as classroom walk-throughs of procedures and scenario-based discussions, support knowledge retention and improve performance.^{14,806 819-830,832-844,848-853} Such environments are also ideal for discussing ethically complex decisions, like escalation or withdrawal of care, outside of emergency time constraints. In parallel, the use of quality dashboards with clearly defined metrics can guide performance improvement efforts.^{1,11,816-818,845-847}

Every tMCS programme should include a dedicated quality assurance team responsible for monitoring outcomes, reviewing structural processes, and standardizing care.^{1,11,14,806,816-841} This team should convene regularly (e.g., monthly) to evaluate outcomes and discuss dashboard data.

ELSO recommends comprehensive training modules incorporating didactics, wet laboratories, animal models, clinical experience, and written assessments.^{1,817,845,846} The curriculum spans nearly 20 topics, from ECLS history and system components to cannulation techniques, complications, and ethical considerations. Core cognitive content is provided in the ELSO Red Book and the ECLS Specialist Training Manual.^{845 846}

Several studies have shown that simulation-based tMCS training reduces errors, improves team dynamics, strengthens the safety culture, and enhances patient outcomes.^{14,806,819-830,832-841,848-853} High-fidelity simulations also improve psychological readiness by familiarizing teams with realistic emergencies, potentially reducing stress in actual clinical settings.⁸⁴⁷

Simulation is recognized as a valid complement to bedside learning, enabling knowledge transfer into real clinical environments. Both initial training and continuing education programs should emphasize multidisciplinary collaboration and technical mastery to reduce

device- and patient-related risks.^{14,806,819-830,832-840,848-853} Pre- and post-training questionnaires can help assess gains in knowledge, technical skills, behaviour, and confidence.

The first formal attempts to define required specialization and experience for tMCS teams date back to the 1990s.^{1,11,816-818,845-847} Modern interdisciplinary teams commonly include perfusionists, nurses, respiratory therapists, and intensivists. Formal designation of tMCS team members has been associated with improved survival, particularly when care is managed by full-time intensivist teams.^{14,806,819-830,832-841,848-853} This finding aligns with broader evidence supporting ICU organization models that include continuous physician presence and structured care processes. Finally, to assess practical experience, many hospitals use “ECLS care hours per year” as a proxy for hands-on competence, with a target of 75 hours annually per team member for credentialing.^{1,11,816-818,845-847}

Although continuing education programs may appear costly, these investments are justified by their prevention of medical errors, reduced morbidity, and improved rescue outcomes. Data from other high-risk industries underscore that the returns in performance and safety far outweigh the costs.

Recommendation Table 29: Recommendations for the tMCS Teams

Recommendations	Class ^a	Level ^b	Ref ^c
Routine didactic and simulation-based training is recommended.	I	B	(822, 827, 831, 838, 839, 844, 851, 853)
Time-out procedures and checklists are recommended to increase safety.	I	C	(778)
Institutional protocols for tailored device selection and treatment pathways are recommended to increase safety and outcomes.	I	B	(14, 833)
Routine team meetings and regular post-procedural debriefings are recommended to promote a culture of safety.	I	C	-

It is recommended that all ECLS programmes have a performance and quality improvement committee that meets at least monthly.	I	C	-
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Abbreviations: ECLS, extracorporeal life support; tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

21. ETHICS AND END-OF-LIFE CARE

Optimal end-of-life (EOL) care begins prior to implanting a tMCS device and should continue throughout the duration of support, particularly in patients with increasing comorbidities.^{815,854,855} The primary goals of palliative care (PC) in this context include symptom control, psychosocial support, and addressing spiritual concerns. EOL care in patients supported with tMCS involves multiple key components, including (1) discussion within the ICU team and engagement of palliative care specialists, (2) discussion with the patient (where possible), (3) preparation and involvement of the family and caregivers, and (4) planning and implementation of support withdrawal.

Long-term survival and QoL are difficult to predict in patients requiring tMCS. Fewer than 50% of adult patients supported with tMCS survive to hospital discharge,²³⁶ and a substantial proportion experience major complications.⁸⁵⁶ In many cases, clinical teams must weigh the patient's clinical condition, likely trajectory, any documented or verbalized advance directives, and family preferences when deciding on further treatment.⁸⁵⁷ Self-directed decision making is often not possible.⁸⁵⁸⁻⁸⁶⁰ Although the initiation of tMCS is typically a medical decision made in urgent situations, continuation or withdrawal of support should, where feasible, include input from the patient and surrogate decision makers.

Consequently, the care team should make a proactive effort to guide family members to ensure that treatment decisions align with the patient's values and care goals.^{861,862} Early communication about the potential risks, benefits, and limitations of tMCS is essential. Families should receive regular updates on the clinical status and be encouraged to take part

in care-planning discussions.^{863,864} A timeline for reassessment should be established, with clear objective indicators for evaluating recovery or futility.⁸⁶⁵

The uncertainty of prognosis, lack of familiarity with the technology, and the emotional burden of decision making contribute to significant psychological stress in families. Many experience long-lasting distress, and some develop symptoms of post-traumatic stress disorder years after the event.⁸⁶⁶⁻⁸⁶⁸

EOL care, including withdrawal of tMCS, becomes appropriate when organ recovery fails or when candidacy for HTx or dMCS cannot be achieved.^{869,870} Integrating PC professionals into the care team can support challenging conversations and facilitate communication with families.⁸⁷¹⁻⁸⁷⁴ Early, structured family meetings involving PC specialists and ethics consultants should be encouraged. Family presence and emotional support during the dying process must be prioritized.^{861,863,875,876}

When discussing care futility with family members, efforts should focus on shared decision making and supporting surrogate decision makers. If disagreements arise, ethics consultations may be necessary.^{861,875,877} The PC team should also offer emotional and logistical support to families. Appropriate use of sedatives, anxiolytics, and analgesics is essential to ensure patient comfort. The PAL-HF trial demonstrated that interdisciplinary PC for advanced heart failure significantly improves QoL, spiritual well-being, anxiety, and depression compared to conventional care.⁸⁷⁸

In countries where it is legally permitted, ECLS patients at EOL may be considered for organ donation, including donation after circulatory death.⁸⁷⁹

Recommendation Table 30: Recommendations for Decision Making about End-of-life Care

Recommendations	Class ^a	Level ^b	Ref ^c
Pre-emptive involvement of palliative care			
In patients planned for interventional or operative procedures with a high risk for LCOS, obtaining specific informed consent for the potential periprocedural use of tMCS is recommended.	I	C	-

An early advanced/palliative care team consultation should be considered.	Ila	B	(815, 855, 862)
End-of-life care			
An interdisciplinary model of shared decision-making care, including the patient and/or their representatives, is recommended.	I	C	(872, 875)
It is recommended that an early advanced/palliative care team consultation be obtained.	I	B	(862, 864, 872, 873, 876)
In hospitals offering tMCS, the development of institution-specific protocols for the end-of-life period should be considered.	Ila	B	(854, 874)
In countries where it is permitted, patients who are at the end of life and are on ECLS may be considered for organ donation, including DCD.	Ilb	C	(879)

Abbreviations: DCD, donation after circulatory death; ECLS, extracorporeal life support; LCOS, low cardiac output syndrome; tMCS, temporary mechanical circulatory support.

^aClass of recommendation.

^bLevel of evidence.

^cReferences.

22. GAPS IN KNOWLEDGE

Despite significant advancements in tMCS, several important areas of uncertainty remain. High-quality RCTs directly comparing different tMCS strategies, particularly in patients with CS of varying aetiologies, are lacking. Critical issues such as optimal timing for implanting the device, standardized weaning strategies, and refined patient selection criteria remain inadequately defined. Additionally, the role of institutional experience and infrastructure in influencing patient outcomes has not been systematically evaluated. Data on long-term outcomes beyond acute survival, such as functional recovery, QoL, and neurocognitive function, are sparse. Sex-specific differences in response to tMCS are also markedly under-

reported. Integration of tMCS with durable HF therapies, including long-term LVADs or HTx, is not well characterized. Furthermore, there is an urgent need for standardized definitions and uniform reporting of complications (e.g., haemolysis, limb ischaemia, device-related infections) to facilitate benchmarking, quality assurance, and outcome comparisons across centres.

23. CONCLUSIONS

These clinical-practice guidelines are the product of a broad, multidisciplinary collaboration aimed at optimizing the care of patients who require tMCS for a wide range of life-threatening conditions. Drawing on a rigorous appraisal of the current evidence base and intensive cross-specialty deliberation, the task force has translated complex clinical realities into clear, actionable recommendations. tMCS is now indispensable in the management of cardiogenic shock and in selected perioperative or high-risk interventional settings. Safe, effective therapy demands individualized strategies anchored in multidisciplinary teamwork, standardized protocols, and robust institutional infrastructure. Device selection and timing remain key determinants of outcome, underscoring the importance of early consultation with experienced centres.

Rapid advances in technology and shifting clinical landscapes will continue to create both opportunities and challenges. Accordingly, clinical decision making must stay dynamic, evidence driven, and centred on patient goals. To close persisting knowledge gaps, the field needs harmonized data standards, transparent outcome reporting, and sustained investment in multicentre research. Building collaborative networks and registries will be crucial for benchmarking results, driving quality improvement, and enabling equitable evaluation of future innovations. Finally, these guidelines are intended not only to inform present-day practice but also to catalyse ongoing progress in the delivery of tMCS, and, most importantly, to enhance the outcomes and QoL of the patients we serve.

24. SUPPLEMENTARY DATA

Supplementary data are available online.

DATA AVAILABILITY STATEMENT

No new data were generated or analysed in support of this document.

CONFLICT OF INTEREST

Rakesh Arora reports receiving consulting fees for sitting on advisory boards for BioPorto, HLS Therapeutics Inc., Edwards Lifesciences, Renibus Therapeutics Inc., and Alexion. Rakesh Arora reports receiving travel support for attending a scientific advisory board meeting for Renibus Therapeutics Inc., and from Alexion for a study investigative meeting. Rakesh Arora reports participating in advisory boards for the SCOPE Trial and DECIDE Trial. Rakesh Arora reports receiving payment for lectures from Medtronic, BioPorto, Edwards Lifesciences, Renibus Therapeutics Inc., HLS Therapeutics, and Alexion. Rakesh Arora reports a leadership position on the executive board of the ERAS Cardiac Surgery Society and being the chair of the STS Workforce for Stand Alone Meetings. E.E.C. de Waal reports receiving honoraria for lectures from Abbott and CytoSorbents and research grants from Abbott, CytoSorbents, and SmartQare. E.E.C. de Waal reports a leadership position as the co-chair of the EACTAIC Subcommittee for Transplantation and Assist Devices. Pia Lanmüller reports receiving grants or contracts from Abiomed (J&J). Pia Lanmüller reports receiving consulting fees from Abiomed (J&J). Pia Lanmüller reports receiving payments for lectures from Abiomed (J&J) and Abbott. Antonio Loforte reports receiving payments or honoraria for lectures from Abbott and Abiomed (J&J). Antonio Loforte reports receiving support from Abbott and Abiomed (J&J) for attending meetings. Antonio Loforte reports leadership positions as immediate past-president of ISMCS, ISMCS executive committee member, and member of the board of trustees. Antonio Loforte reports leadership positions on the ISHLT Cardiogenic Shock task force, ISHLT MCS operating committee, and on the ISHLT guidelines for the MCS task force. Roberto Lorusso reports receiving research grants from Medtronic and LivaNova. Roberto Lorusso reports receiving consulting fees from Medtronic and LivaNova (payments to the institution). Roberto Lorusso reports payment for lectures from Abiomed. Roberto Lorusso reports participation on advisory boards for Eurosets (not currently ongoing), HemoCue, ChinaBridge Medical, and Xenios (all payments made to the institution). Roberto Lorusso reports leadership roles as ELSO research committee chair, EuroELSO Honorary Secretary, and ESC Cardiovascular Surgery Working Group nucleus chair. Roberto Lorusso reports receiving ECMO-related material for animal experiments from Medtronic. David Morales reports

receiving consulting fees from Abbott, SynCardia, Azyo, and PECA Labs. David Morales reports participating on advisory boards as a member of the PediMacs and Intermacs Committees (unpaid). Ivan Netuka reports receiving grants or contracts as a consultant and investigator for Abbott, Carmat SA, and FineHeart SA. Ivan Netuka reports receiving consulting fees from Abbott, Carmat SA, and Abiomed. Ivan Netuka reports receiving payment or honoraria for lectures from Abbott, Carmat SA, and Abiomed. Ivan Netuka reports receiving travel support for attending meetings from Abbott, Carmat SA, and Abiomed. Ivan Netuka reports participation on advisory boards for Leviticus Cardio, Carmat SA, and FineHeart SA. Ivan Netuka reports a leadership role as a board member for Leviticus Cardio. Ivan Netuka reports holding stock options from Leviticus Cardio. Ivan Netuka reports holding mutual funds or individual stocks of medical companies as part of a private family investment portfolio. Milan Milojevic reports a leadership position as Director of the Clinical Practice Guidelines Programme at the European Association for Cardio-Thoracic Surgery. Milan Milojevic reports receiving a speaker fee from Corcym in relation to ERAS. Francis Pagani reports receiving support from BrioHealth Solutions for attending meetings. Francis Pagani reports being the National Principal Investigator for the INNOVATE Clinical Trial. Evgenij Potapov reports receiving consulting fees from Abbott, Abiomed, and Recovery Therapeutics. Evgenij Potapov reports receiving payment or honoraria for lectures from Abbott, Abiomed, and Recovery Therapeutics. Evgenij Potapov reports participating in an advisory board for Abbott. Ranjit John reports holding stock options from Calyan Technologies. Andrew Schaffer reports receiving consulting fees as a data safety monitoring board member for a clinical trial from Aria CV DSMB. Scott Silvestry reports receiving consulting fees from Abbott and Abiomed. Scott Silvestry reports receiving payment for lectures from Abbott and Abiomed. Scott Silvestry reports receiving payment for expert testimony from Abbott. Scott Silvestry reports participation on an advisory board for Carmat. Steven Tsui reports receiving grants or contracts as chief medical officer for Life Science Ltd (paid to the institution). Steven Tsui reports holding patent number 12350484 (Implantable co-pulsatile epi-ventricular circulatory support system with sutureless flow cannula assembly) and patent number 12233252 (Sutureless inflow cannula assembly for connecting ventricular assist devices to human circulation). Zuzana Tucanova reports receiving a study grant from Abbott Laboratories S.R.O. (payment to the institution). Glenn Whitman reports receiving consulting fees from Cellphire Therapeutics for a national RCT on cryopreserved platelets in cardiac surgery. Glenn Whitman

reports being in the design phase of a patent, 50% owned, for a safe cannulation assist device. Leora Yarboro reports receiving grants or contracts from Abbott, Medtronic, and Evaheart (all paid to the institution). Leora Yarboro reports receiving reimbursement for travel from Arthrex for attending meetings. Leora Yarboro reports a pending patent for a universal heart port. Leora Yarboro reports leadership positions on the Society of Thoracic Surgeons' national workforce on surgical treatment of end stage cardiopulmonary disease, as 2023 chair of the South Atlantic Cardiovascular Society, as secretary of the Southern Thoracic Surgical Association, and as secretary of the 21st Cardiothoracic Surgical Society. Daniel Zimpfer reports receiving grants or contracts from Edwards, Abbott, Berlin Heart, Corcym, Abiomed, and Atricure. Daniel Zimpfer reports receiving consulting fees from Abiomed and Abbott. Daniel Zimpfer reports receiving payment or honoraria for lectures from Edwards, Corcym, Abbott, Abiomed, Berlin Heart, and Atricure. Daniel Zimpfer reports receiving support for attending meetings from Medtronic, Edwards, Novomed, Fa. Duschek, Biotronik, Bimed, Abbott, St. Jude Medical, Maquet, HeartWare, Sorin Medical, Janssen, Johnson, J.C. General, B. Braun, Boston Scientific, Baxter, Euromed, Getinge, MIN Medical Innovation Network, Actelion, CryoLife, Vascutek, HSEC, Neucomed, Biomedica, Hirmann, Artivion, Takeda, C. Brady, Atricure, CytoSorbents, Covidien, Guidant, ATS, LSI Solutions, Böck, Nycomed, Ducest, Pfizer, Comesa, Tyco, Santec, W. L. Gore, MAC's Medical, Philips, Intramed, and Roche. The other authors report no conflicts of interest.

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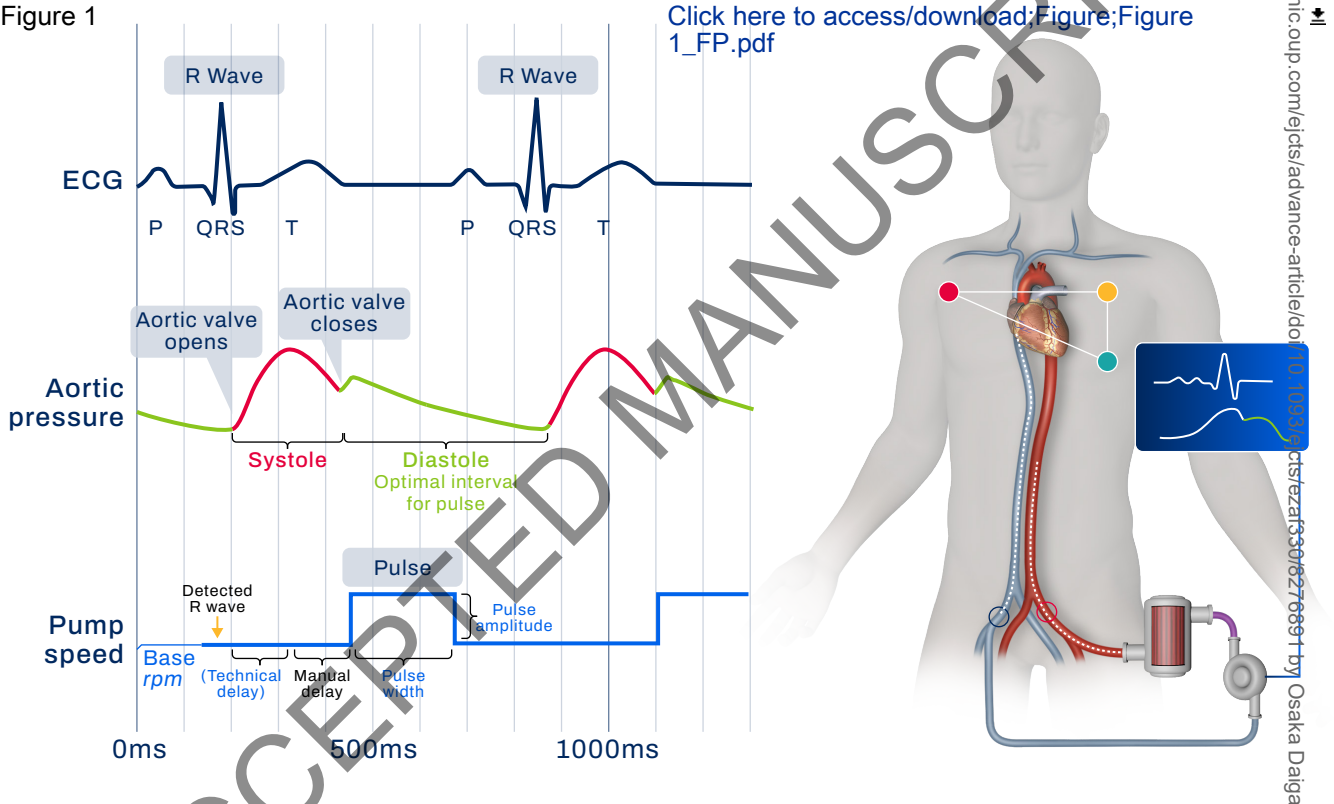
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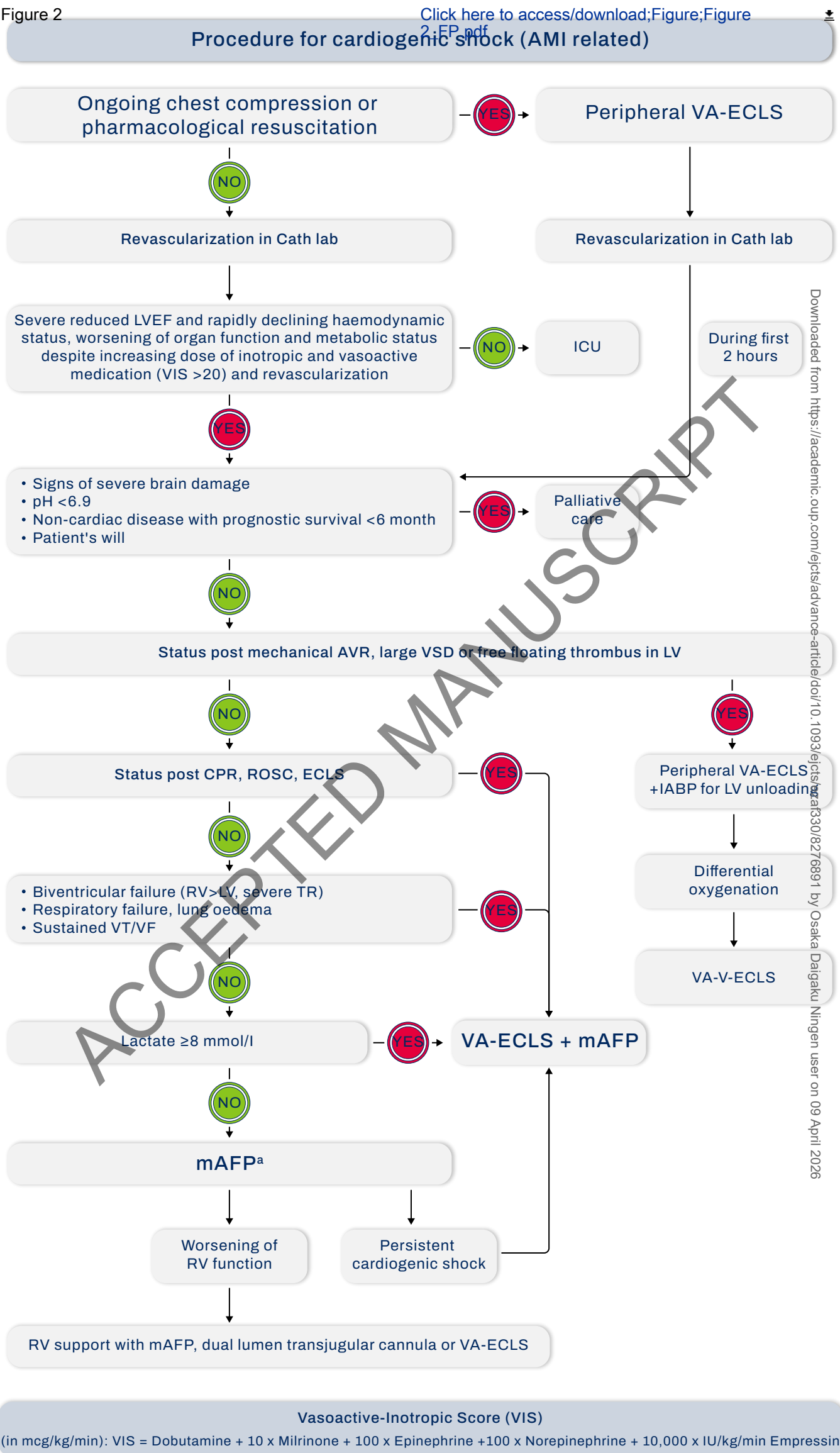
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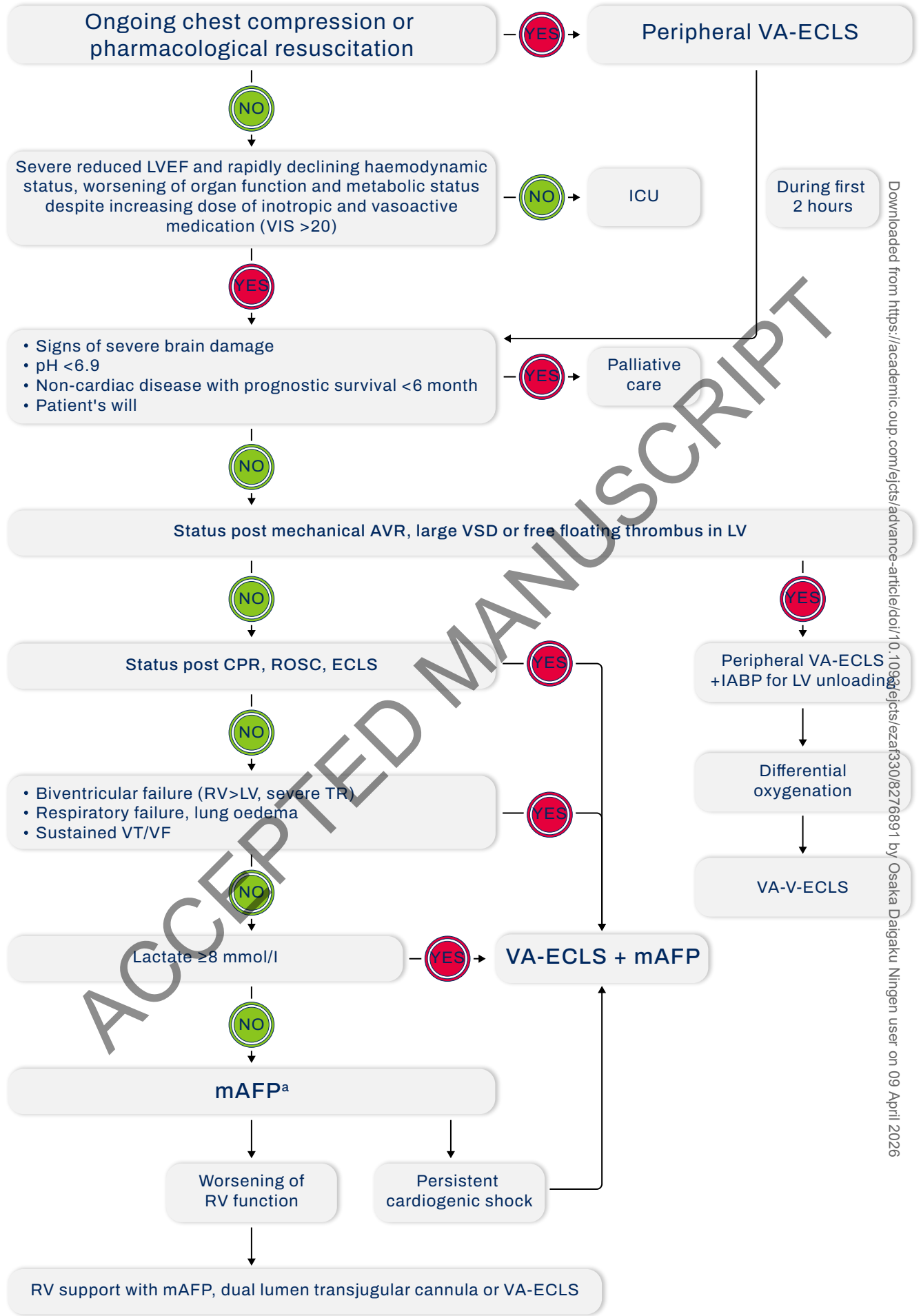
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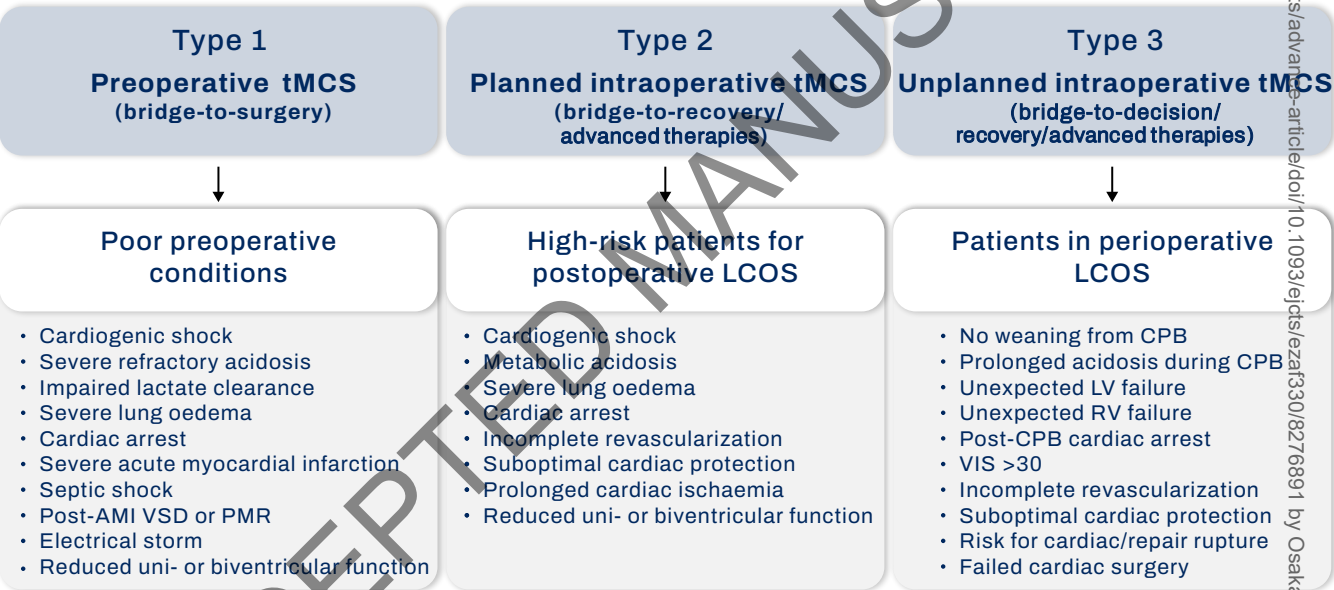
Figure 1

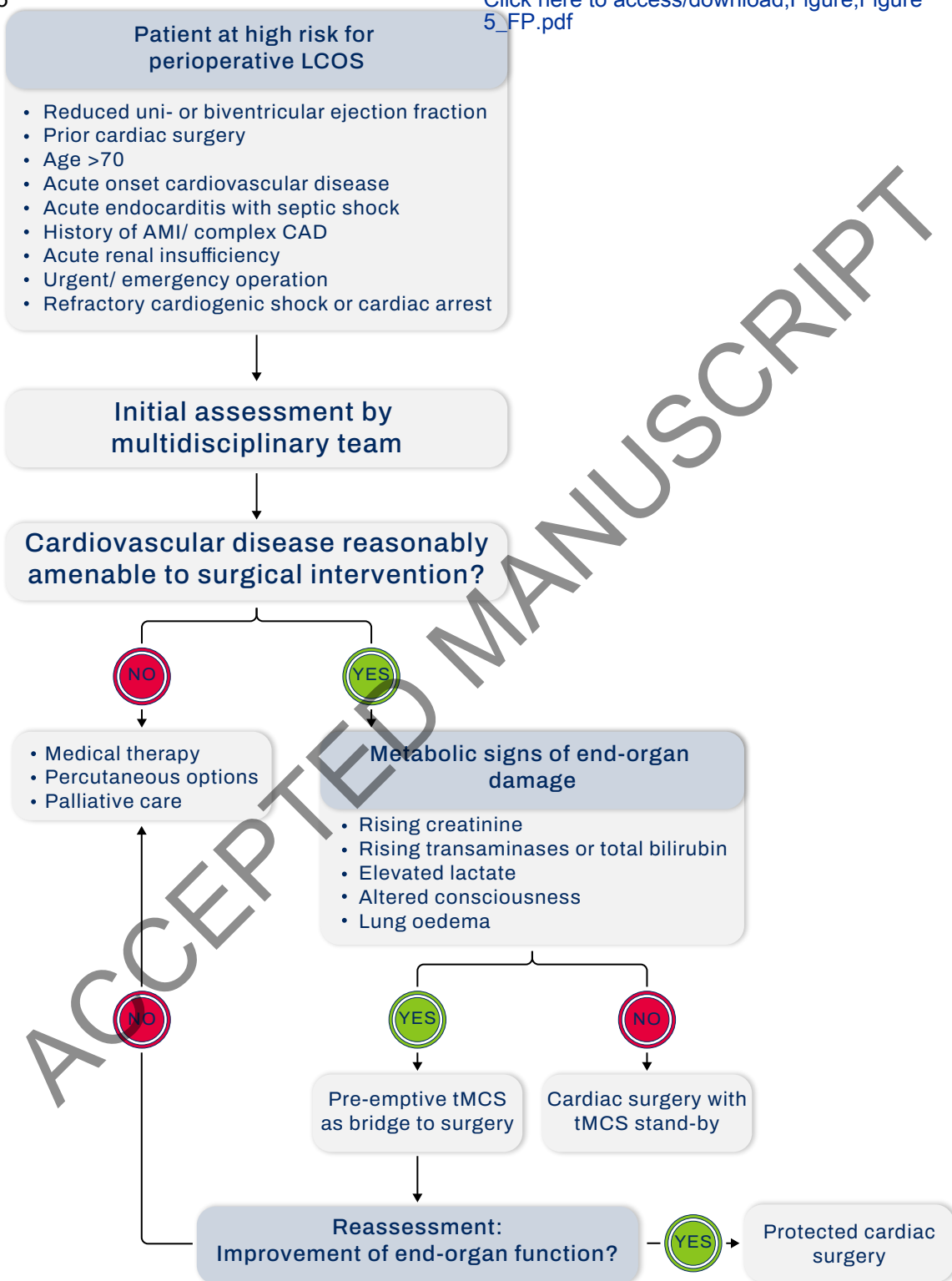


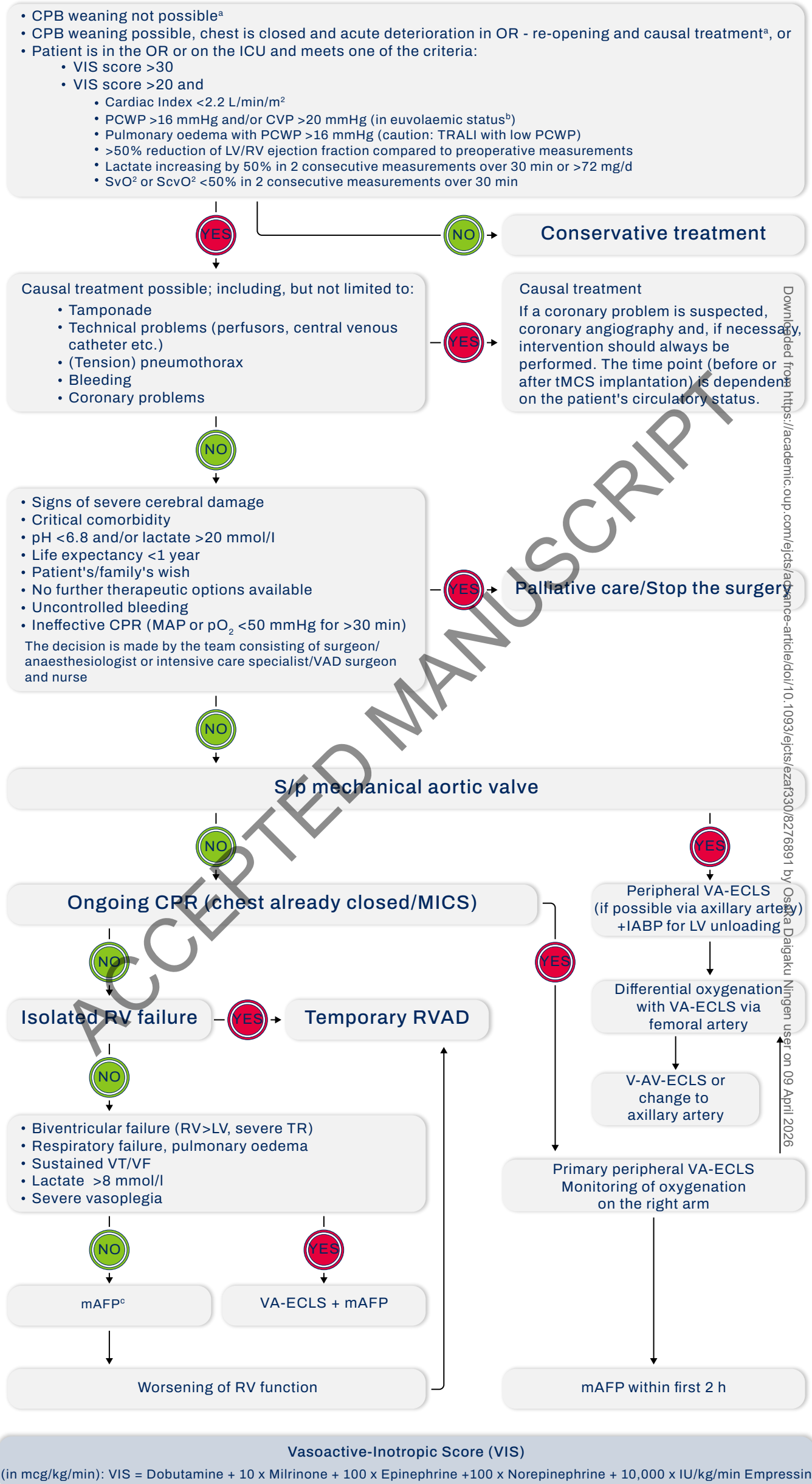


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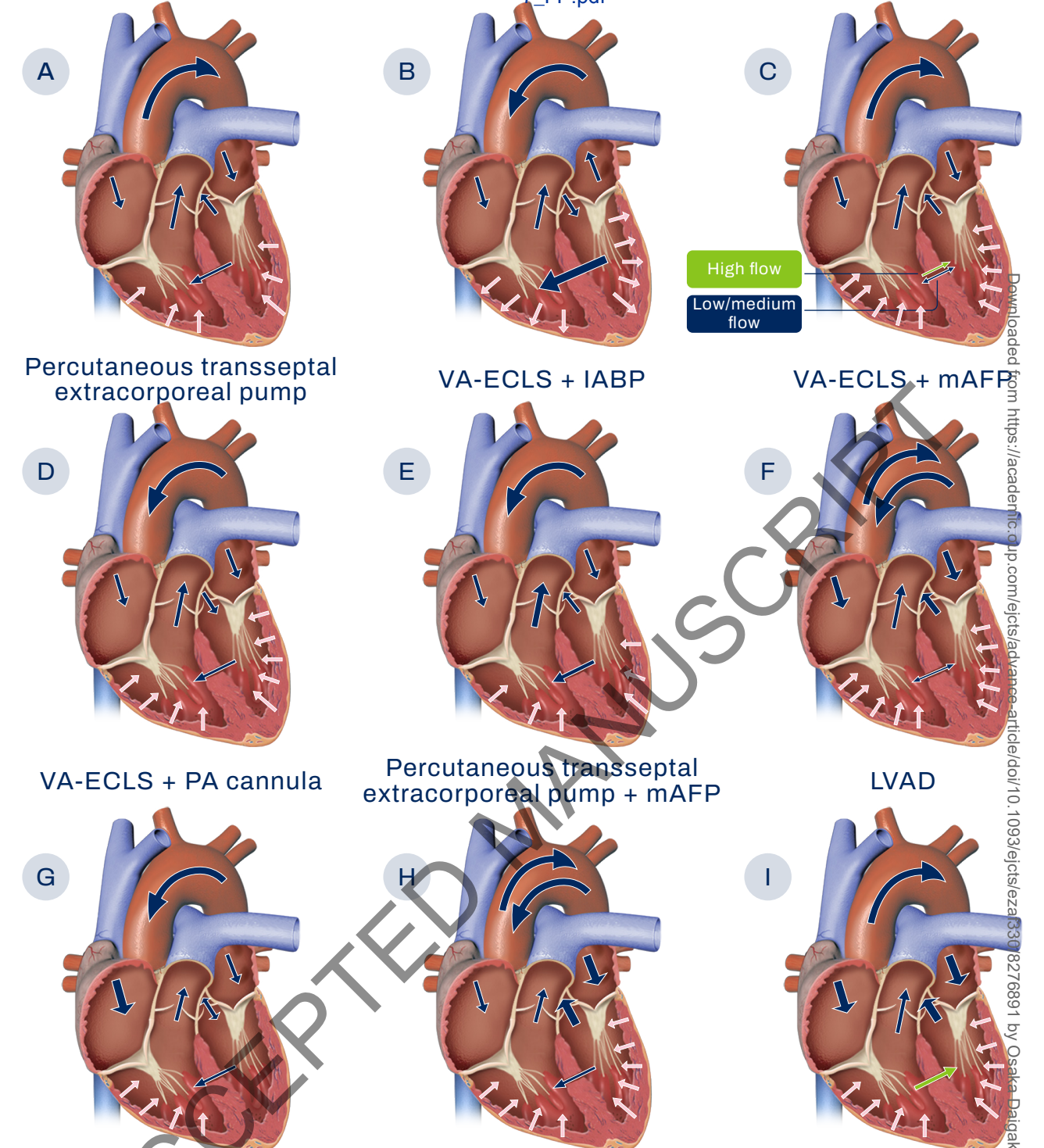








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		Afterload	Preload (PCWP)	PAP	CVP	CO	LV wall stress	RV wall stress	Shunt
A	IABP	↓↓	↓	↓	↓	↑	↓	↓	↓
B	VA-ECLS	↑↑↑	↑↑	↑↑↑	↓	↓	↑↑↑	↑↑↑	↑↑↑
C	mAFP	↓↓↓	↓↓↓	↓↓↓	↓↓	↑↑↑	↓↓↓	↓↓↓	↓↓↓ or inversion
D	Percutaneous transseptal extracorporeal pump	↑↑↑	↓↓↓	↓↓↓	↓↓	↓	↓↓	↓↓↓	↓↓
E	VA-ECLS + IABP	↓	↓	↑	↓↓	↑	↓	↓	↑
F	VA-ECLS + mAFP	↓ or ↑	↓↓↓	↓↓↓	↓↓↓	↑↑	↓↓↓	↓↓↓	↓↓↓
G	VA-ECLS + PA cannula	↓	↓↓↓	↓↓↓	↓↓↓	↑	↓↓	↓↓↓	↑
H	Percutaneous transseptal extracorporeal pump + mAFP	↓ or ↑	↓↓↓	↓↓↓	↓↓↓	↑↑	↓↓↓	↓↓↓	inversion
I	LVAD	↓↓↓	↓↓↓	↓↓↓	↓↓	↑↑↑	↓↓↓	↓↓↓	inversion

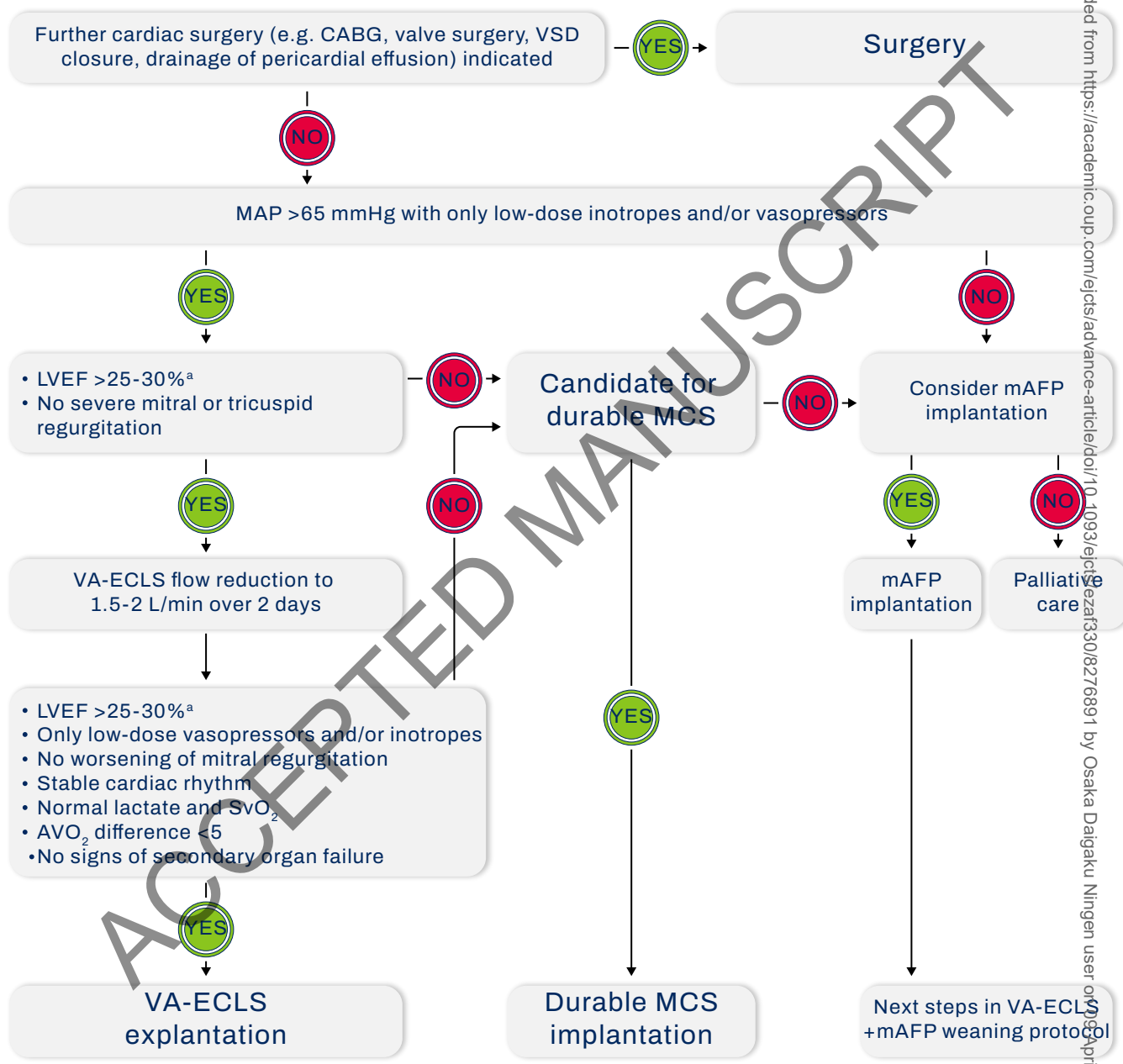


Figure 9

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Weaning protocol of combination of VA-ECLS and mAFP

